



SAPIENZA
UNIVERSITÀ DI ROMA

Linear Algebra

Vectors, matrices, norms and linear systems

Lecture notes with definitions, results and proofs, examples worked out step by step and exercises with solutions at the end of each chapter. Sums and products; vectors and linear independence; matrices, determinants and rank; elementary operations and Gaussian elimination; inverse matrix, LU factorization, eigenvalues; norms; linear systems.

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Online version and downloadable material:

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Part I

Sums and products

Sums

1.1 Definition

Definition 1.1: sum

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **sum**

$$a_1 + a_2 + \dots + a_n$$

can be indicated in compact form with the sum symbol:

$$\sum_{j=1}^n a_j$$

which reads: “sum for j from 1 to n of a_j ”. The symbol j is called **sum index**.

- The sum symbol is therefore very useful when the terms a_j are explicitly defined as a function of the sum index j .

Example 1.1: sums

$$\sum_{j=1}^{10} \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{j=3}^n j^2 = 3^2 + 4^2 + 5^2 + \dots + n^2$$

- The sum index is a **dummy index**. This means that if j is replaced with i , k or any other index in all its occurrences the value of the sum does not change.

Example 1.2: dummy index

We have:

$$\sum_{j=1}^n j^2 = \sum_{i=1}^n i^2 = \sum_{k=1}^n k^2$$

as the two symbols indicate the sum of the squares of the first n natural numbers (without zero). On the contrary, we have:

$$\sum_{j=1}^n j^2 \neq \sum_{j=1}^m j^2$$

as the two symbols indicate the sum, respectively, of the first n and the first m squares of the natural numbers (without zero). Clearly, if $n \neq m$, the result is different.

1.2 Properties

Observation 1.2: sum with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\sum_{j=1}^n c = n c \quad (1.1)$$

Proof. We have:

$$\underbrace{c + c + \dots + c}_{= \sum_{j=1}^n c, \quad c \text{ summed up } n \text{ times}} = n c$$

□

Observation 1.3: product by a constant

For every sum $\sum_{j=1}^n a_j$ and real number c , we have:

$$\sum_{j=1}^n (c a_j) = c \sum_{j=1}^n a_j \quad (1.2)$$

Proof. From the distributive property, we have:

$$\underbrace{c a_1 + c a_2 + \dots + c a_n}_{= \sum_{j=1}^n (c a_j)} = \underbrace{c (a_1 + a_2 + \dots + a_n)}_{= c \sum_{j=1}^n a_j}$$

□

Observation 1.4: union of sums

For every pair of sums $\sum_{j=1}^n a_j$ and $\sum_{j=1}^n b_j$, we have:

$$\sum_{j=1}^n a_j + \sum_{j=1}^n b_j = \sum_{j=1}^n (a_j + b_j) \quad (1.3)$$

Proof. We have:

$$\underbrace{a_1 + a_2 + \dots + a_n + b_1 + b_2 + \dots + b_n}_{=\sum_{j=1}^n a_j + \sum_{j=1}^n b_j} = \underbrace{a_1 + b_1 + a_2 + b_2 + \dots + a_n + b_n}_{=\sum_{j=1}^n (a_j + b_j)}$$

□

Observation 1.5: decomposition

For every sum $\sum_{j=1}^{n+m} a_j$, we have:

$$\sum_{j=1}^{n+m} a_j = \sum_{j=1}^n a_j + \sum_{j=n+1}^{n+m} a_j \quad (1.4)$$

Observation 1.6: index translation

For every sum $\sum_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1+m}^{n+m} a_{j-m} = \sum_{j=1-m}^{n-m} a_{j+m} \quad (1.5)$$

Observation 1.7: index reflexion

For every sum $\sum_{j=1}^n a_j$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1}^n a_{n-j+1} = \sum_{j=0}^{n-1} a_{n-j} \quad (1.6)$$

Proof. The three properties are simply different notations and/or arrangements of the terms in the sums. □

1.3 Some important sums

Observation 1.8: sum of the first n positive natural numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n j = \frac{n^2 + n}{2} \quad (1.7)$$

Proof. We have:

$$\begin{aligned} \sum_{j=1}^n j &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n j \right) \\ &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n (n - j + 1) \right) \\ &= \frac{1}{2} \sum_{j=1}^n (j + n - j + 1) \\ &= \frac{1}{2} \sum_{j=1}^n (n + 1) = \frac{n(n + 1)}{2} = \frac{n^2 + n}{2} \end{aligned}$$

□

Observation 1.9: sum of the first n positive even numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n 2j = n^2 + n \quad (1.8)$$

Proof. We have:

$$\sum_{j=1}^n 2j = 2 \sum_{j=1}^n j = 2 \left(\frac{n^2 + n}{2} \right) = n^2 + n$$

□

Observation 1.10: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=0}^{n-1} (2j+1) = n^2 \quad \text{or equivalently} \quad \sum_{j=1}^n (2j-1) = n^2$$

Proof. We have:

$$\begin{aligned} \sum_{j=0}^{n-1} (2j+1) &= 2 \sum_{j=0}^{n-1} j + \sum_{j=0}^{n-1} 1 = 2 \sum_{j=1}^n (j-1) + \sum_{j=1}^n 1 \\ &= 2 \left(\sum_{j=1}^n j - \sum_{j=1}^n 1 \right) + n \\ &= 2 \left(\frac{n^2+n}{2} - n \right) + n = n^2 + n - 2n + n = n^2 \end{aligned}$$

$$\begin{aligned} \sum_{j=1}^n (2j-1) &= 2 \sum_{j=1}^n j - \sum_{j=1}^n 1 \\ &= 2 \left(\frac{n^2+n}{2} \right) - n = n^2 + n - n = n^2 \end{aligned}$$

□

1.4 Exercises

Exercise 1.1

Prove, using the properties of the sums, that for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{6} \quad (1.9)$$

$= \frac{n(n+1)(2n+1)}{6}$

(sum of the squares of the first n natural numbers without zero).

Solution

We start by writing $\sum_{k=0}^n (k+1)^3$ in two different ways:

$$\begin{aligned}
 1) \quad \sum_{k=0}^n (k+1)^3 &= \sum_{k=0}^n (k^3 + 3k^2 + 3k + 1) \\
 &= \left(\sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + 3 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\
 &= \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \frac{3(n^2 + n)}{2} + n + 1
 \end{aligned}$$

$$2) \quad \sum_{k=0}^n (k+1)^3 = \sum_{k=1}^{n+1} k^3 = \sum_{k=1}^n k^3 + (n+1)^3$$

(in 1) the term with $k = 0$, equal to 1, has been separated from the others). Equating the expressions and canceling the sum with k^3 we get:

$$3 \sum_{k=1}^n k^2 + \frac{3(n^2 + n)}{2} + n + 1 = (n+1)^3$$

Isolating the sum with k^2 we get:

$$\begin{aligned}
 3 \sum_{k=1}^n k^2 &= (n+1)^3 - \frac{3(n^2 + n)}{2} - n - 1 \\
 &= n^3 + 3n^2 + 3n + 1 - \frac{3n^2 + 3n}{2} - n - 1 \\
 &= \frac{2n^3 + 3n^2 + n}{2}
 \end{aligned}$$

Then we have:

$$\sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{6}$$

Exercise 1.2

Compute the sum of the squares of the first 10, 100 and 1000 natural numbers (without zero).

Solution

Using formula (1.9), the sum of squares of the first 10 natural numbers (without zero) is:

$$\sum_{k=1}^{10} k^2 = \frac{2 \cdot 10^3 + 3 \cdot 10^2 + 10}{6} = 385$$

The sum of squares of the first 100 natural numbers (without zero) is:

$$\sum_{k=1}^{100} k^2 = \frac{2 \cdot 100^3 + 3 \cdot 100^2 + 100}{6} = 338.350$$

The sum of squares of the first 1000 natural numbers (without zero) is:

$$\sum_{k=1}^{1000} k^2 = \frac{2 \cdot 1000^3 + 3 \cdot 1000^2 + 1000}{6} = 333.833.500$$

Exercise 1.3

Prove, using the properties of the sums, that for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4} \quad (1.10)$$

$$= \frac{n^2(n+1)^2}{4} = \frac{(n^2+n)^2}{4}$$

(sum of the cubes of the first n natural numbers without zero).

Solution

We start by writing $\sum_{k=0}^n (k+1)^4$ in two different ways, using formula (1.9) for the sum of the squares:

$$\begin{aligned} 1) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=0}^n (k^4 + 4k^3 + 6k^2 + 4k + 1) \\ &= \left(\sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\ &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 2n^3 + 3n^2 + n + 2n^2 + 2n + n + 1 \\ &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 \\ 2) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=1}^{n+1} k^4 = \sum_{k=1}^n k^4 + (n+1)^4 \end{aligned}$$

Equating the expressions and canceling the sum with k^4 we get:

$$4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 = (n+1)^4$$

Isolating the sum with k^3 we get:

$$4 \sum_{k=1}^n k^3 = n^4 + 4n^3 + 6n^2 + 4n + 1 - 2n^3 - 5n^2 - 4n - 1 = n^4 + 2n^3 + n^2$$

Then we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4}$$

Exercise 1.4

Compute the sum of the cubes of the first 10, 100 and 1000 natural numbers (without zero).

Solution

Using formula (1.10), the sum of cubes of the first 10 natural numbers (without zero) is:

$$\sum_{k=1}^{10} k^3 = \frac{10^4 + 2 \cdot 10^3 + 10^2}{4} = 3025$$

The sum of cubes of the first 100 natural numbers (without zero) is:

$$\sum_{k=1}^{100} k^3 = \frac{100^4 + 2 \cdot 100^3 + 100^2}{4} = 25.502.500$$

The sum of cubes of the first 1000 natural numbers (without zero) is:

$$\sum_{k=1}^{1000} k^3 = \frac{1000^4 + 2 \cdot 1000^3 + 1000^2}{4} = 250.500.250.000$$

Exercise 1.5

Prove that the sum of the cubes of the first n natural numbers (without zero) equals the square of the sum of the first n natural numbers (without zero):

$$\sum_{k=1}^n k^3 = \left(\sum_{k=1}^n k \right)^2$$

Solution

Using formula (1.10) and the formula $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2(n^2 + 2n + 1)}{4} = \frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2}\right)^2 = \left(\sum_{k=1}^n k\right)^2$$

Exercise 1.6

Using the index translation, rewrite the following sums so that the sum index starts from 1, and compute them:

$$\text{a) } \sum_{j=3}^{12} (j-2)^2, \quad \text{b) } \sum_{j=0}^{n-1} (j+1), \quad \text{c) } \sum_{j=0}^{49} (2j+1)$$

Solution

a) With the index translation by $m = 2$ (setting $i = j - 2$, when j goes from 3 to 12, i goes from 1 to 10) and formula (1.9):

$$\sum_{j=3}^{12} (j-2)^2 = \sum_{i=1}^{10} i^2 = \frac{2 \cdot 10^3 + 3 \cdot 10^2 + 10}{6} = 385$$

b) With the index translation by $m = 1$ (setting $i = j + 1$):

$$\sum_{j=0}^{n-1} (j+1) = \sum_{i=1}^n i = \frac{n^2 + n}{2}$$

c) With the index translation by $m = 1$ (setting $i = j + 1$, so that $2j + 1 = 2i - 1$), this is the sum of the first 50 odd numbers:

$$\sum_{j=0}^{49} (2j+1) = \sum_{i=1}^{50} (2i-1) = 50^2 = 2500$$

Exercise 1.7

Find a closed formula for the sum

$$\sum_{j=1}^n (3j^2 - 2j + 5)$$

and use it to compute the value for $n = 10$.

Solution

Using the union of sums, the product by a constant and the sum with constant term, we have:

$$\begin{aligned}\sum_{j=1}^n (3j^2 - 2j + 5) &= 3 \sum_{j=1}^n j^2 - 2 \sum_{j=1}^n j + \sum_{j=1}^n 5 \\ &= 3 \frac{2n^3 + 3n^2 + n}{6} - 2 \frac{n^2 + n}{2} + 5n \\ &= n^3 + \frac{3}{2}n^2 + \frac{1}{2}n - n^2 - n + 5n = \frac{2n^3 + n^2 + 9n}{2}\end{aligned}$$

For $n = 10$ we get:

$$\sum_{j=1}^{10} (3j^2 - 2j + 5) = \frac{2000 + 100 + 90}{2} = 1095$$

As a check: $3 \cdot 385 - 2 \cdot 55 + 5 \cdot 10 = 1155 - 110 + 50 = 1095$.

Exercise 1.8

Using the decomposition property, compute:

$$\text{a) } \sum_{j=11}^{20} j^2, \quad \text{b) } \sum_{j=n+1}^{2n} j \quad (n \geq 1)$$

Solution

a) From the decomposition $\sum_{j=1}^{20} j^2 = \sum_{j=1}^{10} j^2 + \sum_{j=11}^{20} j^2$ and formula (1.9):

$$\sum_{j=11}^{20} j^2 = \sum_{j=1}^{20} j^2 - \sum_{j=1}^{10} j^2 = \frac{2 \cdot 20^3 + 3 \cdot 20^2 + 20}{6} - 385 = 2870 - 385 = 2485$$

b) From the decomposition $\sum_{j=1}^{2n} j = \sum_{j=1}^n j + \sum_{j=n+1}^{2n} j$:

$$\sum_{j=n+1}^{2n} j = \frac{(2n)^2 + 2n}{2} - \frac{n^2 + n}{2} = \frac{4n^2 + 2n - n^2 - n}{2} = \frac{3n^2 + n}{2}$$

Exercise 1.9

a) Prove that for every sequence of real numbers $a_1, a_2, \dots, a_{n+1}$ we have (**telescoping sum**):

$$\sum_{j=1}^n (a_{j+1} - a_j) = a_{n+1} - a_1$$

b) Using a), compute $\sum_{j=1}^n \frac{1}{j(j+1)}$ and its value for $n = 99$.

Solution

a) Using the union of sums, the index translation and the decomposition, we have:

$$\begin{aligned} \sum_{j=1}^n (a_{j+1} - a_j) &= \sum_{j=1}^n a_{j+1} - \sum_{j=1}^n a_j = \sum_{j=2}^{n+1} a_j - \sum_{j=1}^n a_j \\ &= \left(\sum_{j=2}^n a_j + a_{n+1} \right) - \left(a_1 + \sum_{j=2}^n a_j \right) = a_{n+1} - a_1 \end{aligned}$$

b) Since

$$\frac{1}{j(j+1)} = \frac{(j+1) - j}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}$$

we can apply a) with $a_j = -\frac{1}{j}$:

$$\sum_{j=1}^n \frac{1}{j(j+1)} = \sum_{j=1}^n \left(-\frac{1}{j+1} - \left(-\frac{1}{j} \right) \right) = -\frac{1}{n+1} + 1 = \frac{n}{n+1}$$

For $n = 99$ we get $\frac{99}{100}$.

Exercise 1.10

Compute $\sum_{j=1}^n ((j+1)^2 - j^2)$ in two different ways and deduce again the formula

$$\sum_{j=1}^n j = \frac{n^2 + n}{2}$$

Solution

First way: it is a telescoping sum (Exercise 1.4) with $a_j = j^2$, then:

$$\sum_{j=1}^n ((j+1)^2 - j^2) = (n+1)^2 - 1 = n^2 + 2n$$

Second way: since $(j+1)^2 - j^2 = 2j + 1$, we have:

$$\sum_{j=1}^n ((j+1)^2 - j^2) = \sum_{j=1}^n (2j + 1) = 2 \sum_{j=1}^n j + n$$

Equating the two expressions:

$$2 \sum_{j=1}^n j + n = n^2 + 2n \quad \text{that is} \quad \sum_{j=1}^n j = \frac{n^2 + n}{2}$$

Exercise 1.11

A **double sum** $\sum_{i=1}^n \sum_{j=1}^m a_{ij}$ is the sum for i from 1 to n of the inner sums $\sum_{j=1}^m a_{ij}$. Compute, for every pair of natural numbers $n, m \geq 1$:

$$\text{a) } \sum_{i=1}^n \sum_{j=1}^m (i+j), \quad \text{b) } \sum_{i=1}^n \sum_{j=1}^n i j$$

and the value of a) for $n = 3$ and $m = 4$.

Solution

a) For the inner sum (i is constant with respect to j):

$$\sum_{j=1}^m (i+j) = \sum_{j=1}^m i + \sum_{j=1}^m j = m i + \frac{m^2 + m}{2}$$

then

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m (i+j) &= \sum_{i=1}^n \left(m i + \frac{m^2 + m}{2} \right) = m \frac{n^2 + n}{2} + n \frac{m^2 + m}{2} \\ &= \frac{n m (n+1) + n m (m+1)}{2} = \frac{n m (n+m+2)}{2} \end{aligned}$$

For $n = 3$ and $m = 4$ we get $\frac{3 \cdot 4 \cdot 9}{2} = 54$.

b) Using twice the product by a constant:

$$\sum_{i=1}^n \sum_{j=1}^n i j = \sum_{i=1}^n \left(i \sum_{j=1}^n j \right) = \left(\sum_{j=1}^n j \right) \sum_{i=1}^n i = \left(\frac{n^2 + n}{2} \right)^2$$

Exercise 1.12

Compute, for every natural number $n \geq 1$, the following double sums in which the upper index of the inner sum depends on i :

$$\text{a) } \sum_{i=1}^n \sum_{j=1}^i 1, \quad \text{b) } \sum_{i=1}^n \sum_{j=1}^i j$$

and the value of b) for $n = 10$.

Solution

a) Since $\sum_{j=1}^i 1 = i$, we have:

$$\sum_{i=1}^n \sum_{j=1}^i 1 = \sum_{i=1}^n i = \frac{n^2 + n}{2}$$

b) Since $\sum_{j=1}^i j = \frac{i^2 + i}{2}$, using formula (1.9) we have:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^i j &= \sum_{i=1}^n \frac{i^2 + i}{2} = \frac{1}{2} \left(\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right) = \frac{1}{2} \left(\frac{2n^3 + 3n^2 + n}{6} + \frac{n^2 + n}{2} \right) \\ &= \frac{2n^3 + 6n^2 + 4n}{12} = \frac{n(n+1)(n+2)}{6} \end{aligned}$$

For $n = 10$ we get $\frac{10 \cdot 11 \cdot 12}{6} = 220$.

Products

2.1 Definition

Definition 2.1: product

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **product**

$$a_1 a_2 \dots a_n$$

can be indicated in compact form with the product symbol:

$$\prod_{j=1}^n a_j$$

which reads: “product for j from 1 to n of a_j ”. The symbol j is called **product index**.

- The product symbol is therefore very useful when the terms a_j are explicitly defined as a function of the product index j .

Example 2.1: products

$$\prod_{j=1}^{10} \frac{1}{j} = 1 \frac{1}{2} \frac{1}{3} \frac{1}{4} \frac{1}{5} \frac{1}{6} \frac{1}{7} \frac{1}{8} \frac{1}{9} \frac{1}{10}$$

$$\prod_{j=3}^n j^2 = 3^2 4^2 5^2 \dots n^2$$

2.2 Properties

Observation 2.2: product with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\prod_{j=1}^n c = c^n \quad (2.1)$$

Proof. We have:

$$\underbrace{c \ c \ \dots \ c}_{c \text{ multiplied } n \text{ times}} = c^n$$

□

Observation 2.3: product by a constant

For every product $\prod_{j=1}^n a_j$ and real number c , we have:

$$\prod_{j=1}^n (c a_j) = c^n \prod_{j=1}^n a_j \quad (2.2)$$

Proof. From the commutative and associative properties of the product, we have:

$$\underbrace{c a_1 \ c a_2 \ \dots \ c a_n}_{=\prod_{j=1}^n (c a_j)} = \underbrace{\underbrace{(c \ c \ \dots \ c)}_{=\prod_{j=1}^n c, \ c \text{ multiplied } n \text{ times}} (a_1 \ a_2 \ \dots \ a_n)}_{=c^n \prod_{j=1}^n a_j}$$

□

Observation 2.4: product of products

For every pair of products $\prod_{j=1}^n a_j$ and $\prod_{j=1}^n b_j$, we have:

$$\prod_{j=1}^n a_j \prod_{j=1}^n b_j = \prod_{j=1}^n (a_j b_j) \quad (2.3)$$

Proof. We have:

$$\underbrace{a_1 \ a_2 \ \dots \ a_n}_{=\prod_{j=1}^n a_j} \underbrace{b_1 \ b_2 \ \dots \ b_n}_{=\prod_{j=1}^n b_j} = \underbrace{a_1 \ b_1 \ a_2 \ b_2 \ \dots \ a_n \ b_n}_{=\prod_{j=1}^n (a_j b_j)}$$

□

Observation 2.5: decomposition

For every product $\prod_{j=1}^{n+m} a_j$, we have:

$$\prod_{j=1}^{n+m} a_j = \prod_{j=1}^n a_j \prod_{j=n+1}^{n+m} a_j \quad (2.4)$$

Observation 2.6: index translation

For every product $\prod_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1+m}^{n+m} a_{j-m} = \prod_{j=1-m}^{n-m} a_{j+m} \quad (2.5)$$

Observation 2.7: index reflexion

For every product $\prod_{j=1}^n a_j$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1}^n a_{n-j+1} = \prod_{j=0}^{n-1} a_{n-j} \quad (2.6)$$

Proof. The three properties are simply different notations and/or arrangements of the terms in the products. $\square$

2.3 Factorial

Definition 2.8: Factorial

The **factorial** of a natural number n , denoted as $n!$, is the product of the first n positive natural numbers. By convention, the factorial of 0 is defined as 1.

- Formally, the factorial is defined as follows:

$$n! = \begin{cases} 1 & \text{if } n = 0, \\ \prod_{j=1}^n j & \text{if } n \geq 1. \end{cases} \quad (2.7)$$

- A closed formula for the factorial is not known but Stirling's approximation provides an asymptotic approximation.

Observation 2.9

For every pair of natural numbers $n \geq 0$ and $k \in \{0, 1, \dots, n\}$, we have:

$$\frac{n!}{(n-k)!} = \prod_{j=1}^k (n-j+1) = \prod_{j=n-k+1}^n j \quad (2.8)$$

- Formula (2.8) is the product of k factors, from n and decreasing by one unit at a time.
- By convention, a product with no factors (i.e., whose upper index is smaller than its lower index, as $\prod_{j=1}^0 a_j$) is equal to 1. This convention is consistent with $0! = 1$ and makes formula (2.8) valid also for $k = 0$ and $k = n$.

Proof. We have:

$$\frac{n!}{(n-k)!} = \frac{\prod_{j=1}^n j}{\prod_{j=1}^{n-k} j} = \frac{\prod_{j=1}^{n-k} j \cdot \prod_{j=n-k+1}^n j}{\prod_{j=1}^{n-k} j} = \prod_{j=n-k+1}^n j = \prod_{j=1}^k (n-j+1)$$

□

Example 2.2: use of the observation

$$\frac{100!}{98!} = \frac{100!}{(100-2)!} = \prod_{j=1}^2 (100-j+1) = 100 \cdot 99 = 9.900$$

2.4 Exercises

Exercise 2.1

Compute:

$$\text{a) } 5!, \quad \text{b) } \frac{7!}{5!}, \quad \text{c) } \frac{10!}{7!}, \quad \text{d) } \frac{10!}{7! \cdot 3!}$$

Solution

a) $5! = \prod_{j=1}^5 j = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120.$

b) Using the formula $\frac{n!}{(n-k)!} = \prod_{j=1}^k (n-j+1)$ with $n = 7$ and $k = 2$: $\frac{7!}{5!} = 7 \cdot 6 = 42.$

c) With $n = 10$ and $k = 3$: $\frac{10!}{7!} = 10 \cdot 9 \cdot 8 = 720.$

d) From c): $\frac{10!}{7! \cdot 3!} = \frac{720}{3!} = \frac{720}{6} = 120.$

Exercise 2.2

Prove that for every natural number $n \geq 1$ the product of the first n positive even numbers is:

$$\prod_{j=1}^n 2j = 2^n n!$$

and compute its value for $n = 5$.

Solution

Using the product by a constant (with $c = 2$ and $a_j = j$) we have:

$$\prod_{j=1}^n 2j = 2^n \prod_{j=1}^n j = 2^n n!$$

For $n = 5$: $\prod_{j=1}^5 2j = 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 = 2^5 \cdot 5! = 32 \cdot 120 = 3840$.

Exercise 2.3

Prove that for every natural number $n \geq 1$ the product of the first n odd numbers is:

$$\prod_{j=1}^n (2j-1) = \frac{(2n)!}{2^n n!}$$

and compute its value for $n = 4$.

Solution

The factors of $(2n)! = \prod_{j=1}^{2n} j$ are the n even numbers $2, 4, \dots, 2n$ and the n odd numbers $1, 3, \dots, 2n-1$. Rearranging the factors and using Exercise 2.4, we have:

$$(2n)! = \prod_{j=1}^n 2j \prod_{j=1}^n (2j-1) = 2^n n! \prod_{j=1}^n (2j-1)$$

and then

$$\prod_{j=1}^n (2j-1) = \frac{(2n)!}{2^n n!}$$

For $n = 4$: $1 \cdot 3 \cdot 5 \cdot 7 = 105$ and indeed $\frac{8!}{2^4 \cdot 4!} = \frac{40320}{16 \cdot 24} = \frac{40320}{384} = 105$.

Exercise 2.4

a) Prove that for every sequence of non-zero real numbers $a_1, a_2, \dots, a_{n+1}$ we have (telescoping product):

$$\prod_{j=1}^n \frac{a_{j+1}}{a_j} = \frac{a_{n+1}}{a_1}$$

b) Using a), compute $\prod_{j=1}^n \frac{j+1}{j}$.

Solution

a) Using the product of products, the index translation and the decomposition, we have:

$$\prod_{j=1}^n \frac{a_{j+1}}{a_j} = \frac{\prod_{j=1}^n a_{j+1}}{\prod_{j=1}^n a_j} = \frac{\prod_{j=2}^{n+1} a_j}{\prod_{j=1}^n a_j} = \frac{\left(\prod_{j=2}^n a_j\right) a_{n+1}}{a_1 \prod_{j=2}^n a_j} = \frac{a_{n+1}}{a_1}$$

b) Applying a) with $a_j = j$:

$$\prod_{j=1}^n \frac{j+1}{j} = \frac{n+1}{1} = n+1$$

Exercise 2.5

Prove that for every natural number $n \geq 2$ we have:

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \frac{n+1}{2n}$$

and compute its value for $n = 10$.

Solution

Since

$$1 - \frac{1}{j^2} = \frac{j^2 - 1}{j^2} = \frac{(j-1)(j+1)}{j^2} = \frac{j-1}{j} \cdot \frac{j+1}{j}$$

from the product of products we have:

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \prod_{j=2}^n \frac{j-1}{j} \cdot \prod_{j=2}^n \frac{j+1}{j}$$

Both products are telescoping (Exercise 2.4, with the index starting from 2): with $a_j = \frac{1}{j}$ we have $\frac{a_j}{a_{j-1}} = \frac{j-1}{j}$, and with $a_j = j$ we have $\frac{a_{j+1}}{a_j} = \frac{j+1}{j}$. Then:

$$\prod_{j=2}^n \frac{j-1}{j} = \frac{1}{2} \cdot \frac{2}{3} \cdots \frac{n-1}{n} = \frac{1}{n}, \quad \prod_{j=2}^n \frac{j+1}{j} = \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2}$$

and therefore

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \frac{1}{n} \cdot \frac{n+1}{2} = \frac{n+1}{2n}$$

For $n = 10$ we get $\frac{11}{20}$.

Exercise 2.6

Using the index translation and the index reflection, prove that:

$$\text{a) } \prod_{j=3}^n (j-2) = (n-2)! \quad (n \geq 3), \quad \text{b) } \prod_{j=0}^{n-1} (n-j) = n! \quad (n \geq 1)$$

and compute the product in a) for $n = 8$.

Solution

a) With the index translation by $m = 2$ (setting $i = j - 2$, when j goes from 3 to n , i goes from 1 to $n - 2$):

$$\prod_{j=3}^n (j-2) = \prod_{i=1}^{n-2} i = (n-2)!$$

For $n = 8$: $\prod_{j=3}^8 (j-2) = 6! = 720$.

b) With the index reflection (with $a_j = j$, so that $a_{n-j} = n - j$):

$$\prod_{j=0}^{n-1} (n-j) = \prod_{j=0}^{n-1} a_{n-j} = \prod_{j=1}^n a_j = \prod_{j=1}^n j = n!$$

Exercise 2.7

Prove that for every natural number $n \geq 1$ we have:

$$\prod_{j=1}^n 2^j = 2^{\frac{n^2+n}{2}}$$

and compute its value for $n = 4$.

Solution

Since the product of powers with the same base is the power with the sum of the exponents ($2^a 2^b = 2^{a+b}$), we have:

$$\prod_{j=1}^n 2^j = 2^1 \cdot 2^2 \cdot \dots \cdot 2^n = 2^{1+2+\dots+n} = 2^{\sum_{j=1}^n j} = 2^{\frac{n^2+n}{2}}$$

For $n = 4$: $2^1 \cdot 2^2 \cdot 2^3 \cdot 2^4 = 2^{10} = 1024$.

Exercise 2.8

Prove that for every natural number $n \geq 1$ we have:

$$2^{n-1} \leq n! \leq n^n$$

Solution

We use the fact that if $0 \leq a_j \leq b_j$ for every $j \in \{1, \dots, n\}$, then $\prod_{j=1}^n a_j \leq \prod_{j=1}^n b_j$ (the inequalities between non-negative numbers can be multiplied member to member).

- Upper bound: since $j \leq n$ for every $j \in \{1, \dots, n\}$, using the product with constant term we have:

$$n! = \prod_{j=1}^n j \leq \prod_{j=1}^n n = n^n$$

- Lower bound: for $n = 1$ we have $2^0 = 1 = 1!$. For $n \geq 2$, using the decomposition and since $j \geq 2$ for every $j \in \{2, \dots, n\}$, we have:

$$n! = 1 \cdot \prod_{j=2}^n j \geq \prod_{j=2}^n 2 = 2^{n-1}$$

Exercise 2.9

Prove that $(n+1)! = (n+1) n!$ for every natural number $n \geq 0$, and use it to simplify, for $n \geq 1$, the expression

$$\frac{(n+1)! - n!}{(n-1)!}$$

Then compute its value for $n = 6$.

Solution

For $n = 0$ we have $1! = 1 = 1 \cdot 0!$. For $n \geq 1$, from the decomposition:

$$(n+1)! = \prod_{j=1}^{n+1} j = \left(\prod_{j=1}^n j \right) (n+1) = (n+1) n!$$

Then, using it twice ($n! = n (n-1)!$ holds for $n \geq 1$):

$$\frac{(n+1)! - n!}{(n-1)!} = \frac{(n+1) n! - n!}{(n-1)!} = \frac{n n!}{(n-1)!} = \frac{n n (n-1)!}{(n-1)!} = n^2$$

For $n = 6$: $\frac{7! - 6!}{5!} = \frac{5040 - 720}{120} = \frac{4320}{120} = 36 = 6^2$.

Part II

Vectors and matrices



Vectors

3.1 Definition and transpose

Definition 3.1: Vector

A **vector** is a column of real numbers with n rows.

Given a vector $\mathbf{a} \in \mathbb{R}^n$, we write:

$$\mathbf{a} = \begin{pmatrix} [\mathbf{a}]_1 \\ [\mathbf{a}]_2 \\ \vdots \\ [\mathbf{a}]_n \end{pmatrix} \quad \text{or, for brevity,} \quad \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

- The integer n is the **dimension** of the vector (the number of rows).
- The quantity $[\mathbf{a}]_i$ (or a_i) denotes the **i -th entry** (or **i -th component**) of $\mathbf{a}$, for $i \in \{1, 2, \dots, n\}$.

Example 3.1: Vector

The vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

is a column vector with $n = 4$ rows, where $a_1 = 2$, $a_2 = -5$, $a_3 = 3$, and $a_4 = 1$.

Definition 3.2: Transpose of a vector

Given a column vector $\mathbf{a} \in \mathbb{R}^n$, the **transpose** of $\mathbf{a}$, denoted by $\mathbf{a}'$, is the row vector:

$$\mathbf{a}' = (a_1 \quad a_2 \quad \dots \quad a_n) \in \mathbb{R}^{1 \times n}. \quad (3.1)$$

Example 3.2: Transpose of a vector

Given the column vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4,$$

its transpose is the row vector

$$\mathbf{a}' = (2 \quad -5 \quad 3 \quad 1) \in \mathbb{R}^{1 \times 4}.$$

3.2 Sum of vectors and scalar multiplication

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

their **sum** is the vector

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} p_1 + w_1 \\ p_2 + w_2 \\ \vdots \\ p_n + w_n \end{pmatrix} \in \mathbb{R}^n \quad (3.2)$$

Given a scalar $\lambda \in \mathbb{R}$ and a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

the **scalar multiplication** of $\mathbf{p}$ by λ is the vector

$$\lambda \mathbf{p} = \begin{pmatrix} \lambda p_1 \\ \lambda p_2 \\ \vdots \\ \lambda p_n \end{pmatrix} \in \mathbb{R}^n \quad (3.3)$$

Example 3.3: Sum of vectors and scalar multiplication

Consider the following two vectors and a scalar:

$$\mathbf{p} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2, \quad \lambda = 2.$$

Then:

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 1+4 \\ 4+1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \quad \lambda \mathbf{p} = 2 \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}.$$

3.2.1 Properties

The properties of vector addition and scalar multiplication are:

$$\mathbf{p} + \mathbf{w} = \mathbf{w} + \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (3.4)$$

$$(\mathbf{p} + \mathbf{w}) + \mathbf{u} = \mathbf{p} + (\mathbf{w} + \mathbf{u}), \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (3.5)$$

$$\lambda(\mathbf{p} + \mathbf{w}) = \lambda \mathbf{p} + \lambda \mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (3.6)$$

$$(\lambda + \mu)\mathbf{p} = \lambda \mathbf{p} + \mu \mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (3.7)$$

$$\lambda(\mu \mathbf{p}) = (\lambda \mu) \mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (3.8)$$

$$1 \cdot \mathbf{p} = \mathbf{p}, \quad \forall \mathbf{p} \in \mathbb{R}^n. \quad (3.9)$$

3.3 Scalar product

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the expression

$$\mathbf{p}' \mathbf{w} = \sum_{j=1}^n p_j w_j \quad (3.10)$$

is called the **scalar product** of $\mathbf{p}$ and $\mathbf{w}$.

Example 3.4: Scalar product

Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$. Their scalar product is:

$$(1 \ 4) \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 1 \cdot 4 + 4 \cdot 1 = 4 + 4 = 8.$$

3.3.1 Properties

The properties of the scalar product are:

$$\mathbf{p}' \mathbf{w} = \mathbf{w}' \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (3.11)$$

$$\mathbf{p}' (\mathbf{w} + \mathbf{u}) = \mathbf{p}' \mathbf{w} + \mathbf{p}' \mathbf{u}, \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (3.12)$$

$$\lambda (\mathbf{p}' \mathbf{w}) = (\lambda \mathbf{p})' \mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (3.13)$$

From the definition we have:

$$\mathbf{p}' \mathbf{p} \geq 0, \quad \forall \mathbf{p} \in \mathbb{R}^n \quad \text{and} \quad \mathbf{p}' \mathbf{p} = 0 \iff \mathbf{p} = \mathbf{0} \quad (3.14)$$

3.4 Linear combinations and linear independence**Definition 3.3: Linear combination**

Given k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and k scalars $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R}$, the vector

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \sum_{i=1}^k \lambda_i \mathbf{v}_i \in \mathbb{R}^n \quad (3.15)$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ with **coefficients** $\lambda_1, \lambda_2, \dots, \lambda_k$.

- We denote by $\mathbf{0} \in \mathbb{R}^n$ the **zero vector**, i.e., the vector whose entries are all equal to 0.
- Choosing $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$ we always obtain $\mathbf{0}$: this is called the **trivial** linear combination.

Example 3.5: Linear combination

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^3,$$

and the coefficients $\lambda_1 = 2$ and $\lambda_2 = -3$. The corresponding linear combination is

$$2\mathbf{v}_1 - 3\mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

Definition 3.4: Linearly independent vectors

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are **linearly independent** if the only linear combination equal to the zero vector is the trivial one, i.e.,

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0} \implies \lambda_1 = \lambda_2 = \dots = \lambda_k = 0. \quad (3.16)$$

Otherwise, i.e., if there exist coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$, **not all zero**, such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$, the vectors are **linearly dependent**.

Example 3.6: Linearly independent vectors in $\mathbb{R}^2$

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \in \mathbb{R}^2$. The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + 3\lambda_2 = 0 \\ 2\lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first equation $\lambda_1 = -3\lambda_2$; substituting in the second one we get $-6\lambda_2 + \lambda_2 = -5\lambda_2 = 0$, hence $\lambda_2 = 0$ and $\lambda_1 = 0$. Therefore, $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

Example 3.7: Linearly dependent vectors in $\mathbb{R}^2$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \in \mathbb{R}^2.$$

With the coefficients $\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = -1$ (not all zero) we get

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \begin{pmatrix} 1 + 3 - 4 \\ 2 + 1 - 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore, $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ are linearly dependent.

Example 3.8: Linearly independent vectors in $\mathbb{R}^3$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^3.$$

The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = 0 \\ \lambda_2 + \lambda_3 = 0 \\ \lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first two equations $\lambda_1 = -\lambda_3$ and $\lambda_2 = -\lambda_3$; substituting in the third one we get $-2\lambda_3 = 0$. Hence $\lambda_1 = \lambda_2 = \lambda_3 = 0$, and the three vectors are linearly independent.

Example 3.9: Linearly dependent vectors in $\mathbb{R}^3$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

With the coefficients $\lambda_1 = 1$, $\lambda_2 = -2$, $\lambda_3 = 1$ we get

$$\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} 1 - 8 + 7 \\ 2 - 10 + 8 \\ 3 - 12 + 9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, the three vectors are linearly dependent. Equivalently, $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Observation 3.5: Sets containing the zero vector

Any set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ containing the zero vector $\mathbf{0}$ is linearly dependent.

Proof. Assume, without loss of generality, that $\mathbf{v}_1 = \mathbf{0}$. Choosing $\lambda_1 = 1$ and $\lambda_2 = \dots = \lambda_k = 0$ we get $1 \cdot \mathbf{0} + 0 \cdot \mathbf{v}_2 + \dots + 0 \cdot \mathbf{v}_k = \mathbf{0}$, which is a linear combination equal to $\mathbf{0}$ with coefficients not all zero. $\square$

Observation 3.6: Characterization of linear dependence

Let $k \geq 2$. The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are linearly dependent if and only if at least one of them is a linear combination of the others.

Proof. $(\Rightarrow)$ If the vectors are linearly dependent, there exist coefficients, not all zero, such that $\sum_{i=1}^k \lambda_i \mathbf{v}_i = \mathbf{0}$. Let j be an index with $\lambda_j \neq 0$. Then

$$\mathbf{v}_j = \sum_{i \neq j} \left(-\frac{\lambda_i}{\lambda_j} \right) \mathbf{v}_i,$$

i.e., $\mathbf{v}_j$ is a linear combination of the others.

$(\Leftarrow)$ If $\mathbf{v}_j = \sum_{i \neq j} \mu_i \mathbf{v}_i$ for some scalars μ_i , then $\sum_{i \neq j} \mu_i \mathbf{v}_i + (-1) \mathbf{v}_j = \mathbf{0}$, which is a linear combination equal to $\mathbf{0}$ whose coefficient of $\mathbf{v}_j$ is $-1 \neq 0$. $\square$

Observation 3.7: More than n vectors of $\mathbb{R}^n$

Any $k > n$ vectors of $\mathbb{R}^n$ are linearly dependent.

- We state this result without proof. For instance, any three vectors of $\mathbb{R}^2$ are linearly dependent (see Example 3.7).

3.4.1 Span and basis

Definition 3.8: Span

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is the set of all their linear combinations:

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \left\{ \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k : \lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R} \right\}. \quad (3.17)$$

Definition 3.9: Basis of $\mathbb{R}^n$

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ form a **basis** of $\mathbb{R}^n$ if

- they are linearly independent, and
- $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$, i.e., every vector of $\mathbb{R}^n$ is a linear combination of them.

Observation 3.10: Coordinates with respect to a basis

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ form a basis of $\mathbb{R}^n$, then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written in a **unique** way as $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k$. The coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$ are called the **coordinates** of $\mathbf{x}$ with respect to the basis.

Proof. Such coefficients exist because the span of the basis is $\mathbb{R}^n$. If $\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{v}_i = \sum_{i=1}^k \mu_i \mathbf{v}_i$, subtracting the two expressions we get $\sum_{i=1}^k (\lambda_i - \mu_i) \mathbf{v}_i = \mathbf{0}$. Since the vectors are linearly independent, $\lambda_i - \mu_i = 0$ for all $i \in \{1, 2, \dots, k\}$. $\square$

Definition 3.11: Canonical basis

For $i \in \{1, 2, \dots, n\}$, we denote by $\mathbf{e}_i \in \mathbb{R}^n$ the vector whose i -th entry is equal to 1 and all other entries are equal to 0:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form the **canonical basis** of $\mathbb{R}^n$.

Observation 3.12: The canonical basis is a basis

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form a basis of $\mathbb{R}^n$, and every $\mathbf{x} \in \mathbb{R}^n$ satisfies

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \dots + x_n\mathbf{e}_n. \quad (3.18)$$

Proof. For any scalars $\lambda_1, \dots, \lambda_n$ we have

$$\lambda_1\mathbf{e}_1 + \dots + \lambda_n\mathbf{e}_n = (\lambda_1 \ \lambda_2 \ \dots \ \lambda_n)'.$$

This vector is equal to $\mathbf{0}$ only if $\lambda_1 = \dots = \lambda_n = 0$, hence the vectors are linearly independent; choosing $\lambda_i = x_i$ we obtain (3.18), hence their span is $\mathbb{R}^n$. $\square$

- It can be shown that every basis of $\mathbb{R}^n$ consists of exactly n vectors.

Example 3.10: Canonical basis of $\mathbb{R}^3$

The vector $\mathbf{x} = (2 \ -5 \ 3)' \in \mathbb{R}^3$ can be written as

$$\mathbf{x} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 5 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2\mathbf{e}_1 - 5\mathbf{e}_2 + 3\mathbf{e}_3.$$

Example 3.11: Coordinates with respect to a basis of $\mathbb{R}^2$

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

- They are linearly independent: $\lambda_1 + \lambda_2 = 0$ and $\lambda_1 - \lambda_2 = 0$ give $\lambda_1 = \lambda_2 = 0$.
- Their span is $\mathbb{R}^2$: for every $(a \ b)' \in \mathbb{R}^2$,

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{a-b}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence $\mathbf{v}_1, \mathbf{v}_2$ form a basis of $\mathbb{R}^2$. For instance, the coordinates of $\mathbf{x} = (3 \ 1)'$ are $\lambda_1 = \frac{3+1}{2} = 2$ and $\lambda_2 = \frac{3-1}{2} = 1$:

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

3.5 Exercises

Exercise 3.1

Consider the vectors

$$\mathbf{p} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} \in \mathbb{R}^3.$$

Compute $\mathbf{p} + \mathbf{w}$, $3\mathbf{p}$ and $2\mathbf{p} - \mathbf{w}$, and verify that $\mathbf{p} + \mathbf{w} = \mathbf{w} + \mathbf{p}$.

Solution

Sum and scalar multiplication are computed entry by entry:

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 2 + (-1) \\ -1 + 4 \\ 3 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}, \quad 3\mathbf{p} = \begin{pmatrix} 3 \cdot 2 \\ 3 \cdot (-1) \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix}.$$

For the last vector, $2\mathbf{p} - \mathbf{w} = 2\mathbf{p} + (-1)\mathbf{w}$:

$$2\mathbf{p} - \mathbf{w} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 + 1 \\ -2 - 4 \\ 6 - 0 \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \\ 6 \end{pmatrix}.$$

Finally, $\mathbf{w} + \mathbf{p} = (-1 + 2 \quad 4 - 1 \quad 0 + 3)' = (1 \quad 3 \quad 3)' = \mathbf{p} + \mathbf{w}$, as stated by the commutative property of the sum.

Exercise 3.2

Consider the vector $\mathbf{a} = (1 \quad -2 \quad 0 \quad 5)'$.

1. Write $\mathbf{a}$ as a column vector and give its dimension.
2. Write $\mathbf{a}'$ and $(\mathbf{a}')'$.
3. Compute $\mathbf{a}'\mathbf{a}$.

Solution

1. The vector is the transpose of a row vector with 4 entries, hence

$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 5 \end{pmatrix} \in \mathbb{R}^4,$$

and its dimension is $n = 4$.

2. Transposing a column vector gives a row vector, and transposing twice gives back the

original vector:

$$\mathbf{a}' = (1 \quad -2 \quad 0 \quad 5) \in \mathbb{R}^{1 \times 4}, \quad (\mathbf{a}')' = \mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 5 \end{pmatrix}.$$

3. By definition of scalar product,

$$\mathbf{a}'\mathbf{a} = 1 \cdot 1 + (-2) \cdot (-2) + 0 \cdot 0 + 5 \cdot 5 = 1 + 4 + 0 + 25 = 30.$$

As expected, $\mathbf{a}'\mathbf{a} \geq 0$, and $\mathbf{a}'\mathbf{a} \neq 0$ since $\mathbf{a} \neq \mathbf{0}$.

Exercise 3.3

Consider the vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^3.$$

1. Compute $\mathbf{p}'\mathbf{w}$ and $\mathbf{w}'\mathbf{p}$.
2. Compute $\mathbf{p}'(\mathbf{w} + \mathbf{u})$ and verify that it is equal to $\mathbf{p}'\mathbf{w} + \mathbf{p}'\mathbf{u}$.
3. Compute $(2\mathbf{p})'\mathbf{w}$ and compare it with $2(\mathbf{p}'\mathbf{w})$.

Solution

1.

$$\mathbf{p}'\mathbf{w} = 1 \cdot 4 + 2 \cdot (-1) + (-3) \cdot 2 = 4 - 2 - 6 = -4, \quad \mathbf{w}'\mathbf{p} = 4 \cdot 1 + (-1) \cdot 2 + 2 \cdot (-3) = -4.$$

The two values coincide (symmetry of the scalar product).

2. We have $\mathbf{w} + \mathbf{u} = (4 \quad 2 \quad 3)'$, hence

$$\mathbf{p}'(\mathbf{w} + \mathbf{u}) = 1 \cdot 4 + 2 \cdot 2 + (-3) \cdot 3 = 4 + 4 - 9 = -1.$$

On the other hand, $\mathbf{p}'\mathbf{u} = 1 \cdot 0 + 2 \cdot 3 + (-3) \cdot 1 = 3$, so $\mathbf{p}'\mathbf{w} + \mathbf{p}'\mathbf{u} = -4 + 3 = -1$, as expected.

3. We have $2\mathbf{p} = (2 \quad 4 \quad -6)'$, hence

$$(2\mathbf{p})'\mathbf{w} = 2 \cdot 4 + 4 \cdot (-1) + (-6) \cdot 2 = 8 - 4 - 12 = -8 = 2 \cdot (-4) = 2(\mathbf{p}'\mathbf{w}).$$

Exercise 3.4

Two vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ are called **orthogonal** if their scalar product is zero, i.e., $\mathbf{p}'\mathbf{w} = 0$.

1. Verify that $(1 \quad 2 \quad 2)'$ and $(2 \quad -2 \quad 1)'$ are orthogonal.

2. Find the value of $\alpha \in \mathbb{R}$ such that $\mathbf{p} = (1 \ \alpha \ 2)'$ and $\mathbf{w} = (3 \ -1 \ \alpha)'$ are orthogonal.

Solution

1. $(1 \ 2 \ 2) \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 1 \cdot 2 + 2 \cdot (-2) + 2 \cdot 1 = 2 - 4 + 2 = 0$, hence the two vectors are orthogonal.

2. We compute the scalar product as a function of α :

$$\mathbf{p}'\mathbf{w} = 1 \cdot 3 + \alpha \cdot (-1) + 2 \cdot \alpha = 3 + \alpha.$$

Imposing $\mathbf{p}'\mathbf{w} = 0$ we get $3 + \alpha = 0$, i.e., $\alpha = -3$. Check: with $\alpha = -3$, $\mathbf{p} = (1 \ -3 \ 2)'$, $\mathbf{w} = (3 \ -1 \ -3)'$ and $\mathbf{p}'\mathbf{w} = 3 + 3 - 6 = 0$.

Exercise 3.5

Consider the vectors $\mathbf{v}_1 = (1 \ 0 \ 2)'$ and $\mathbf{v}_2 = (-1 \ 3 \ 1)'$. Compute the linear combinations $3\mathbf{v}_1 - 2\mathbf{v}_2$ and $-\mathbf{v}_1 + \mathbf{v}_2$.

Solution

$$3\mathbf{v}_1 - 2\mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix} + \begin{pmatrix} 2 \\ -6 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \\ 4 \end{pmatrix}, \quad -\mathbf{v}_1 + \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}.$$

Exercise 3.6

Consider the vectors $\mathbf{v}_1 = (1 \ 1 \ 0)'$ and $\mathbf{v}_2 = (0 \ 1 \ 2)'$. Determine whether the following vectors are linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$ and, if so, find the coefficients:

$$\mathbf{w} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Solution

We look for $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2$ equals the given vector. Since $\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 = (\lambda_1 \ \lambda_1 + \lambda_2 \ 2\lambda_2)'$, we compare the entries.

- For w :

$$\begin{cases} \lambda_1 = 2 \\ \lambda_1 + \lambda_2 = 5 \\ 2\lambda_2 = 6 \end{cases}$$

The first equation gives $\lambda_1 = 2$, the second one $\lambda_2 = 3$, and the third one is satisfied ($2 \cdot 3 = 6$). Hence $w = 2v_1 + 3v_2$.

- For u :

$$\begin{cases} \lambda_1 = 1 \\ \lambda_1 + \lambda_2 = 2 \\ 2\lambda_2 = 3 \end{cases}$$

The first two equations give $\lambda_1 = 1$ and $\lambda_2 = 1$, but then $2\lambda_2 = 2 \neq 3$. The system has no solution, hence u is **not** a linear combination of v_1 and v_2 .

Exercise 3.7

Determine whether the following pairs of vectors of $\mathbb{R}^2$ are linearly independent:

$$(a) \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \end{pmatrix}; \quad (b) \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Solution

- (a) The second vector is twice the first one: $\begin{pmatrix} 4 \\ 2 \end{pmatrix}' = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}'$. Hence, with $\lambda_1 = 2$ and $\lambda_2 = -1$,

$$2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

a linear combination equal to $\mathbf{0}$ with coefficients not all zero: the vectors are linearly **dependent**.

- (b) The condition $\lambda_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix}' + \lambda_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix}' = \mathbf{0}$ reads

$$\begin{cases} 2\lambda_1 + \lambda_2 = 0 \\ \lambda_1 + 3\lambda_2 = 0 \end{cases}$$

From the second equation $\lambda_1 = -3\lambda_2$; substituting in the first one, $-6\lambda_2 + \lambda_2 = -5\lambda_2 = 0$, hence $\lambda_2 = 0$ and $\lambda_1 = 0$. The vectors are linearly **independent**.

Exercise 3.8

Determine whether the following sets of vectors of $\mathbb{R}^3$ are linearly independent. If they are dependent, find a linear combination equal to $\mathbf{0}$ with coefficients not all zero.

1. $v_1 = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix}', v_2 = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}', v_3 = \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}'$.
2. $w_1 = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}', w_2 = \begin{pmatrix} 1 & 1 & 0 \end{pmatrix}', w_3 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}'$.

Solution

1. The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = 0 \\ 2\lambda_1 + \lambda_2 + 3\lambda_3 = 0 \\ \lambda_1 + \lambda_2 + 2\lambda_3 = 0 \end{cases}$$

From the first equation $\lambda_1 = -\lambda_3$. Substituting in the second one: $-2\lambda_3 + \lambda_2 + 3\lambda_3 = 0$, i.e., $\lambda_2 = -\lambda_3$. The third equation becomes $-\lambda_3 - \lambda_3 + 2\lambda_3 = 0$, which holds for every λ_3 . Choosing $\lambda_3 = 1$ we get $\lambda_1 = \lambda_2 = -1$:

$$-\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} -1 - 0 + 1 \\ -2 - 1 + 3 \\ -1 - 1 + 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The vectors are linearly **dependent** (indeed $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$).

2. The condition $\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2 + \lambda_3 \mathbf{w}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_2 + \lambda_3 = 0 \\ \lambda_2 + \lambda_3 = 0 \\ \lambda_3 = 0 \end{cases}$$

Solving from the last equation upwards: $\lambda_3 = 0$, then $\lambda_2 = 0$, then $\lambda_1 = 0$. The vectors are linearly **independent**.

Exercise 3.9

Without solving any system, explain why the following sets of vectors are linearly dependent, and exhibit a linear combination equal to $\mathbf{0}$ with coefficients not all zero.

- $(1 \ 0 \ 0)', (0 \ 1 \ 0)', (0 \ 0 \ 1)', (1 \ 2 \ 3)' \in \mathbb{R}^3$.
- $(1 \ 2)', (0 \ 0)' \in \mathbb{R}^2$.
- $(1 \ 2)', (-1 \ -2)' \in \mathbb{R}^2$.

Solution

1. These are $4 > 3$ vectors of $\mathbb{R}^3$, hence they are linearly dependent. Explicitly, the first three vectors are the canonical basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $(1 \ 2 \ 3)' = 1\mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3$, so

$$1\mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3 - 1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \mathbf{0}.$$

2. The set contains the zero vector, hence it is linearly dependent: $0(1 \ 2)' + 1(0 \ 0)' = \mathbf{0}$.

3. The second vector is a linear combination of the first one, $(-1 \ -2)' = -1(1 \ 2)'$, hence $1(1 \ 2)' + 1(-1 \ -2)' = \mathbf{0}$.

Exercise 3.10

Consider the vectors

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^3.$$

1. Prove that $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ form a basis of $\mathbb{R}^3$.
2. Find the coordinates of $\mathbf{x} = (4 \ 3 \ 5)'$ with respect to this basis and with respect to the canonical basis.

Solution

1. We must show that the vectors are linearly independent and that their span is $\mathbb{R}^3$. Given $\mathbf{y} = (a \ b \ c)' \in \mathbb{R}^3$, the equation $\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \lambda_3 \mathbf{b}_3 = \mathbf{y}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = a \\ \lambda_1 + \lambda_2 = b \\ \lambda_2 + \lambda_3 = c \end{cases}$$

Summing the three equations: $2(\lambda_1 + \lambda_2 + \lambda_3) = a + b + c$, i.e., $\lambda_1 + \lambda_2 + \lambda_3 = s$ with $s = \frac{a+b+c}{2}$. Subtracting from this each equation we get the *unique* solution

$$\lambda_1 = s - c, \quad \lambda_2 = s - a, \quad \lambda_3 = s - b.$$

- Since a solution exists for every $\mathbf{y}$, the span is $\mathbb{R}^3$.
- For $\mathbf{y} = \mathbf{0}$ ($a = b = c = 0$) we get $s = 0$ and $\lambda_1 = \lambda_2 = \lambda_3 = 0$: the vectors are linearly independent.

Hence $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ form a basis of $\mathbb{R}^3$.

2. For $\mathbf{x}$ we have $a = 4, b = 3, c = 5$, so $s = \frac{4+3+5}{2} = 6$ and $\lambda_1 = 6 - 5 = 1, \lambda_2 = 6 - 4 = 2, \lambda_3 = 6 - 3 = 3$. Check:

$$1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+0+3 \\ 1+2+0 \\ 0+2+3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

With respect to the canonical basis the coordinates are simply the entries of $\mathbf{x}$: $\mathbf{x} = 4\mathbf{e}_1 + 3\mathbf{e}_2 + 5\mathbf{e}_3$.

Matrices

4.1 Definition and properties

4.1.1 Definition and transpose

Definition 4.1: Matrix

A **matrix** is a rectangular table of real numbers with m rows and n columns.

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, we write:

$$\mathbf{A} = \begin{pmatrix} [\mathbf{A}]_{1,1} & [\mathbf{A}]_{1,2} & \cdots & [\mathbf{A}]_{1,n} \\ [\mathbf{A}]_{2,1} & [\mathbf{A}]_{2,2} & \cdots & [\mathbf{A}]_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ [\mathbf{A}]_{m,1} & [\mathbf{A}]_{m,2} & \cdots & [\mathbf{A}]_{m,n} \end{pmatrix} \quad \text{or, for brevity,} \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

- The integers m and n are the **dimensions** of the matrix: m is the number of rows and n is the number of columns.
- The quantity $[\mathbf{A}]_{i,j}$ (or a_{ij}) denotes the **entry** of $\mathbf{A}$ in row i and column j , for $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$.

Example 4.1: Matrix

The matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

is a matrix with $m = 2$ rows and $n = 3$ columns, where $a_{11} = 1$, $a_{12} = 2$, $a_{13} = 3$, $a_{21} = 4$, $a_{22} = 5$, and $a_{23} = 6$.

Definition 4.2: Transpose of a matrix

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **transpose** of $\mathbf{A}$, denoted by $\mathbf{A}'$, is the $n \times m$ matrix

obtained by interchanging rows and columns:

$$\mathbf{A}' = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{n \times m}. \quad (4.1)$$

- The entry in position (i, j) of the transpose matrix $\mathbf{A}'$ is:

$$a'_{ij} = a_{ji}, \quad \forall i \in \{1, 2, \dots, n\}, \quad \forall j \in \{1, 2, \dots, m\}.$$

Example 4.2: Transpose of a matrix

Given the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3},$$

its transpose is

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

4.1.1.1 Rows and columns

- The i -th row of a matrix $\mathbf{A}$ is denoted by

$$R_i(\mathbf{A}) \quad \text{or} \quad \mathbf{A}_i$$

- The j -th column of a matrix $\mathbf{A}$ is denoted by

$$C_j(\mathbf{A}) \quad \text{or} \quad \mathbf{A}_j$$

- In both cases, we can simply use R_i or C_j when the matrix is clear from the context.

4.1.1.2 Minor matrices

- Let $\mathbf{A} \in \mathbb{R}^{n \times m}$ be a matrix and let $i \in \{1, 2, \dots, n\}$, $j \in \{1, 2, \dots, m\}$. We denote by $\mathbf{A}_{ij}$ the $(n-1) \times (m-1)$ matrix obtained from $\mathbf{A}$ by deleting row i and column j . The matrix $\mathbf{A}_{ij}$ is called the **minor matrix** associated with the entry a_{ij} .

Example 4.3: Minor matrix

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 & 5 \\ 3 & 4 & 7 & -1 \\ 2 & 0 & -3 & 6 \\ -4 & 1 & 2 & 8 \end{pmatrix} \in \mathbb{R}^{4 \times 4}.$$

For $i = 2$ and $j = 3$, the minor matrix $\mathbf{A}_{23} \in \mathbb{R}^{3 \times 3}$ is obtained by deleting row 2 and column 3:

$$\mathbf{A}_{23} = \begin{pmatrix} 1 & -2 & 5 \\ 2 & 0 & 6 \\ -4 & 1 & 8 \end{pmatrix}.$$

4.1.2 Sum of matrices and scalar multiplication

Given two matrices

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_{\mathbf{A}} \in \mathbb{R}^{m \times n} \quad \text{and} \quad \underbrace{\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}}_{\mathbf{B}} \in \mathbb{R}^{m \times n}$$

their sum is the matrix

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix} \quad (4.2)$$

Given a scalar $\lambda \in \mathbb{R}$ and a matrix

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_{\mathbf{A}} \in \mathbb{R}^{m \times n}$$

the **scalar multiplication** of $\mathbf{A}$ by λ is the matrix

$$\lambda \mathbf{A} = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{pmatrix} \quad (4.3)$$

Example 4.4: Sum of matrices and scalar multiplication

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 4 & 5 \\ -1 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1+4 & -2+5 \\ 3+(-1) & 0+2 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 2 & 2 \end{pmatrix},$$

and, for $\lambda = -2$,

$$\lambda \mathbf{A} = -2 \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ -6 & 0 \end{pmatrix}.$$

4.1.2.1 Properties

The main properties of matrix sum and scalar multiplication are:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (4.4)$$

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}), \quad \forall \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{m \times n} \quad (4.5)$$

$$\mathbf{A} + \mathbf{0} = \mathbf{A}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (4.6)$$

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (4.7)$$

$$\lambda(\mathbf{A} + \mathbf{B}) = \lambda \mathbf{A} + \lambda \mathbf{B}, \quad \forall \lambda \in \mathbb{R}, \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (4.8)$$

$$(\lambda + \mu)\mathbf{A} = \lambda \mathbf{A} + \mu \mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (4.9)$$

$$\lambda(\mu \mathbf{A}) = (\lambda\mu)\mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (4.10)$$

$$1 \cdot \mathbf{A} = \mathbf{A} \quad \text{and} \quad 0 \cdot \mathbf{A} = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n}. \quad (4.11)$$

4.1.3 Matrix product

Given two matrices

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_{\mathbf{A}} \in \mathbb{R}^{m \times n} \quad \text{and} \quad \underbrace{\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix}}_{\mathbf{B}} \in \mathbb{R}^{n \times p}$$

their multiplication is the matrix

$$\mathbf{AB} = \begin{pmatrix} \sum_{j=1}^n a_{1j}b_{j1} & \sum_{j=1}^n a_{1j}b_{j2} & \cdots & \sum_{j=1}^n a_{1j}b_{jp} \\ \sum_{j=1}^n a_{2j}b_{j1} & \sum_{j=1}^n a_{2j}b_{j2} & \cdots & \sum_{j=1}^n a_{2j}b_{jp} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n a_{mj}b_{j1} & \sum_{j=1}^n a_{mj}b_{j2} & \cdots & \sum_{j=1}^n a_{mj}b_{jp} \end{pmatrix} \in \mathbb{R}^{m \times p} \quad (4.12)$$

where the entry in position (i, k) , i.e., $[\mathbf{AB}]_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$, is the scalar product of the i -th row of $\mathbf{A}$ and the k -th column of $\mathbf{B}$.

Example 4.5: Matrix product

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

The product $\mathbf{AB}$ is:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} \\ &= \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \end{aligned}$$

The product $\mathbf{BA}$ is:

$$\begin{aligned} \mathbf{BA} &= \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix} \end{aligned}$$

Example 4.6: Matrix product

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 3} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 2}$$

The product $\mathbf{AB}$ is:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 3 & 1 \cdot 1 + 2 \cdot 0 + (-1) \cdot 5 \\ 0 \cdot 2 + 3 \cdot (-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 5 \end{pmatrix} \\ &= \begin{pmatrix} -3 & -4 \\ 9 & 20 \end{pmatrix} \end{aligned}$$

The product $\mathbf{BA}$ is:

$$\begin{aligned} \mathbf{BA} &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot 3 & 2 \cdot (-1) + 1 \cdot 4 \\ -1 \cdot 1 + 0 \cdot 0 & -1 \cdot 2 + 0 \cdot 3 & -1 \cdot (-1) + 0 \cdot 4 \\ 3 \cdot 1 + 5 \cdot 0 & 3 \cdot 2 + 5 \cdot 3 & 3 \cdot (-1) + 5 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 7 & 2 \\ -1 & -2 & 1 \\ 3 & 21 & 17 \end{pmatrix} \end{aligned}$$

In particular, $\mathbf{AB} \neq \mathbf{BA}$.

4.1.3.1 Properties

The main properties of the matrix product (when dimensions are compatible) are:

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}), \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}, \mathbf{C} \in \mathbb{R}^{p \times q} \quad (4.13)$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times p} \quad (4.14)$$

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}, \mathbf{C} \in \mathbb{R}^{n \times p} \quad (4.15)$$

$$\lambda(\mathbf{AB}) = (\lambda\mathbf{A})\mathbf{B} = \mathbf{A}(\lambda\mathbf{B}), \quad \forall \lambda \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p} \quad (4.16)$$

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} \quad \text{and} \quad \mathbf{A} \mathbf{I}_n = \mathbf{A}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (4.17)$$

In general, the matrix product is **not commutative**:

$$\mathbf{AB} \neq \mathbf{BA} \quad \text{in general.} \quad (4.18)$$

4.1.4 Special matrices

Definition 4.3: Square matrix

A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is called a **square matrix** if it has the same number of rows and columns ($m = n$). In this case, $\mathbf{A} \in \mathbb{R}^{n \times n}$ and it has the form:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}. \quad (4.19)$$

Example 4.7: Square matrix

An example of a square matrix of size 3 is:

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \\ 1 & 0 & 7 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Definition 4.4: Identity matrix

A square matrix $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ is called the **identity matrix** if it has ones on the main diagonal and zeros elsewhere. It has the form:

$$\mathbf{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \quad (4.20)$$

- Equivalently, the identity matrix satisfies

$$[\mathbf{I}_n]_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases} \quad \forall i, j \in \{1, 2, \dots, n\}.$$

Example 4.8: Identity matrix

The identity matrix of size 3 is:

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

- For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, we have:

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} \quad \text{and} \quad \mathbf{A} \mathbf{I}_n = \mathbf{A}.$$

Definition 4.5: Diagonal matrix

A square matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ is called **diagonal** if all its off-diagonal entries are zero. It has the form:

$$\mathbf{D} = \begin{pmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{pmatrix}. \quad (4.21)$$

- Equivalently, a diagonal matrix satisfies

$$d_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i \neq j.$$

Example 4.9: Diagonal matrix

The matrix

$$\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is diagonal.

Definition 4.6: Symmetric matrix

A square matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is called **symmetric** if it has the same entries with respect to the main diagonal (i.e., the entry in position (i, j) equals the entry in position (j, i)). It has the form:

$$\mathbf{Q} = \begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{12} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{1n} & q_{2n} & \cdots & q_{nn} \end{pmatrix}. \quad (4.22)$$

- Equivalently, a symmetric matrix satisfies

$$q_{ij} = q_{ji}, \quad \forall i, j \in \{1, 2, \dots, n\}.$$

- In matrix form, symmetry can be written as

$$Q' = Q.$$

Definition 4.7: Upper triangular matrix

A square matrix $U \in \mathbb{R}^{n \times n}$ is called **upper triangular** if all its entries below the main diagonal are zero. It has the form:

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}. \quad (4.23)$$

- Equivalently, an upper triangular matrix satisfies

$$u_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i > j.$$

Example 4.10: Upper triangular matrix

The matrix

$$U = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & -2 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is upper triangular.

Definition 4.8: Lower triangular matrix

A square matrix $L \in \mathbb{R}^{n \times n}$ is called **lower triangular** if all its entries above the main diagonal are zero. It has the form:

$$L = \begin{pmatrix} \ell_{11} & 0 & \cdots & 0 \\ \ell_{21} & \ell_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{nn} \end{pmatrix}. \quad (4.24)$$

- Equivalently, a lower triangular matrix satisfies

$$\ell_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i < j.$$

Example 4.11: Lower triangular matrix

The matrix

$$\mathbf{L} = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is lower triangular.

4.1.4.1 Permutation matrices

- Permutation matrices are square matrices obtained by permuting the rows (or equivalently, the columns) of the identity matrix. They encode permutations of indices and implement row/column reorderings through matrix multiplication.

Given an ordering (permutation) π of $\{1, \dots, n\}$, we denote by $\mathbf{P}_\pi$ the **permutation matrix** obtained from $\mathbf{I}_n$ by reordering its rows according to π , i.e., placing row $\pi(1)$ first, then row $\pi(2)$, ..., and finally row $\pi(n)$.

- **Row reordering by left multiplication.** Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{m \times m}$, the matrix

$$\mathbf{P}_\pi \mathbf{A}$$

consists of the rows of $\mathbf{A}$ reordered according to the permutation π (row $\pi(1)$ becomes the first row, row $\pi(2)$ becomes the second row, etc.).

- **Column reordering by right multiplication.** Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{n \times n}$, the matrix

$$\mathbf{A} \mathbf{P}'_\pi$$

(note the transpose: $\mathbf{P}'_\pi$ is obtained from $\mathbf{I}_n$ by reordering its *columns* according to π) consists of the columns of $\mathbf{A}$ reordered according to the permutation π (column $\pi(1)$ becomes the first column, column $\pi(2)$ becomes the second column, etc.).

Example 4.12: Row reordering via permutation matrix

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

Given the desired row ordering $\pi = (3, 1, 2)$ (i.e., row 3 first, then row 1, then row 2), we use the permutation matrix $\mathbf{P}_\pi$ obtained from $\mathbf{I}_3$ by reordering its rows as $(3, 1, 2)$:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then:

$$\mathbf{P}_\pi \mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The rows of $\mathbf{A}$ have been reordered according to $\pi = (3, 1, 2)$.

Example 4.13: Column reordering via permutation matrix

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

Given the desired column ordering $\pi = (2, 3, 1)$ (i.e., column 2 first, then column 3, then column 1), we use the permutation matrix $\mathbf{P}_\pi$ obtained from $\mathbf{I}_3$ by reordering its rows as $(2, 3, 1)$, and its transpose:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{P}'_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then:

$$\mathbf{A} \mathbf{P}'_\pi = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{pmatrix}.$$

The columns of $\mathbf{A}$ have been reordered according to $\pi = (2, 3, 1)$.

4.1.4.2 Inverse matrices

Definition 4.9: inverse matrix

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $\mathbf{A}^{-1} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{A} \mathbf{A}^{-1} = \mathbf{I} \quad \text{and} \quad \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}.$$

Such a matrix $\mathbf{A}^{-1}$ is called the **inverse** of $\mathbf{A}$.

- If it exists, the inverse matrix $\mathbf{A}^{-1}$ is unique.

4.1.4.3 Semidefinite matrices

Definition 4.10: semidefinite matrices

A square **symmetric** matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is

- **positive semidefinite** if

$$\mathbf{x}'\mathbf{Q}\mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- **positive definite** if

$$\mathbf{x}'\mathbf{Q}\mathbf{x} > 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$$

- **negative semidefinite** if

$$\mathbf{x}'\mathbf{Q}\mathbf{x} \leq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- **negative definite** if

$$\mathbf{x}'\mathbf{Q}\mathbf{x} < 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$$

Example 4.14: positive semidefinite matrix

- The identity matrix

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + x_2^2 \geq 0, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

- The matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{aligned} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 \\ &= x_1^2 + (x_1 - x_2)^2 + x_2^2 \geq 0, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2. \end{aligned}$$

4.1.5 Determinant

- **Determinant of a 1×1 matrix.** Let

$$\mathbf{A} = (a) \in \mathbb{R}^{1 \times 1}.$$

Then

$$\det(\mathbf{A}) = a. \quad (4.25)$$

- **Determinant of a 2×2 matrix.** Let

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then

$$\det(\mathbf{A}) = ad - bc. \quad (4.26)$$

- **Determinant of a 3×3 matrix (Sarrus rule).** Let

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then

$$\det(\mathbf{A}) = aei + bfg + cdh - ceg - bdi - afh. \quad (4.27)$$

Example 4.15: determinant of a 2×2 matrix

Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

Then

$$\det(\mathbf{A}) = 2 \cdot 4 - (-1) \cdot 3 = 8 + 3 = 11.$$

Example 4.16: determinant of a 3×3 matrix

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}.$$

Using Sarrus:

$$\det(\mathbf{A}) = 1 \cdot 1 \cdot 1 + 2 \cdot 4 \cdot 2 + 0 \cdot 3 \cdot (-1) - 0 \cdot 1 \cdot 2 - 2 \cdot 3 \cdot 1 - 1 \cdot 4 \cdot (-1).$$

Hence

$$\det(\mathbf{A}) = 1 + 16 + 0 - 0 - 6 + 4 = 15.$$

4.1.5.1 Laplace expansion

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The determinant $\det(\mathbf{A})$ can be computed recursively by **Laplace expansion** along any row or column.

For any row $i \in \{1, 2, \dots, n\}$, we have:

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} C_{ij} \quad (4.28)$$

For any column $j \in \{1, 2, \dots, n\}$, we have:

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} C_{ij} \quad (4.29)$$

Where C_{ij} is the **cofactor** of the entry a_{ij} , defined by

$$C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij}) \quad (4.30)$$

where $\mathbf{A}_{ij}$ is the minor matrix obtained from $\mathbf{A}$ by deleting row i and column j .

Example 4.17: Determinant via Laplace expansion

Let

$$\mathbf{A} = \begin{pmatrix} -2 & 2 & -3 \\ -1 & 1 & 3 \\ 2 & 0 & -1 \end{pmatrix}.$$

We expand along the second column (it contains a zero):

$$\det(\mathbf{A}) = 2 C_{12} + 1 C_{22} + 0 \cdot C_{32}.$$

Compute the cofactors:

$$C_{12} = (-1)^{1+2} \det \begin{pmatrix} -1 & 3 \\ 2 & -1 \end{pmatrix} = -((-1)(-1) - 2 \cdot 3) = -(1 - 6) = 5,$$

$$C_{22} = (-1)^{2+2} \det \begin{pmatrix} -2 & -3 \\ 2 & -1 \end{pmatrix} = ((-2)(-1) - 2(-3)) = 2 + 6 = 8.$$

Therefore,

$$\det(\mathbf{A}) = 2 \cdot 5 + 1 \cdot 8 = 18.$$

4.1.5.2 Determinant of triangular and diagonal matrices

- For triangular matrices (upper or lower), the determinant has a particularly simple form: it is the product of the diagonal entries.

If $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**, then

$$\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}. \quad (4.31)$$

If $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix**, then

$$\det(\mathbf{L}) = \prod_{i=1}^n \ell_{ii}. \quad (4.32)$$

- In particular, for a **diagonal matrix** $\mathbf{D} \in \mathbb{R}^{n \times n}$:

$$\det(\mathbf{D}) = \prod_{i=1}^n d_{ii}.$$

- For the **identity matrix** $\mathbf{I}_n$:

$$\det(\mathbf{I}_n) = 1.$$

Example 4.18: Determinant of an upper triangular matrix

Let

$$\mathbf{U} = \begin{pmatrix} 2 & 3 & -1 \\ 0 & -4 & 5 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{U}) = 2 \cdot (-4) \cdot 3 = -24.$$

Example 4.19: Determinant of a lower triangular matrix

Let

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 4 & -2 \end{pmatrix}.$$

Then

$$\det(\mathbf{L}) = 1 \cdot 3 \cdot (-2) = -6.$$

Example 4.20: Determinant of a diagonal matrix

Let

$$\mathbf{D} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{D}) = 5 \cdot (-2) \cdot 3 = -30.$$

4.1.5.3 Properties of determinants

- The determinant satisfies several important algebraic properties that make its computation and application more efficient.

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ be square matrices and $\lambda \in \mathbb{R}$ be a scalar. Then:

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B}) \quad (4.33)$$

$$\det(\mathbf{A}') = \det(\mathbf{A}) \quad (4.34)$$

$$\det(\lambda \mathbf{A}) = \lambda^n \det(\mathbf{A}) \quad (4.35)$$

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}, \quad \text{if } \mathbf{A} \text{ is invertible} \quad (4.36)$$

$$\det(\mathbf{I}_n) = 1 \quad (4.37)$$

- From properties (4.33) and (4.37), if $\mathbf{A}$ is invertible:

$$\det(\mathbf{A}) \cdot \det(\mathbf{A}^{-1}) = \det(\mathbf{AA}^{-1}) = \det(\mathbf{I}_n) = 1,$$

which gives property (4.36).

- Property (4.35) shows that scalar multiplication affects the determinant by the n -th power of the scalar (not linearly).
- Property (4.34) implies that row operations and column operations have symmetric effects on the determinant.

Example 4.21: Determinant of a product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 0 = 6, \quad \det(\mathbf{B}) = 1 \cdot 2 - (-1) \cdot 4 = 6.$$

The product is:

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 12 & 6 \end{pmatrix}.$$

Then:

$$\det(\mathbf{AB}) = 6 \cdot 6 - 0 \cdot 12 = 36 = \det(\mathbf{A}) \cdot \det(\mathbf{B}).$$

Example 4.22: Determinant of a transpose

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{A}' = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\mathbf{A}') = 1 \cdot 4 - 3 \cdot 2 = -2.$$

Therefore, $\det(\mathbf{A}') = \det(\mathbf{A})$.

Example 4.23: Determinant of a scalar multiple

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad \lambda = 2.$$

Then:

$$\lambda \mathbf{A} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\lambda \mathbf{A}) = 2 \cdot 8 - 4 \cdot 6 = -8.$$

Since $n = 2$, we verify:

$$\det(\lambda \mathbf{A}) = -8 = 2^2 \cdot (-2) = \lambda^2 \det(\mathbf{A}).$$

Example 4.24: Determinant of an inverse

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 1 = 1.$$

The inverse matrix is:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Then:

$$\det(\mathbf{A}^{-1}) = 1 \cdot 2 - (-1) \cdot (-1) = 1 = \frac{1}{\det(\mathbf{A})}.$$

A matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

4.1.6 Rank

Definition 4.11: rank of a matrix

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The **rank** of $\mathbf{A}$, denoted by $\text{rank}(\mathbf{A})$, is the maximum number of linearly independent columns of $\mathbf{A}$.

- The columns of $\mathbf{A}$ are vectors of $\mathbb{R}^m$, and linear independence is meant in the sense of the definition given in the chapter on vectors: $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$ implies $\lambda_1 = \dots = \lambda_k = 0$.

- Equivalently, $\text{rank}(\mathbf{A})$ is the maximum number of linearly independent rows of $\mathbf{A}$.
- We always have

$$\text{rank}(\mathbf{A}) \leq \min\{m, n\}.$$

- If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, then

$$\text{rank}(\mathbf{A}) = n \iff \det(\mathbf{A}) \neq 0.$$

Example 4.25: rank of a matrix

Let

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

The second row is a multiple of the first one:

$$(2 \quad 4 \quad 6) = 2(1 \quad 2 \quad 3).$$

Hence the two rows are linearly dependent, so there is only one linearly independent row. Therefore,

$$\text{rank}(\mathbf{B}) = 1.$$

Example 4.26: rank of a 3×3 matrix

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

The third column is the sum of the first two:

$$C_3(\mathbf{A}) = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = C_1(\mathbf{A}) + C_2(\mathbf{A}),$$

hence the three columns are linearly dependent and $\text{rank}(\mathbf{A}) \leq 2$. On the other hand, the first two columns are linearly independent: $\lambda_1 C_1(\mathbf{A}) + \lambda_2 C_2(\mathbf{A}) = \mathbf{0}$ gives $\lambda_1 = 0$ (first entry) and $\lambda_2 = 0$ (third entry). Therefore,

$$\text{rank}(\mathbf{A}) = 2.$$

Consistently, expanding along the first row, $\det(\mathbf{A}) = 1 \cdot (1 \cdot 1 - 3 \cdot 1) - 0 + 1 \cdot (2 \cdot 1 - 1 \cdot 0) = -2 + 2 = 0$.

4.1.7 Exercises

A square symmetric matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is positive semidefinite if:

$$\mathbf{x}'\mathbf{Q}\mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

Exercise 4.1

Using the definition, prove that the matrix

$$\mathbf{Q}_1 = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

is positive semidefinite.

Solution

We have:

$$\begin{aligned} (x_1 \ x_2) \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= ((\frac{1}{3}x_1 + 0 \cdot x_2) \ (0 \cdot x_1 + \frac{1}{2}x_2)) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= \left(\frac{1}{3}x_1\right)x_1 + \left(\frac{1}{2}x_2\right)x_2 \\ &= \frac{1}{3}x_1^2 + \frac{1}{2}x_2^2 \geq 0, \quad \forall (x_1, x_2) \in \mathbb{R}^2 \end{aligned}$$

Accordingly, the matrix $\mathbf{Q}_1$ is positive semidefinite.

Exercise 4.2

Using the definition, prove that the matrix

$$\mathbf{Q}_2 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

is positive semidefinite.

Solution

We have:

$$\begin{aligned}
 (x_1 \ x_2) \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= ((3x_1 + x_2) \ (x_1 + 3x_2)) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
 &= (3x_1 + x_2)x_1 + (x_1 + 3x_2)x_2 \\
 &= 3x_1^2 + x_1x_2 + x_1x_2 + 3x_2^2 \\
 &= 3x_1^2 + 2x_1x_2 + 3x_2^2 \\
 &= (x_1 + x_2)^2 + 2x_1^2 + 2x_2^2 \geq 0, \quad \forall (x_1, x_2) \in \mathbb{R}^2
 \end{aligned}$$

accordingly the matrix $\mathbf{Q}_2$ is positive semidefinite.

Exercise 4.3

Using the definition, prove that the matrix

$$\mathbf{Q}_3 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is positive semidefinite.

Solution

We have:

$$\begin{aligned}
 (x_1 \ x_2 \ x_3) \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
 &= ((2x_1 - x_2) \ (-x_1 + 2x_2 - x_3) \ (-x_2 + 2x_3)) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
 &= (2x_1 - x_2)x_1 + (-x_1 + 2x_2 - x_3)x_2 + (-x_2 + 2x_3)x_3 \\
 &= 2x_1^2 - x_2x_1 - x_1x_2 + 2x_2^2 - x_3x_2 - x_2x_3 + 2x_3^2 \\
 &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 \\
 &= x_1^2 + x_1^2 - 2x_1x_2 + x_2^2 + x_2^2 - 2x_2x_3 + x_3^2 + x_3^2 \\
 &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2 \geq 0, \quad \forall (x_1, x_2, x_3) \in \mathbb{R}^3
 \end{aligned}$$

accordingly the matrix Q_3 is positive semidefinite.

Exercise 4.4

Consider the matrices

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 0 \\ -3 & 2 & 5 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

Compute $A + B$, $2A - B$, A' , and verify that $(A + B)' = A' + B'$.

Solution

Sum and scalar multiplication are computed entry by entry:

$$A + B = \begin{pmatrix} 1+2 & -2+1 & 3+0 \\ 0-3 & 4+2 & -1+5 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 3 \\ -3 & 6 & 4 \end{pmatrix},$$

$$2A - B = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 8 & -2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 0 \\ -3 & 2 & 5 \end{pmatrix} = \begin{pmatrix} 0 & -5 & 6 \\ 3 & 6 & -7 \end{pmatrix}.$$

The transpose is obtained by interchanging rows and columns ($A' \in \mathbb{R}^{3 \times 2}$):

$$A' = \begin{pmatrix} 1 & 0 \\ -2 & 4 \\ 3 & -1 \end{pmatrix}, \quad B' = \begin{pmatrix} 2 & -3 \\ 1 & 2 \\ 0 & 5 \end{pmatrix}.$$

Finally,

$$A' + B' = \begin{pmatrix} 3 & -3 \\ -1 & 6 \\ 3 & 4 \end{pmatrix} = (A + B)'.$$

Exercise 4.5

Consider the matrices

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 3}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ -1 & 3 \end{pmatrix} \in \mathbb{R}^{3 \times 2}, \quad C = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

1. Among the products AB , BA , AC , CA , BC , CB , say which ones are defined and give the dimensions of the result.
2. Compute CA , AB and BC .

Solution

1. The product of a matrix in $\mathbb{R}^{m \times n}$ and a matrix in $\mathbb{R}^{q \times p}$ is defined only if $n = q$ (number of columns of the first = number of rows of the second), and the result is in $\mathbb{R}^{m \times p}$. Hence:

- **AB** : $(2 \times 3)(3 \times 2)$, defined, result in $\mathbb{R}^{2 \times 2}$;
 - **BA** : $(3 \times 2)(2 \times 3)$, defined, result in $\mathbb{R}^{3 \times 3}$;
 - **AC** : $(2 \times 3)(2 \times 2)$, **not** defined ($3 \neq 2$);
 - **CA** : $(2 \times 2)(2 \times 3)$, defined, result in $\mathbb{R}^{2 \times 3}$;
 - **BC** : $(3 \times 2)(2 \times 2)$, defined, result in $\mathbb{R}^{3 \times 2}$;
 - **CB** : $(2 \times 2)(3 \times 2)$, **not** defined ($2 \neq 3$).
2. Each entry (i, k) is the scalar product of row i of the first matrix and column k of the second one:

$$CA = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 3 & 1 \cdot 0 + 2 \cdot (-1) & 1 \cdot 2 + 2 \cdot 1 \\ 0 \cdot 1 + (-1) \cdot 3 & 0 \cdot 0 + (-1) \cdot (-1) & 0 \cdot 2 + (-1) \cdot 1 \end{pmatrix} = \begin{pmatrix} 7 & -2 & 4 \\ -3 & 1 & -1 \end{pmatrix},$$

$$AB = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 2 + 2 \cdot (-1) & 1 \cdot 0 + 0 \cdot 1 + 2 \cdot 3 \\ 3 \cdot 1 + (-1) \cdot 2 + 1 \cdot (-1) & 3 \cdot 0 + (-1) \cdot 1 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} -1 & 6 \\ 0 & 2 \end{pmatrix},$$

$$BC = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 0 \cdot (-1) \\ 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot (-1) \\ -1 \cdot 1 + 3 \cdot 0 & -1 \cdot 2 + 3 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ -1 & -5 \end{pmatrix}.$$

Exercise 4.6

1. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Compute AB and BA . Is the product commutative?
2. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Compute AB . What do you observe?

Solution

1.

$$AB = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

$$BA = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Since $AB \neq BA$, the product is not commutative (even for square matrices of the same size).

2.

$$AB = \begin{pmatrix} 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The product of two non-zero matrices can be the zero matrix: unlike real numbers, $AB = \mathbf{0}$ does **not** imply $A = \mathbf{0}$ or $B = \mathbf{0}$.

Exercise 4.7

1. For each of the following matrices, say whether it is square, diagonal, symmetric, upper triangular, lower triangular:

$$\mathbf{M}_1 = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

$$\mathbf{M}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 3 & 4 \end{pmatrix}, \quad \mathbf{M}_4 = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

2. Compute the product of the upper triangular matrices $\mathbf{U}_1 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{U}_2 = \begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix}$. What kind of matrix do you obtain?
3. Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute $\mathbf{DA}$ and $\mathbf{AD}$ and describe the effect of the diagonal matrix.

Solution

1. • $\mathbf{M}_1$: square and symmetric ($m_{12} = m_{21} = 2$); it is neither diagonal nor triangular.
 • $\mathbf{M}_2$: square and diagonal; hence it is also symmetric, upper triangular and lower triangular.
 • $\mathbf{M}_3$: square and lower triangular (all entries above the main diagonal are zero); it is not symmetric (e.g., $m_{21} = 5 \neq 0 = m_{12}$).
 • $\mathbf{M}_4 \in \mathbb{R}^{2 \times 3}$: not square, hence none of the other properties applies.

2.

$$\mathbf{U}_1 \mathbf{U}_2 = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 0 & 1 \cdot (-1) + 2 \cdot 4 \\ 0 \cdot 2 + 3 \cdot 0 & 0 \cdot (-1) + 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 0 & 12 \end{pmatrix},$$

which is again upper triangular, with diagonal entries $1 \cdot 2 = 2$ and $3 \cdot 4 = 12$.

3.

$$\mathbf{DA} = \begin{pmatrix} 2 \cdot 1 & 2 \cdot 2 \\ 3 \cdot 3 & 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 9 & 12 \end{pmatrix}, \quad \mathbf{AD} = \begin{pmatrix} 1 \cdot 2 & 2 \cdot 3 \\ 3 \cdot 2 & 4 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 6 & 12 \end{pmatrix}.$$

Left multiplication by $\mathbf{D}$ multiplies the i -th **row** of $\mathbf{A}$ by d_{ii} ; right multiplication multiplies the j -th **column** of $\mathbf{A}$ by d_{jj} .

Exercise 4.8

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

and let $\pi = (2, 3, 1)$.

1. Write the permutation matrix $\mathbf{P}_\pi$ and compute $\mathbf{P}_\pi \mathbf{A}$.
2. Compute $\mathbf{A} \mathbf{P}'_\pi$ and check that the columns of $\mathbf{A}$ are reordered according to π .
3. Verify that $\mathbf{P}_\pi \mathbf{P}'_\pi = \mathbf{I}_3$.

Solution

1. $\mathbf{P}_\pi$ is obtained from $\mathbf{I}_3$ by placing row 2 first, then row 3, then row 1:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{P}_\pi \mathbf{A} = \begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{pmatrix}.$$

The rows of $\mathbf{A}$ appear in the order 2, 3, 1.

- 2.

$$\mathbf{A} \mathbf{P}'_\pi = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \\ 8 & 9 & 7 \end{pmatrix}.$$

The columns of $\mathbf{A}$ appear in the order 2, 3, 1.

- 3.

$$\mathbf{P}_\pi \mathbf{P}'_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}_3.$$

Hence $\mathbf{P}_\pi$ is invertible and $\mathbf{P}_\pi^{-1} = \mathbf{P}'_\pi$ (one checks in the same way that $\mathbf{P}'_\pi \mathbf{P}_\pi = \mathbf{I}_3$).

Exercise 4.9

Compute the determinant of

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & -2 \\ 1 & 4 & 5 \end{pmatrix}$$

using the Laplace expansion along the most convenient row or column, and check the result with the Sarrus rule.

Solution

The second column contains two zeros, so we expand along it: only the entry $a_{32} = 4$

contributes,

$$\det(\mathbf{A}) = 0 \cdot C_{12} + 0 \cdot C_{22} + 4 \cdot C_{32}.$$

The cofactor is

$$C_{32} = (-1)^{3+2} \det \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix} = -(2 \cdot (-2) - 1 \cdot 3) = -(-4 - 3) = 7.$$

Therefore $\det(\mathbf{A}) = 4 \cdot 7 = 28$.

Check with Sarrus ($a = 2, b = 0, c = 1, d = 3, e = 0, f = -2, g = 1, h = 4, i = 5$):

$$\det(\mathbf{A}) = 2 \cdot 0 \cdot 5 + 0 \cdot (-2) \cdot 1 + 1 \cdot 3 \cdot 4 - 1 \cdot 0 \cdot 1 - 0 \cdot 3 \cdot 5 - 2 \cdot (-2) \cdot 4 = 0 + 0 + 12 - 0 - 0 + 16 = 28.$$

Exercise 4.10

Compute the determinant of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 2 & 0 \\ 4 & 1 & 0 & 2 \\ 1 & 0 & 1 & 3 \end{pmatrix} \in \mathbb{R}^{4 \times 4}$$

choosing the row or column that minimizes the number of computations.

Solution

The second row has only one non-zero entry, $a_{23} = 2$. Expanding along the second row:

$$\det(\mathbf{A}) = a_{23} C_{23} = 2 \cdot (-1)^{2+3} \det(\mathbf{A}_{23}) = -2 \det(\mathbf{A}_{23}),$$

where $\mathbf{A}_{23}$ is obtained by deleting row 2 and column 3:

$$\mathbf{A}_{23} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}.$$

We expand $\det(\mathbf{A}_{23})$ along its third row, which contains a zero:

$$\begin{aligned} \det(\mathbf{A}_{23}) &= 1 \cdot (-1)^{3+1} \det \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + 0 + 3 \cdot (-1)^{3+3} \det \begin{pmatrix} 1 & 2 \\ 4 & 1 \end{pmatrix} \\ &= 1 \cdot (4 - 3) + 3 \cdot (1 - 8) = 1 - 21 = -20. \end{aligned}$$

Therefore

$$\det(\mathbf{A}) = -2 \cdot (-20) = 40.$$

Exercise 4.11

Consider the matrices

$$U = \begin{pmatrix} 2 & 5 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 7 & 2 & 0 \\ -3 & 4 & 5 \end{pmatrix}.$$

1. Compute $\det(U)$ and $\det(L)$.
2. Without computing any product, find $\det(UL)$, $\det(U')$, $\det(3U)$ and $\det(U^{-1})$. Why does U^{-1} exist?
3. Verify that $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$ is the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, and check that $\det(A^{-1}) = 1/\det(A)$.

Solution

1. Both matrices are triangular, so their determinant is the product of the diagonal entries:

$$\det(U) = 2 \cdot (-3) \cdot 1 = -6, \quad \det(L) = 1 \cdot 2 \cdot 5 = 10.$$

2. Using the properties of determinants (with $n = 3$):

$$\det(UL) = \det(U) \det(L) = -6 \cdot 10 = -60, \quad \det(U') = \det(U) = -6,$$

$$\det(3U) = 3^3 \det(U) = 27 \cdot (-6) = -162.$$

Since $\det(U) = -6 \neq 0$, the matrix U is invertible, and $\det(U^{-1}) = \frac{1}{\det(U)} = -\frac{1}{6}$.

- 3.

$$\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & -3+3 \\ 10-10 & -5+6 \end{pmatrix} = I_2,$$

$$\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & 2-2 \\ -15+15 & -5+6 \end{pmatrix} = I_2.$$

Hence it is the inverse of A . Moreover $\det(A) = 3 \cdot 2 - 1 \cdot 5 = 1$ and $\det(A^{-1}) = 2 \cdot 3 - (-1) \cdot (-5) = 1 = \frac{1}{1}$.

Exercise 4.12

Compute the rank of the following matrices:

$$A_1 = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Solution

- $\mathbf{A}_1$: the second column is twice the first one, $(2 \ 6)' = 2(1 \ 3)'$, so the two columns are linearly dependent; the first column is non-zero, hence it is linearly independent by itself. Therefore $\text{rank}(\mathbf{A}_1) = 1$ (consistently, $\det(\mathbf{A}_1) = 6 - 6 = 0$).
- $\mathbf{A}_2$: the third column is the sum of the first two, $(3 \ 1 \ 4)' = (1 \ 0 \ 1)' + (2 \ 1 \ 3)'$, so the three columns are linearly dependent and $\text{rank}(\mathbf{A}_2) \leq 2$. The first two columns are linearly independent: $\lambda_1(1 \ 0 \ 1)' + \lambda_2(2 \ 1 \ 3)' = \mathbf{0}$ gives $\lambda_2 = 0$ (second entry) and then $\lambda_1 = 0$ (first entry). Hence $\text{rank}(\mathbf{A}_2) = 2$. Consistently, expanding along the first row, $\det(\mathbf{A}_2) = 1 \cdot (4 - 3) - 2 \cdot (0 - 1) + 3 \cdot (0 - 1) = 1 + 2 - 3 = 0$.

- $\mathbf{A}_3$: expanding along the first row,

$$\det(\mathbf{A}_3) = 1 \cdot (1 \cdot 0 - 1 \cdot 1) - 0 \cdot (0 \cdot 0 - 1 \cdot 1) + 2 \cdot (0 \cdot 1 - 1 \cdot 1) = -1 - 0 - 2 = -3 \neq 0.$$

Since $\mathbf{A}_3$ is square and $\det(\mathbf{A}_3) \neq 0$, we have $\text{rank}(\mathbf{A}_3) = 3$.

4.2 Matrix operations

4.2.1 Elementary row and column operations

- Elementary row and column operations are fundamental tools in linear algebra. They are used in many algorithms, including Gaussian elimination, computation of the inverse matrix, and matrix factorizations (such as the LU decomposition)

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **elementary row operations** are:

1. **Row swap:** Exchange rows i and k .

$$R_i \leftrightarrow R_k$$

2. **Row scaling:** Multiply row i by a nonzero scalar $\lambda \neq 0$.

$$R_i \leftarrow \lambda R_i$$

3. **Row addition:** Replace row i by the sum of row i and λ times row k (with $i \neq k$).

$$R_i \leftarrow R_i + \lambda R_k$$

Example 4.27: Row swap

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After swapping rows 1 and 3, i.e., $R_1 \leftrightarrow R_3$, we obtain:

$$\begin{pmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Example 4.28: Row scaling

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After multiplying row 2 by $\lambda = -2$, i.e., $R_2 \leftarrow -2R_2$, we obtain:

$$\begin{pmatrix} 1 & 2 & 3 \\ -8 & -10 & -12 \\ 7 & 8 & 9 \end{pmatrix}.$$

Example 4.29: Row addition

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After replacing row 2 with row 2 minus 4 times row 1, i.e., $R_2 \leftarrow R_2 - 4R_1$, we obtain:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Indeed:

$$\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ -6 \end{pmatrix}.$$

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **elementary column operations** are:

1. **Column swap:** Exchange columns j and k .

$$C_j \leftrightarrow C_k$$

2. **Column scaling:** Multiply column j by a nonzero scalar $\lambda \neq 0$.

$$C_j \leftarrow \lambda C_j$$

3. **Column addition:** Replace column j by the sum of column j and λ times column k (with $j \neq k$).

$$C_j \leftarrow C_j + \lambda C_k$$

Example 4.30: Column swap

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After swapping columns 1 and 3, i.e., $C_1 \leftrightarrow C_3$, we obtain:

$$\begin{pmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{pmatrix}.$$

Example 4.31: Column scaling

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After multiplying column 2 by $\lambda = 3$, i.e., $C_2 \leftarrow 3C_2$, we obtain:

$$\begin{pmatrix} 1 & 6 & 3 \\ 4 & 15 & 6 \\ 7 & 24 & 9 \end{pmatrix}.$$

Example 4.32: Column addition

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After replacing column 3 with column 3 plus 2 times column 1, i.e., $C_3 \leftarrow C_3 + 2C_1$, we obtain:

$$\begin{pmatrix} 1 & 2 & 5 \\ 4 & 5 & 14 \\ 7 & 8 & 23 \end{pmatrix}.$$

Indeed:

$$\begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 5 \\ 14 \\ 23 \end{pmatrix}.$$

4.2.2 Effect of elementary operations on the determinant

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The elementary row operations affect the determinant as follows:

1. **Row swap:** $R_i \leftrightarrow R_k$ changes the sign of the determinant.

$$\det(\text{new matrix}) = -\det(\mathbf{A})$$

2. **Row scaling:** $R_i \leftarrow \lambda R_i$ (with $\lambda \neq 0$) multiplies the determinant by λ .

$$\det(\text{new matrix}) = \lambda \det(\mathbf{A})$$

3. **Row addition:** $R_i \leftarrow R_i + \lambda R_k$ (with $i \neq k$) does not change the determinant.

$$\det(\text{new matrix}) = \det(\mathbf{A})$$

Example 4.33: Effect of row swap on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

After swapping rows 1 and 2, i.e., $R_1 \leftrightarrow R_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 3 \cdot 2 - 4 \cdot 1 = 2 = -\det(\mathbf{A}).$$

Example 4.34: Effect of row scaling on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 0 = 6.$$

After multiplying row 1 by $\lambda = 2$, i.e., $R_1 \leftarrow 2R_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 4 & 2 \\ 0 & 3 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 4 \cdot 3 - 2 \cdot 0 = 12 = 2 \cdot \det(\mathbf{A}) = \lambda \det(\mathbf{A}).$$

Example 4.35: Effect of row addition on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

After replacing row 2 with row 2 minus 3 times row 1, i.e., $R_2 \leftarrow R_2 - 3R_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot (-2) - 2 \cdot 0 = -2 = \det(\mathbf{A}).$$

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The elementary column operations affect the determinant as follows:

1. **Column swap:** $C_j \leftrightarrow C_k$ changes the sign of the determinant.

$$\det(\text{new matrix}) = -\det(\mathbf{A})$$

2. **Column scaling:** $C_j \leftarrow \lambda C_j$ (with $\lambda \neq 0$) multiplies the determinant by λ .

$$\det(\text{new matrix}) = \lambda \det(\mathbf{A})$$

3. **Column addition:** $C_j \leftarrow C_j + \lambda C_k$ (with $j \neq k$) does not change the determinant.

$$\det(\text{new matrix}) = \det(\mathbf{A})$$

Example 4.36: Effect of column swap on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

After swapping columns 1 and 2, i.e., $C_1 \leftrightarrow C_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 2 \cdot 3 - 1 \cdot 4 = 2 = -\det(\mathbf{A}).$$

Example 4.37: Effect of column scaling on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

After multiplying column 2 by $\lambda = 3$, i.e., $C_2 \leftarrow 3C_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 6 \\ 3 & 12 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot 12 - 6 \cdot 3 = -6 = 3 \cdot (-2) = \lambda \det(\mathbf{A}).$$

Example 4.38: Effect of column addition on determinant

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

After replacing column 2 with column 2 plus 2 times column 1, i.e., $C_2 \leftarrow C_2 + 2C_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot 10 - 4 \cdot 3 = -2 = \det(\mathbf{A}).$$

- These properties follow from the fact that the determinant of a matrix is equal to the determinant of its transpose (see the properties of determinants in the chapter on matrices), and a column operation on a matrix is the corresponding row operation on its transpose. Hence column operations have the same effect on the determinant as the corresponding row operations.
- These results are fundamental for understanding how determinants behave under elementary operations, which is crucial in algorithms such as Gaussian elimination and matrix factorization.

4.2.3 Gaussian elimination and row echelon form

- Elementary row operations can be used to transform any matrix into a matrix with a “staircase” structure, from which many properties of the original matrix (for example, its determinant) can be read easily. The systematic procedure that does this is called **Gaussian elimination**.

Definition 4.12: Row echelon form

A matrix $U \in \mathbb{R}^{m \times n}$ is in **row echelon form** if:

1. all its zero rows (if any) are below all its nonzero rows;
2. the first nonzero entry of each nonzero row (called the **pivot** of the row) is strictly to the right of the pivot of the row above it.

- As a consequence, all the entries below a pivot are equal to zero.
- A square matrix in row echelon form is upper triangular.

Gaussian elimination. Given $A \in \mathbb{R}^{m \times n}$, we process the columns from left to right. At each step, considering the rows not yet used as pivot rows:

1. find the first column containing a nonzero entry in these rows; if needed, bring such an entry to the top of these rows with a row swap $R_k \leftrightarrow R_i$ (this nonzero entry is the pivot);
2. for each row i below the pivot row k , create a zero below the pivot with the row addition

$$R_i \leftarrow R_i - \frac{a_{ij}}{a_{kj}} R_k,$$

where a_{kj} is the pivot and a_{ij} is the entry to be eliminated (in the current matrix);

3. repeat on the remaining rows.

At the end, the matrix is in row echelon form. Only row swaps and row additions are used.

Observation 4.13: Determinant by Gaussian elimination

Let $A \in \mathbb{R}^{n \times n}$ and let U be a row echelon form of A obtained by Gaussian elimination with s row swaps (and any number of row additions). Then

$$\det(A) = (-1)^s \prod_{i=1}^n u_{ii}.$$

In particular, $\det(A) = 0$ if and only if U has a zero on its diagonal.

Proof. Each row addition does not change the determinant and each row swap changes its sign. Hence $\det(U) = (-1)^s \det(A)$, i.e., $\det(A) = (-1)^s \det(U)$. Since U is square and in row echelon form, it is upper triangular, and its determinant is the product of its diagonal entries. $\square$

Example 4.39: Determinant by Gaussian elimination

Let us consider

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}.$$

We reduce $\mathbf{A}$ to row echelon form with Gaussian elimination.

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix} \\ \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad (R_1 \leftrightarrow R_2) \\ \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1) \\ \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_2)$$

The matrix is now in row echelon form. We used $s = 1$ row swap, so

$$\det(\mathbf{A}) = (-1)^1 \cdot 1 \cdot 1 \cdot 3 = -3.$$

Indeed, by Laplace expansion along the first row: $\det(\mathbf{A}) = 0 \cdot (3-1) - 1 \cdot (3-2) + 2 \cdot (1-2) = -3$.

4.2.4 Pivoting and partial pivoting

- In numerical algorithms such as matrix factorizations and row reduction methods, pivoting is a technique used to improve numerical stability and avoid division by zero.
- The pivot element is the element used as the divisor in the elimination process.

4.2.4.1 Pivoting

Pivoting is the process of selecting a suitable pivot element in a matrix to perform elimination steps in row reduction algorithms.

When performing row elimination on a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, at step k :

- The **pivot element** is the entry a_{kk} (the diagonal element in position (k, k)).
- If $a_{kk} = 0$, we cannot use it as a divisor, so we must swap row k with a row $i > k$ such that $a_{ik} \neq 0$.

- This row swap is called **pivoting**.

Example 4.40: Pivoting in row elimination

Consider the matrix:

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -1 & 0 \\ 2 & 1 & 3 \end{pmatrix}.$$

We cannot use $a_{11} = 0$ as the first pivot. We need to swap row 1 with a row that has a nonzero element in the first column.

We swap rows 1 and 2 (i.e., $R_1 \leftrightarrow R_2$):

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}.$$

Now $a_{11} = 1 \neq 0$ and we can proceed with the elimination:

$$R_3 \leftarrow R_3 - 2R_1 : \quad \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & 3 & 3 \end{pmatrix}.$$

Now we use $a_{22} = 2$ as the second pivot:

$$R_3 \leftarrow R_3 - \frac{3}{2}R_2 : \quad \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}.$$

The matrix is now in upper triangular form.

4.2.4.2 Partial pivoting

Partial pivoting is a strategy to improve numerical stability by choosing the largest available pivot in absolute value.

At step k of Gaussian elimination:

1. Find the row $i \geq k$ such that $|a_{ik}|$ is maximum among all $|a_{jk}|$ for $j \geq k$.
2. Swap row k with row i (i.e., $R_k \leftrightarrow R_i$).
3. Use the new a_{kk} as the pivot.

This ensures that:

- The pivot is the largest element in absolute value in its column, among the entries on or below the diagonal.
- Division by a larger number reduces round-off errors in floating-point arithmetic.
- The method is numerically more stable.

Example 4.41: Partial pivoting in row elimination

Consider the matrix:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & -1 & 2 \\ 2 & 3 & -1 \end{pmatrix}.$$

Step 1: Find the largest element in absolute value in the first column.

$$|a_{11}| = 1, \quad |a_{21}| = 3, \quad |a_{31}| = 2.$$

The maximum is $|a_{21}| = 3$, so we swap rows 1 and 2:

$$R_1 \leftrightarrow R_2 : \begin{pmatrix} 3 & -1 & 2 \\ 1 & 2 & 1 \\ 2 & 3 & -1 \end{pmatrix}.$$

Now eliminate below the pivot $a_{11} = 3$:

$$R_2 \leftarrow R_2 - \frac{1}{3}R_1 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{7}{3} & \frac{1}{3} \\ 2 & 3 & -1 \end{pmatrix},$$

$$R_3 \leftarrow R_3 - \frac{2}{3}R_1 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{7}{3} & \frac{1}{3} \\ 0 & \frac{11}{3} & -\frac{7}{3} \end{pmatrix}.$$

Step 2: Find the largest element in absolute value in the second column (rows 2 and 3).

$$\left| \frac{7}{3} \right| = \frac{7}{3}, \quad \left| \frac{11}{3} \right| = \frac{11}{3}.$$

The maximum is $|\frac{11}{3}|$, so we swap rows 2 and 3:

$$R_2 \leftrightarrow R_3 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{11}{3} & -\frac{7}{3} \\ 0 & \frac{7}{3} & \frac{1}{3} \end{pmatrix}.$$

Eliminate below the pivot $a_{22} = \frac{11}{3}$:

$$R_3 \leftarrow R_3 - \frac{7}{11}R_2 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{11}{3} & -\frac{7}{3} \\ 0 & 0 & \frac{20}{11} \end{pmatrix}.$$

The matrix is now in upper triangular form (row echelon form).

Example 4.42: Comparison: without and with partial pivoting

Consider the matrix:

$$A = \begin{pmatrix} \frac{1}{10000} & 1 \\ 1 & 1 \end{pmatrix}.$$

Without partial pivoting:

Using $a_{11} = \frac{1}{10000}$ as the pivot:

$$R_2 \leftarrow R_2 - 10000 R_1.$$

This gives:

$$\begin{pmatrix} \frac{1}{10000} & 1 \\ 0 & 1 - 10000 \end{pmatrix} = \begin{pmatrix} \frac{1}{10000} & 1 \\ 0 & -9999 \end{pmatrix}.$$

The large multiplier 10000 can cause significant round-off errors in floating-point arithmetic.

With partial pivoting:

Since $|a_{21}| = 1 > |a_{11}| = \frac{1}{10000}$, we swap rows:

$$R_1 \leftrightarrow R_2 : \begin{pmatrix} 1 & 1 \\ \frac{1}{10000} & 1 \end{pmatrix}.$$

Now eliminate using $a_{11} = 1$ as the pivot:

$$R_2 \leftarrow R_2 - \frac{1}{10000}R_1 : \begin{pmatrix} 1 & 1 \\ 0 & 1 - \frac{1}{10000} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & \frac{9999}{10000} \end{pmatrix}.$$

The multiplier $\frac{1}{10000}$ is much smaller, leading to better numerical stability.

- Partial pivoting is crucial in numerical linear algebra to ensure that algorithms produce accurate results, especially when dealing with matrices that have a wide range of magnitudes.
- Most modern numerical software (such as MATLAB, NumPy, CPLEX, Gurobi) automatically uses partial pivoting in Gaussian elimination and LU factorization.

- The computational cost of partial pivoting is minimal compared to the benefit of improved numerical stability.

4.2.5 Exercises

Exercise 4.13

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -2 & 1 & 4 \end{pmatrix}.$$

Apply to $\mathbf{A}$, one after the other, the elementary row operations $R_1 \leftrightarrow R_3$, $R_2 \leftarrow -2R_2$ and $R_3 \leftarrow R_3 + 2R_1$, and write the matrix obtained after each operation.

Solution

Each operation is applied to the matrix produced by the previous one:

$$\begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -2 & 1 & 4 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & 4 \\ 3 & 0 & 2 \\ 1 & 2 & -1 \end{pmatrix} \quad (R_1 \leftrightarrow R_3)$$

$$\begin{pmatrix} -2 & 1 & 4 \\ -6 & 0 & -4 \\ 1 & 2 & -1 \end{pmatrix} \quad (R_2 \leftarrow -2R_2)$$

$$\begin{pmatrix} -2 & 1 & 4 \\ -6 & 0 & -4 \\ -3 & 4 & 7 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 2R_1)$$

Note that in the last step R_1 is the *current* first row $(-2, 1, 4)$: $(1, 2, -1) + 2(-2, 1, 4) = (-3, 4, 7)$.

Exercise 4.14

Consider the matrix

$$\mathbf{B} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}.$$

Apply to $\mathbf{B}$, one after the other, the elementary column operations $C_1 \leftrightarrow C_3$, $C_2 \leftarrow C_2 + C_1$ and $C_3 \leftarrow 2C_3$, and write the matrix obtained after each operation.

Solution

$$\begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & 0 \end{pmatrix} \quad (C_1 \leftrightarrow C_3)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 4 & 0 \end{pmatrix} \quad (C_2 \leftarrow C_2 + C_1)$$

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 4 & 0 \end{pmatrix} \quad (C_3 \leftarrow 2C_3)$$

In the second step: $\begin{pmatrix} -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$.

Exercise 4.15

Let $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ with $\det(\mathbf{A}) = 5$. Without computing any matrix entry, find the determinant of the matrix obtained from $\mathbf{A}$ by:

1. the row swap $R_1 \leftrightarrow R_2$;
2. the row scaling $R_2 \leftarrow -3R_2$;
3. the row addition $R_3 \leftarrow R_3 + 7R_1$;
4. the three operations above, applied one after the other;
5. the column swap $C_1 \leftrightarrow C_3$ followed by the column scaling $C_2 \leftarrow 4C_2$.

Finally, compute $\det(2\mathbf{A})$.

Solution

We use the effect of the elementary operations on the determinant: a swap changes the sign, a scaling by λ multiplies it by λ , an addition does not change it.

1. $\det = -\det(\mathbf{A}) = -5$.
2. $\det = -3\det(\mathbf{A}) = -15$.
3. $\det = \det(\mathbf{A}) = 5$.
4. $\det = (-1) \cdot (-3) \cdot 1 \cdot \det(\mathbf{A}) = 15$.
5. $\det = (-1) \cdot 4 \cdot \det(\mathbf{A}) = -20$ (column operations act as row operations).

Finally, $2\mathbf{A}$ is obtained by scaling each of the 3 rows by 2, so

$$\det(2\mathbf{A}) = 2^3 \det(\mathbf{A}) = 8 \cdot 5 = 40.$$

For instance, all these values can be checked on $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$, which has $\det(\mathbf{A}) = 5$.

Exercise 4.16

Using elementary row operations (and no expansion formula), show that the following matrices have determinant equal to zero:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Solution

Row additions do not change the determinant, and a matrix with a zero row has determinant zero (Laplace expansion along that row).

For $\mathbf{A}$, the third row is twice the first one:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

Hence $\det(\mathbf{A}) = 0$.

For $\mathbf{B}$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 4R_1)$$

$$\rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 7R_1)$$

$$\rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_2)$$

Hence $\det(\mathbf{B}) = 0$.

Exercise 4.17

Which of the following matrices are in row echelon form? Justify your answer.

$$M_1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 2 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 5 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Solution

- M_1 : **yes**. The pivots are in columns 1 and 3 (strictly moving to the right) and the zero row is at the bottom.
- M_2 : **no**. The pivot of row 2 is in column 1, which is to the left of the pivot of row 1 (column 2).
- M_3 : **no**. The zero row (row 2) is above a nonzero row (row 3).
- M_4 : **yes**. The pivots are in columns 1 and 2 and the zero row is at the bottom.

Exercise 4.18

Reduce the following matrix to row echelon form with Gaussian elimination and indicate the pivots:

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \\ 1 & 2 & 2 & 4 \end{pmatrix}.$$

Solution

The first pivot is $a_{11} = 1$; we eliminate the entries below it, then we look for the next pivot:

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \\ 1 & 2 & 2 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & 2 & 2 & 4 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

After the first step, the second column has no nonzero entry in rows 2 and 3, so we move to the third column: the second pivot is the entry 1 in position $(2, 3)$. The last matrix is in row echelon form, with pivots 1 (position $(1, 1)$) and 1 (position $(2, 3)$), and one zero row.

Exercise 4.19

Reduce the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$$

to upper triangular form with Gaussian elimination, and use the result to compute $\det(\mathbf{A})$.

Solution

The multipliers are $\frac{4}{2} = 2$ and $\frac{8}{2} = 4$ for the first column, and $\frac{3}{1} = 3$ for the second column:

$$\begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 8 & 7 & 9 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 4R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_2)$$

Only row additions were used ($s = 0$ row swaps), hence

$$\det(\mathbf{A}) = 2 \cdot 1 \cdot 2 = 4.$$

Exercise 4.20

Compute the determinant of

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$$

by Gaussian elimination. Check the result with the Laplace expansion along the first row.

Solution

Since $a_{11} = 0$, we start with a row swap:

$$\begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 3 & 1 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & -2 & -5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_2)$$

We used $s = 1$ row swap, hence

$$\det(\mathbf{A}) = (-1)^1 \cdot 1 \cdot 2 \cdot (-1) = 2.$$

Check: $\det(\mathbf{A}) = 0 \cdot (1 - 2) - 2 \cdot (1 - 6) + 4 \cdot (1 - 3) = 10 - 8 = 2.$

Exercise 4.21

Compute the determinant of the 4×4 matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

by Gaussian elimination.

Solution

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad (R_4 \leftarrow R_4 - R_2)$$

The matrix is upper triangular and only row additions were used, hence

$$\det(\mathbf{A}) = 1 \cdot 1 \cdot 1 \cdot 3 = 3.$$

(With the Laplace expansion we would need to compute four 3×3 determinants.)

Exercise 4.22

Reduce the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 4 & 2 \end{pmatrix}$$

to upper triangular form using Gaussian elimination with **partial pivoting**. Then compute $\det(\mathbf{A})$ and check that all the multipliers have absolute value at most 1.

Solution

Step 1: in the first column $|1| < |2| < |4|$: the pivot is 4, in row 3. **Step 2:** after the elimination, in the second column (rows 2 and 3) we have $|-1| < |2|$: the pivot is 2, in row

3.

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 4 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 2 & 1 & 3 \\ 1 & 3 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_3)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & -1 & 2 \\ 1 & 3 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{1}{2}R_1)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & -1 & 2 \\ 0 & 2 & \frac{1}{2} \end{pmatrix} \quad (R_3 \leftarrow R_3 - \frac{1}{4}R_1)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & 2 & \frac{1}{2} \\ 0 & -1 & 2 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & 2 & \frac{1}{2} \\ 0 & 0 & \frac{9}{4} \end{pmatrix} \quad (R_3 \leftarrow R_3 + \frac{1}{2}R_2)$$

The multipliers are $\frac{2}{4} = \frac{1}{2}$, $\frac{1}{4}$ and $\frac{-1}{2} = -\frac{1}{2}$: all have absolute value at most 1, because each pivot is the entry of largest absolute value in its column (on or below the diagonal).

We used $s = 2$ row swaps, hence

$$\det(\mathbf{A}) = (-1)^2 \cdot 4 \cdot 2 \cdot \frac{9}{4} = 18.$$

4.3 Inversion of matrices

4.3.1 Properties of the inverse matrix

- Recall (see the chapter on matrices) that a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $\mathbf{A}^{-1} \in \mathbb{R}^{n \times n}$ such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. A square matrix which is not invertible is called **singular**.
- In this chapter, $\mathbf{A}'$ denotes the transpose of $\mathbf{A}$, as in the chapter on matrices.

Proposition 4.14: Uniqueness of the inverse

Let $A \in \mathbb{R}^{n \times n}$ be invertible. If $B, C \in \mathbb{R}^{n \times n}$ are such that $AB = BA = I$ and $AC = CA = I$, then $B = C$.

Proof. Using the associativity of the matrix product, we have:

$$B = BI = B(AC) = (BA)C = IC = C.$$

□

Proposition 4.15: Properties of the inverse

Let $A, B \in \mathbb{R}^{n \times n}$ be invertible matrices. Then:

1. A^{-1} is invertible and $(A^{-1})^{-1} = A$;
2. AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$;
3. A' is invertible and $(A')^{-1} = (A^{-1})'$;
4. $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof. By the uniqueness of the inverse, in each case it is enough to exhibit a matrix that multiplied on the left and on the right gives I .

1. From $A^{-1}A = AA^{-1} = I$, the matrix A is the inverse of A^{-1} .
2. Using associativity:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

3. Recall that $(CD)' = D'C'$ for all $C, D \in \mathbb{R}^{n \times n}$: indeed, the entry (i, j) of $(CD)'$ is the entry (j, i) of CD , i.e., $\sum_{k=1}^n c_{jk}d_{ki} = \sum_{k=1}^n d'_{ik}c'_{kj}$, which is the entry (i, j) of $D'C'$. Then, since $I' = I$:

$$A'(A^{-1})' = (A^{-1}A)' = I' = I, \quad (A^{-1})'A' = (AA^{-1})' = I' = I.$$

4. From the product rule of determinants:

$$\det(A) \det(A^{-1}) = \det(AA^{-1}) = \det(I) = 1.$$

Hence $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$.

□

- Note the reversed order in $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$: since the matrix product is not commutative, in general $(\mathbf{AB})^{-1} \neq \mathbf{A}^{-1}\mathbf{B}^{-1}$.
- Property 4 shows that a matrix with $\det(\mathbf{A}) = 0$ cannot be invertible, i.e., it is singular.

4.3.2 Computation of the inverse

4.3.2.1 Gauss–Jordan method

Gauss–Jordan method (idea). To compute $\mathbf{A}^{-1}$, we build the augmented matrix

$$(\mathbf{A} \mid \mathbf{I})$$

and we apply **elementary row operations** to transform the left block into the identity matrix. If we obtain

$$(\mathbf{I} \mid \mathbf{B}),$$

then $\mathbf{B} = \mathbf{A}^{-1}$.

Example 4.43: inverse matrix via Gauss–Jordan elimination

Let us consider

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

$$\begin{aligned} & \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \\ & \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \quad (R_1 \leftrightarrow R_2) \\ & \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_1) \\ & \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_2 \leftarrow -R_2) \\ & \left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2) \end{aligned}$$

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

We can verify:

$$\mathbf{AA}^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}.$$

Example 4.44: inverse matrix via Gauss–Jordan elimination

Let us consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

$$\begin{pmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 2 & 0 & 1 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & -4 & 1 & | & -2 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & -4 & 1 & | & -2 & 0 & 1 \end{pmatrix} \quad (R_1 \leftarrow R_1 - 2R_2)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 5 & | & -2 & 4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 4R_2)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_3 \leftarrow \frac{1}{5}R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_1 \leftarrow R_1 + 2R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & | & \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_3)$$

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}.$$

Example 4.45: inverse matrix via Gauss–Jordan elimination with a row swap

Let us consider

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan. Since $a_{11} = 0$, we start with a row swap.

$$\begin{pmatrix} 0 & 1 & 2 & | & 1 & 0 & 0 \\ 1 & 0 & 3 & | & 0 & 1 & 0 \\ 4 & -3 & 8 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 4 & -3 & 8 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & -3 & -4 & | & 0 & -4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 4R_1)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 2 & | & 3 & -4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 3R_2)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_3 \leftarrow \frac{1}{2}R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_1 \leftarrow R_1 - 3R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & | & -2 & 4 & -1 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_3)$$

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -9 & 14 & -3 \\ -4 & 8 & -2 \\ 3 & -4 & 1 \end{pmatrix}.$$

Observation 4.16: Singular matrices and the Gauss–Jordan method

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. If, while applying elementary row operations to $(\mathbf{A} \mid \mathbf{I})$, a **zero row appears in the left block**, then $\det(\mathbf{A}) = 0$ and $\mathbf{A}$ is singular: the method stops and $\mathbf{A}^{-1}$ does not exist.

Proof. The left block is obtained from $\mathbf{A}$ by elementary row operations. Each of them multiplies the determinant by a nonzero number (-1 for a row swap, $\lambda \neq 0$ for a row scaling, 1 for a row addition). A matrix with a zero row has zero determinant (Laplace expansion along that row), hence $\det(\mathbf{A}) = 0$. By property 4 of Proposition 4.15, $\mathbf{A}$ is not invertible. $\square$

Example 4.46: Gauss–Jordan elimination on a singular matrix

Let us consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{pmatrix}.$$

We try to compute $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 7 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 0 & 0 & 0 & | & 1 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

The third row of the left block is zero: the left block cannot be transformed into the identity matrix, and the method stops. Hence $\mathbf{A}$ is singular. Indeed, the third row of $\mathbf{A}$ is the difference of the second and the first row, and

$$\det(\mathbf{A}) = 1 \cdot (5 \cdot 4 - 7 \cdot 3) - 2 \cdot (2 \cdot 4 - 7 \cdot 1) + 3 \cdot (2 \cdot 3 - 5 \cdot 1) = -1 - 2 + 3 = 0.$$

4.3.2.2 Inverse via the adjugate matrix

- Recall (see the chapter on matrices) that, given $\mathbf{A} \in \mathbb{R}^{n \times n}$, the **cofactor** of the entry a_{ij} is $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$, where $\mathbf{A}_{ij}$ is the minor matrix obtained from $\mathbf{A}$ by deleting row i and column j .

Definition 4.17: Cofactor matrix and adjugate matrix

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $n \geq 2$. The **cofactor matrix** of $\mathbf{A}$ is the matrix $\mathbf{C} = (C_{ij}) \in \mathbb{R}^{n \times n}$ of the cofactors of $\mathbf{A}$. The **adjugate matrix** of $\mathbf{A}$ is the transpose of the cofactor matrix:

$$\text{adj}(\mathbf{A}) = \mathbf{C}', \quad \text{i.e.,} \quad (\text{adj}(\mathbf{A}))_{ij} = C_{ji}.$$

Theorem 4.18: Adjugate formula

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $n \geq 2$. Then

$$\mathbf{A} \text{adj}(\mathbf{A}) = \text{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}.$$

Consequently, if $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible and

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A}).$$

Proof. The entry (i, j) of $\mathbf{A} \text{adj}(\mathbf{A})$ is

$$\sum_{k=1}^n a_{ik} C_{jk}.$$

- If $i = j$, this is the Laplace expansion of $\det(\mathbf{A})$ along row i .
- If $i \neq j$, this is the Laplace expansion along row j of the matrix $\tilde{\mathbf{A}}$ obtained from $\mathbf{A}$ by replacing row j with row i (the cofactors C_{jk} do not depend on row j). The matrix $\tilde{\mathbf{A}}$ has two equal rows: swapping them leaves $\tilde{\mathbf{A}}$ unchanged and changes the sign of its determinant, so $\det(\tilde{\mathbf{A}}) = -\det(\tilde{\mathbf{A}})$, i.e., $\det(\tilde{\mathbf{A}}) = 0$.

Hence $\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}$. The identity $\text{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}$ is obtained in the same way, using Laplace expansions along columns. If $\det(\mathbf{A}) \neq 0$, dividing by $\det(\mathbf{A})$ shows that $\frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$ is the inverse of $\mathbf{A}$. $\square$

Observation 4.19: Invertibility and determinant

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Proof. If $\mathbf{A}$ is invertible, then $\det(\mathbf{A}) \neq 0$ by property 4 of Proposition 4.15. Conversely, if $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible by Theorem 4.18 (for $n = 1$, $\mathbf{A} = (a)$ with $a \neq 0$ and $\mathbf{A}^{-1} = (1/a)$). $\square$

Observation 4.20: Inverse of a 2×2 matrix

Let

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{with} \quad \det(\mathbf{A}) = ad - bc \neq 0.$$

Then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. The minor matrices of $\mathbf{A}$ are 1×1 , so the cofactors are $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. Hence

$$\mathbf{C} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \quad \text{adj}(\mathbf{A}) = \mathbf{C}' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and the formula follows from Theorem 4.18. $\square$

Inverse of a 2×2 matrix (rule). Swap the two diagonal entries, change the sign of the two off-diagonal entries, and divide by the determinant.

Example 4.47: inverse of a 2×2 matrix with the formula

Let us consider again

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have $\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, hence

$$\mathbf{A}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in Example 4.43.

Example 4.48: inverse of a 3×3 matrix with the adjugate formula

Let us consider again

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

By Laplace expansion along the first row:

$$\det(\mathbf{A}) = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5 \neq 0.$$

The cofactors are:

$$\begin{aligned} C_{11} &= +\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1, & C_{12} &= -\det \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} = 2, & C_{13} &= +\det \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = -2, \\ C_{21} &= -\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = -2, & C_{22} &= +\det \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = 1, & C_{23} &= -\det \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} = 4, \\ C_{31} &= +\det \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = 2, & C_{32} &= -\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -1, & C_{33} &= +\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1. \end{aligned}$$

Hence

$$C = \begin{pmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}, \quad \text{adj}(A) = C' = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix},$$

and

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in Example 4.44.

- For $n = 2$ (and often for $n = 3$) the adjugate formula is convenient for computations by hand. For larger matrices, the method of Gauss–Jordan requires far fewer operations.

4.3.3 Exercises

Exercise 4.23

Using the formula for the inverse of a 2×2 matrix, compute (if it exists) the inverse of

$$A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 2 \\ 3 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}.$$

Solution

Recall that, if $ad - bc \neq 0$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

- $\det(A) = 3 \cdot 2 - 5 \cdot 1 = 1$, hence

$$A^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}. \quad \text{Check: } \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & -15+15 \\ 2-2 & -5+6 \end{pmatrix} = I.$$

- $\det(B) = 4 \cdot 4 - 2 \cdot 3 = 10$, hence

$$B^{-1} = \frac{1}{10} \begin{pmatrix} 4 & -2 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ -\frac{3}{10} & \frac{2}{5} \end{pmatrix}.$$

- $\det(\mathbf{C}) = 2 \cdot 6 - 4 \cdot 3 = 0$: the matrix $\mathbf{C}$ is singular and $\mathbf{C}^{-1}$ does not exist (the second row is $\frac{3}{2}$ times the first one).

Exercise 4.24

Compute the inverse of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

with the method of Gauss–Jordan, and verify the result.

Solution

We apply elementary row operations to $(\mathbf{A} \mid \mathbf{I})$:

$$\begin{pmatrix} 1 & 1 & 0 & | & 1 & 0 & 0 \\ 1 & 2 & 1 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -1 & 1 & 0 \\ 0 & 1 & 2 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 0 & -1 & | & 2 & -1 & 0 \\ 0 & 1 & 1 & | & -1 & 1 & 0 \\ 0 & 1 & 2 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_1 \leftarrow R_1 - R_2)$$

$$\begin{pmatrix} 1 & 0 & -1 & | & 2 & -1 & 0 \\ 0 & 1 & 1 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & 1 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & 3 & -2 & 1 \\ 0 & 1 & 1 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & 1 & -1 & 1 \end{pmatrix} \quad (R_1 \leftarrow R_1 + R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & 3 & -2 & 1 \\ 0 & 1 & 0 & | & -2 & 2 & -1 \\ 0 & 0 & 1 & | & 1 & -1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_3)$$

Hence

$$\mathbf{A}^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Check (row by column): $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 3-2 & -2+2 & 1-1 \\ 3-4+1 & -2+4-1 & 1-2+1 \\ -2+2 & 2-2 & -1+2 \end{pmatrix} = \mathbf{I}.$

Exercise 4.25

Compute the inverse of

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$$

with the method of Gauss–Jordan.

Solution

Since $a_{11} = 0$, we start with a row swap. Later, we swap rows 2 and 3 to get a pivot equal to 1 (and avoid fractions):

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 0 & 2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right) \\ & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftrightarrow R_2) \\ & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_1) \\ & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 2 & 1 & 1 & 0 & 0 \end{array} \right) \quad (R_2 \leftrightarrow R_3) \\ & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 2 & 1 & 1 & 0 & 0 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2) \\ & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & -1 & 1 & 4 & -2 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_2) \\ & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_3 \leftarrow -R_3) \\ & \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_1 \leftarrow R_1 + R_3) \\ & \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_2 \leftarrow R_2 - R_3) \end{aligned}$$

Hence

$$\mathbf{A}^{-1} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}.$$

Exercise 4.26

Apply the method of Gauss–Jordan to

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 3 & 5 & 0 \end{pmatrix}.$$

Is $\mathbf{A}$ invertible?

Solution

$$\begin{pmatrix} 1 & 2 & -1 & | & 1 & 0 & 0 \\ 2 & 3 & 1 & | & 0 & 1 & 0 \\ 3 & 5 & 0 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & -1 & | & 1 & 0 & 0 \\ 0 & -1 & 3 & | & -2 & 1 & 0 \\ 3 & 5 & 0 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & -1 & | & 1 & 0 & 0 \\ 0 & -1 & 3 & | & -2 & 1 & 0 \\ 0 & -1 & 3 & | & -3 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\begin{pmatrix} 1 & 2 & -1 & | & 1 & 0 & 0 \\ 0 & -1 & 3 & | & -2 & 1 & 0 \\ 0 & 0 & 0 & | & -1 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

A zero row appears in the left block: the method stops and $\mathbf{A}$ is **not invertible**. Indeed, the third row of $\mathbf{A}$ is the sum of the first two rows, and

$$\det(\mathbf{A}) = 1 \cdot (0 - 5) - 2 \cdot (0 - 3) + (-1) \cdot (10 - 9) = -5 + 6 - 1 = 0.$$

Exercise 4.27

Consider, for $k \in \mathbb{R}$, the matrix

$$\mathbf{A}(k) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k^2 \end{pmatrix}.$$

1. For which values of k is $\mathbf{A}(k)$ invertible?
2. Compute $\mathbf{A}(0)^{-1}$ with the method of Gauss–Jordan.

Solution

1. We compute $\det(\mathbf{A}(k))$ by Gaussian elimination (row additions only):

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k^2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & k-1 & 0 \\ 1 & 1 & k^2 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & k-1 & 0 \\ 0 & 0 & k^2-1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

The last matrix is upper triangular, hence

$$\det(\mathbf{A}(k)) = 1 \cdot (k-1) \cdot (k^2-1) = (k-1)^2(k+1).$$

Therefore $\mathbf{A}(k)$ is invertible if and only if $k \neq 1$ and $k \neq -1$.

2. For $k = 0$ we have $\det(\mathbf{A}(0)) = 1$, and:

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow -R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array} \right) \quad (R_3 \leftarrow -R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_3)$$

Hence

$$\mathbf{A}(0)^{-1} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Exercise 4.28

Consider, for $k \in \mathbb{R}$, the matrix $\mathbf{A}(k) = \begin{pmatrix} k & 1 \\ 4 & k \end{pmatrix}$. For which values of k is $\mathbf{A}(k)$ invertible? For these values, write $\mathbf{A}(k)^{-1}$, and compute $\mathbf{A}(3)^{-1}$.

Solution

We have $\det(\mathbf{A}(k)) = k^2 - 4 = (k-2)(k+2)$, so $\mathbf{A}(k)$ is invertible if and only if $k \neq 2$ and $k \neq -2$. In this case

$$\mathbf{A}(k)^{-1} = \frac{1}{k^2 - 4} \begin{pmatrix} k & -1 \\ -4 & k \end{pmatrix}.$$

For $k = 3$: $\det(\mathbf{A}(3)) = 5$ and

$$\mathbf{A}(3)^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -4 & 3 \end{pmatrix}.$$

Exercise 4.29

Compute the inverse of

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}$$

with the adjugate (cofactor) formula $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$.

Solution

Laplace expansion along the first row: $\det(\mathbf{A}) = 2 \cdot (1 - 0) - 0 + 1 \cdot (3 - 0) = 5 \neq 0$. The cofactors $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$ are:

$$\begin{aligned} C_{11} &= +\det \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = 1, & C_{12} &= -\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -1, & C_{13} &= +\det \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} = 3, \\ C_{21} &= -\det \begin{pmatrix} 0 & 1 \\ 3 & 1 \end{pmatrix} = 3, & C_{22} &= +\det \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = 2, & C_{23} &= -\det \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = -6, \\ C_{31} &= +\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1, & C_{32} &= -\det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = 1, & C_{33} &= +\det \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = 2. \end{aligned}$$

The adjugate matrix is the transpose of the cofactor matrix:

$$\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} 1 & -1 & 3 \\ 3 & 2 & -6 \\ -1 & 1 & 2 \end{pmatrix}' = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 1 \\ 3 & -6 & 2 \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 1 \\ 3 & -6 & 2 \end{pmatrix}.$$

Exercise 4.30

Let $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$.

1. Compute $\mathbf{A}^{-1}$, $\mathbf{B}^{-1}$ and then $(\mathbf{AB})^{-1}$ using the properties of the inverse. Check the result by inverting $\mathbf{AB}$ directly.
2. Show that $\mathbf{A}^{-1}\mathbf{B}^{-1} \neq (\mathbf{AB})^{-1}$.
3. Compute $(\mathbf{B}')^{-1}$ and $\det((\mathbf{AB})^{-1})$.

Solution

1. $\det(\mathbf{A}) = 1$ and $\det(\mathbf{B}) = 3$, hence

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix},$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 & -7 \\ -1 & 2 \end{pmatrix}.$$

Check: $\mathbf{AB} = \begin{pmatrix} 2 & 7 \\ 1 & 5 \end{pmatrix}$, $\det(\mathbf{AB}) = 10 - 7 = 3$, and the 2×2 formula gives $(\mathbf{AB})^{-1} = \frac{1}{3} \begin{pmatrix} 5 & -7 \\ -1 & 2 \end{pmatrix}$.

$$2. \quad \mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & -3 \\ -3 & 4 \end{pmatrix} \neq (\mathbf{AB})^{-1}.$$

$$3. \quad (\mathbf{B}')^{-1} = (\mathbf{B}^{-1})' = \frac{1}{3} \begin{pmatrix} 3 & 0 \\ -2 & 1 \end{pmatrix} \quad (\text{check with the } 2 \times 2 \text{ formula applied to } \mathbf{B}' = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}). \text{ Moreover}$$

$$\det((\mathbf{AB})^{-1}) = \frac{1}{\det(\mathbf{AB})} = \frac{1}{\det(\mathbf{A})\det(\mathbf{B})} = \frac{1}{3}.$$

Exercise 4.31

1. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible and $\lambda \in \mathbb{R}$, $\lambda \neq 0$. Show that $\lambda\mathbf{A}$ is invertible and $(\lambda\mathbf{A})^{-1} = \frac{1}{\lambda}\mathbf{A}^{-1}$.

2. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \mathbf{0}$. Show that $\mathbf{A}$ is invertible and $\mathbf{A}^{-1} = 3\mathbf{I} - \mathbf{A}$.
Check this on $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

Solution

1. Using the properties of the product by a scalar:

$$(\lambda \mathbf{A}) \left(\frac{1}{\lambda} \mathbf{A}^{-1} \right) = \lambda \cdot \frac{1}{\lambda} \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}, \quad \left(\frac{1}{\lambda} \mathbf{A}^{-1} \right) (\lambda \mathbf{A}) = \frac{1}{\lambda} \cdot \lambda \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}.$$

2. From $\mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \mathbf{0}$ we get $3\mathbf{A} - \mathbf{A}^2 = \mathbf{I}$, i.e.,

$$\mathbf{A}(3\mathbf{I} - \mathbf{A}) = \mathbf{I} \quad \text{and} \quad (3\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{I}.$$

Hence $\mathbf{A}^{-1} = 3\mathbf{I} - \mathbf{A}$. For $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$:

$$\mathbf{A}^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad \mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \begin{pmatrix} 5-6+1 & 3-3 \\ 3-3 & 2-3+1 \end{pmatrix} = \mathbf{0},$$

and $3\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, which is indeed $\mathbf{A}^{-1}$.

Exercise 4.32

Compute the inverse of the upper triangular matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

with the method of Gauss–Jordan. What kind of matrix is $\mathbf{A}^{-1}$?

Solution

The left block is already upper triangular with ones on the diagonal: we only need to

eliminate the entries above the pivots, starting from the last column.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -3 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 3R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -3 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - 4R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & 5 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 2R_2)$$

Hence

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix},$$

which is again upper triangular with ones on the diagonal.

4.4 Factorization of matrices

4.4.1 LU factorization

- The LU factorization (or LU decomposition) is a fundamental matrix decomposition that expresses a matrix as the product of a lower triangular matrix and an upper triangular matrix.

Given a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, a **PLU factorization** of $\mathbf{A}$ is a decomposition of the form

$$\mathbf{PA} = \mathbf{LU},$$

where:

- $\mathbf{P} \in \mathbb{R}^{n \times n}$ is a **permutation matrix** (it accounts for possible row swaps),
- $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix** with ones on the diagonal,
- $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**.

- The factorization is computed by Gaussian elimination.

- If no row swaps are needed, then $\mathbf{P} = \mathbf{I}_n$ and the factorization reduces to the **LU factorization**:

$$\mathbf{A} = \mathbf{L}\mathbf{U}.$$

- The PLU/LU factorization is very useful for solving linear systems $\mathbf{A}\mathbf{x} = \mathbf{b}$ efficiently.

Method to compute the PLU (LU with pivoting) factorization.

Starting from $\mathbf{A}$, we perform Gaussian elimination. Whenever a row swap is required, it is recorded in the permutation matrix $\mathbf{P}$. At the end of the elimination we obtain the upper triangular matrix $\mathbf{U}$, while $\mathbf{L}$ stores the elimination multipliers.

Specifically:

- Each elimination step $R_i \leftarrow R_i - \ell_{ij}R_j$ (with $i > j$) creates a zero in position (i, j) .
- The multiplier ℓ_{ij} is stored in position (i, j) of $\mathbf{L}$.
- The diagonal of $\mathbf{L}$ is filled with ones.
- If at step j rows i and k are swapped ($R_i \leftrightarrow R_k$), the multipliers already stored in rows i and k of $\mathbf{L}$ (in columns $1, \dots, j-1$) are swapped as well, and the same swap is applied to the rows of $\mathbf{P}$ (which starts as $\mathbf{I}_n$).

Example 4.49: LU factorization

Let us compute the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}.$$

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$.

We need to eliminate $a_{21} = 4$. The multiplier is:

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{4}{2} = 2.$$

We perform $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{pmatrix}.$$

We need to eliminate $a_{31} = -2$. The multiplier is:

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{-2}{2} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_1 = R_3 + R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}.$$

Step 2: Eliminate the entries below the second pivot -8 .

We need to eliminate the entry in position $(3, 2)$, which is currently 8. The multiplier is:

$$\ell_{32} = \frac{8}{-8} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_2 = R_3 + R_2$:

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example 4.50: LU factorization (continued)

Step 3: Build the matrix L .

The matrix L has ones on the diagonal and the multipliers below the diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}.$$

Verification: We check that $LU = A$:

$$\begin{aligned} LU &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2-8 & 2-2 \\ -2 & -1+8 & -1+2+1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} = A. \end{aligned}$$

- Given $A \in \mathbb{R}^{n \times n}$ and $k \in \{1, \dots, n\}$, we denote by $A_{[k]}$ the **leading principal submatrix** of order k , i.e., the $k \times k$ matrix formed by the first k rows and the first k columns of A . Its determinant $\det(A_{[k]})$ is called the **leading principal minor** of order k . Note that $A_{[n]} = A$.

Observation 4.21: Existence of the LU factorization without row swaps

Let $A \in \mathbb{R}^{n \times n}$ be a nonsingular matrix. Then A admits a factorization

$$A = LU,$$

with L lower triangular with ones on the diagonal and U upper triangular, if and only if

all its leading principal minors are nonzero:

$$\det(\mathbf{A}_{[k]}) \neq 0, \quad \forall k \in \{1, \dots, n\}.$$

In this case, Gaussian elimination never requires a row swap and the pivots are

$$u_{11} = \det(\mathbf{A}_{[1]}), \quad u_{kk} = \frac{\det(\mathbf{A}_{[k]})}{\det(\mathbf{A}_{[k-1]})}, \quad k \in \{2, \dots, n\}.$$

Proof. ($\Leftarrow$) We perform Gaussian elimination without row swaps. A row addition $R_i \leftarrow R_i - \ell_{ij}R_j$ with $j < i$ does not change any leading principal minor: if $k \geq i$, it acts on $\mathbf{A}_{[k]}$ as a row addition; if $k < i$, it does not modify $\mathbf{A}_{[k]}$. Suppose that the first $k-1$ steps have been performed. Then the leading principal submatrix of order k of the current matrix is upper triangular with diagonal entries $u_{11}, \dots, u_{kk}$, hence

$$\det(\mathbf{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk}.$$

Since $\det(\mathbf{A}_{[k]}) \neq 0$, the pivot $u_{kk} = \det(\mathbf{A}_{[k]}) / \det(\mathbf{A}_{[k-1]})$ is nonzero and step k can be performed without a row swap.

($\Rightarrow$) If $\mathbf{A} = \mathbf{L}\mathbf{U}$, since $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular, the first k rows and columns of the product only involve the first k rows and columns of the factors: $\mathbf{A}_{[k]} = \mathbf{L}_{[k]}\mathbf{U}_{[k]}$. Hence $\det(\mathbf{A}_{[k]}) = 1 \cdot u_{11} \cdots u_{kk}$. Since $\det(\mathbf{A}) = u_{11} \cdots u_{nn} \neq 0$, all u_{ii} are nonzero, and so are all the leading principal minors. $\square$

- For the matrix of Example 4.49, we have $\det(\mathbf{A}_{[1]}) = 2$, $\det(\mathbf{A}_{[2]}) = 2 \cdot (-6) - 1 \cdot 4 = -16$ and $\det(\mathbf{A}_{[3]}) = \det(\mathbf{A}) = -16$, which give the pivots $u_{11} = 2$, $u_{22} = -16/2 = -8$ and $u_{33} = -16/(-16) = 1$.
- If some leading principal minor is zero, a row swap is needed and we must compute a PLU factorization (with $\mathbf{P} \neq \mathbf{I}$). This happens in the next examples.

Example 4.51: PLU factorization with row permutation

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $a_{11} = 0$, we cannot use it as a pivot. We need to swap rows 1 and 2.

Let

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

be the permutation matrix that swaps rows 1 and 2.

Then:

$$\mathbf{PA} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now $\mathbf{PA}$ already has an upper triangular form (no elimination needed below the diagonal), so:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Therefore, the PLU factorization is:

$$\mathbf{PA} = \mathbf{LU},$$

where $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\mathbf{L} = \mathbf{I}$, and $\mathbf{U} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Example 4.52: PLU factorization with a row swap during the elimination

Let us compute the PLU factorization of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Note that $\det(\mathbf{A}_{[2]}) = 2 \cdot 2 - 1 \cdot 4 = 0$: a row swap will be needed.

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$. The multipliers are $\ell_{21} = \frac{4}{2} = 2$ and $\ell_{31} = \frac{-2}{2} = -1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ -2 & 3 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 4 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_1)$$

Step 2: The entry in position (2,2) is 0 and cannot be used as a pivot. We swap rows 2 and 3:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

The multipliers already computed for rows 2 and 3 are swapped as well: now $\ell_{21} = -1$ and $\ell_{31} = 2$. The entry below the new pivot 4 is already zero, so $\ell_{32} = 0$, and the elimination is complete:

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Step 3: Build $\mathbf{P}$ and $\mathbf{L}$. The matrix $\mathbf{P}$ is obtained from $\mathbf{I}_3$ by swapping rows 2 and 3, and $\mathbf{L}$ contains the (swapped) multipliers:

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Verification: We check that $\mathbf{PA} = \mathbf{LU}$:

$$\mathbf{PA} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix},$$

$$\mathbf{LU} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & -1+4 & -1+2 \\ 4 & 2 & 2+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix} = \mathbf{PA}.$$

- Without swapping the multipliers in Step 2 we would get the wrong matrix $\tilde{\mathbf{L}}$ with $\tilde{\ell}_{21} = 2$ and $\tilde{\ell}_{31} = -1$, and $\tilde{\mathbf{L}}\mathbf{U} \neq \mathbf{PA}$ (its second row would be $(4, 6, 4)$).

4.4.2 Determinant from the PLU factorization

Observation 4.22: Determinant from the PLU factorization

Let $\mathbf{PA} = \mathbf{LU}$ be a PLU factorization of $\mathbf{A} \in \mathbb{R}^{n \times n}$, where $\mathbf{P}$ is obtained from $\mathbf{I}_n$ with s row swaps. Then

$$\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii} = \pm \prod_{i=1}^n u_{ii}.$$

In particular, $\mathbf{A}$ is invertible if and only if all the diagonal entries of $\mathbf{U}$ are nonzero.

Proof. Each row swap changes the sign of the determinant, so $\det(\mathbf{P}) = (-1)^s \det(\mathbf{I}_n) = (-1)^s$. The matrices $\mathbf{L}$ and $\mathbf{U}$ are triangular, so $\det(\mathbf{L}) = 1$ and $\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}$. From $\det(\mathbf{PA}) = \det(\mathbf{LU})$ and the product rule of determinants,

$$(-1)^s \det(\mathbf{A}) = \prod_{i=1}^n u_{ii},$$

and we multiply both sides by $(-1)^s$. □

Example 4.53: Determinant from the PLU factorization

- Example 4.49: no row swaps ($s = 0$), so $\det(\mathbf{A}) = 2 \cdot (-8) \cdot 1 = -16$.
- Example 4.51: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(1 \cdot 1) = -1$. Indeed, $\det(\mathbf{A}) = 0 \cdot 1 - 1 \cdot 1 = -1$.
- Example 4.52: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(2 \cdot 4 \cdot 1) = -8$. Indeed, by Laplace expansion along the first row, $\det(\mathbf{A}) = 2 \cdot (2 - 9) - 1 \cdot (4 + 6) + 1 \cdot (12 + 4) = -14 - 10 + 16 = -8$.

4.4.3 Exercises

Exercise 4.33

Compute the LU factorization of $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$ and verify that $\mathbf{LU} = \mathbf{A}$.

Solution

The first pivot is $a_{11} = 4$ and the multiplier is $\ell_{21} = \frac{6}{4} = \frac{3}{2}$:

$$\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} \\ \begin{pmatrix} 4 & 3 \\ 0 & -\frac{3}{2} \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{3}{2}R_1)$$

Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 4 & 3 \\ 0 & -\frac{3}{2} \end{pmatrix}, \quad \mathbf{LU} = \begin{pmatrix} 4 & 3 \\ \frac{3}{2} \cdot 4 & \frac{3}{2} \cdot 3 - \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} = \mathbf{A}.$$

Exercise 4.34

Compute the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix}$$

and verify that $\mathbf{LU} = \mathbf{A}$.

Solution

The multipliers are $\ell_{21} = \frac{2}{1} = 2$, $\ell_{31} = \frac{3}{1} = 3$ and, after the first step, $\ell_{32} = \frac{2}{1} = 2$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 3 & 8 & 13 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_2)$$

Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Verification:

$$\mathbf{LU} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4+1 & 6+1 \\ 3 & 6+2 & 9+2+2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix} = \mathbf{A}.$$

Exercise 4.35

Consider

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix}.$$

1. Check, using the leading principal minors, that $\mathbf{A}$ admits an LU factorization without row swaps.
2. Compute the LU factorization and verify that the pivots are $u_{kk} = \det(\mathbf{A}_{[k]}) / \det(\mathbf{A}_{[k-1]})$.

Solution

1. $\det(\mathbf{A}_{[1]}) = 2$, $\det(\mathbf{A}_{[2]}) = 2 \cdot (-1) - 4 \cdot 1 = -6$ and, by Laplace expansion along the first row,

$$\det(\mathbf{A}_{[3]}) = \det(\mathbf{A}) = 2 \cdot (-1 - 35) - 4 \cdot (1 + 5) + (-2) \cdot (7 - 1) = -72 - 24 - 12 = -108.$$

All the leading principal minors are nonzero, so the LU factorization exists.

2. The multipliers are $\ell_{21} = \frac{1}{2}$, $\ell_{31} = \frac{-1}{2} = -\frac{1}{2}$ and, after the first step, $\ell_{32} = \frac{9}{-3} = -3$:

$$\begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ -1 & 7 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{1}{2}R_1)$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 9 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 + \frac{1}{2}R_1)$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 0 & 18 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 3R_2)$$

Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & -3 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 0 & 18 \end{pmatrix}.$$

Indeed $u_{11} = 2$, $u_{22} = \frac{-6}{2} = -3$ and $u_{33} = \frac{-108}{-6} = 18$. Verification:

$$\mathbf{LU} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & 2-3 & -1+6 \\ -1 & -2+9 & 1-18+18 \end{pmatrix} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix} = \mathbf{A}.$$

Exercise 4.36

Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Show that $\mathbf{A}$ is nonsingular but does not admit an LU factorization without row swaps. Then compute a PLU factorization $\mathbf{PA} = \mathbf{LU}$ and verify it.

Solution

We have $\det(\mathbf{A}) = 1 \cdot (4 - 1) - 2 \cdot (2 - 0) + 0 = -1 \neq 0$, but $\det(\mathbf{A}_{[2]}) = 1 \cdot 4 - 2 \cdot 2 = 0$: the LU factorization without row swaps does not exist.

Gaussian elimination: $\ell_{21} = 2$ and $\ell_{31} = 0$ (the entry a_{31} is already zero). Then the entry in position $(2, 2)$ is zero and we swap rows 2 and 3, swapping also the multipliers (now

$\ell_{21} = 0, \ell_{31} = 2$):

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

The entry below the second pivot is already zero, so $\ell_{32} = 0$. Hence

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Verification:

$$\mathbf{PA} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 4 & 1 \end{pmatrix}, \quad \mathbf{LU} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 4 & 0+1 \end{pmatrix} = \mathbf{PA}.$$

Exercise 4.37

Compute a PLU factorization of

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & 5 & 4 \end{pmatrix},$$

verify that $\mathbf{PA} = \mathbf{LU}$, and compute $\det(\mathbf{A})$ from $\mathbf{U}$.

Solution

Since $a_{11} = 0$, we swap rows 1 and 2 (no multiplier has been computed yet). Then $\ell_{21} =$

$\frac{0}{2} = 0$, $\ell_{31} = \frac{4}{2} = 2$ and, after the first step, $\ell_{32} = \frac{3}{1} = 3$:

$$\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & 5 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 5 & 4 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & -2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_2)$$

Hence

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{pmatrix}.$$

Verification:

$$\mathbf{PA} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 5 & 4 \end{pmatrix}, \quad \mathbf{LU} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 2+3 & 6+3-5 \end{pmatrix} = \mathbf{PA}.$$

One row swap was used ($s = 1$), hence $\det(\mathbf{A}) = -(2 \cdot 1 \cdot (-5)) = 10$.

Exercise 4.38

Compute the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 2 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{pmatrix}$$

and use it to compute $\det(\mathbf{A})$ and $\det(\mathbf{A}^{-1})$.

Solution

The multipliers are $\ell_{21} = \frac{6}{3} = 2$ and $\ell_{31} = \frac{3}{3} = 1$:

$$\begin{pmatrix} 3 & 1 & 2 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 3 & 1 & 5 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

The entry in position $(3, 2)$ is already zero, so $\ell_{32} = 0$ and

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

No row swaps were used, hence

$$\det(\mathbf{A}) = \det(\mathbf{L}) \det(\mathbf{U}) = 1 \cdot (3 \cdot 1 \cdot 3) = 9, \quad \det(\mathbf{A}^{-1}) = \frac{1}{9}.$$

Exercise 4.39

Consider, for $k \in \mathbb{R}$, the matrix

$$\mathbf{A}(k) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

1. For which values of k is $\mathbf{A}(k)$ nonsingular and admits an LU factorization without row swaps? Compute it.
2. For $k = 1$, compute a PLU factorization.

Solution

1. The leading principal minors are

$$\det(\mathbf{A}_{[1]}) = 1, \quad \det(\mathbf{A}_{[2]}) = k - 1, \quad \det(\mathbf{A}_{[3]}) = 1 \cdot (k - 1) - 1 \cdot (1 - 0) + 0 = k - 2.$$

They are all nonzero if and only if $k \neq 1$ and $k \neq 2$. For these values, $\ell_{21} = 1$, $\ell_{31} = 0$

and $\ell_{32} = \frac{1}{k-1}$:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 0 & \frac{k-2}{k-1} \end{pmatrix} \quad (R_3 \leftarrow R_3 - \frac{1}{k-1}R_2)$$

since $1 - \frac{1}{k-1} = \frac{k-2}{k-1}$. Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & \frac{1}{k-1} & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 0 & \frac{k-2}{k-1} \end{pmatrix},$$

and indeed $u_{11}u_{22}u_{33} = k-2 = \det(\mathbf{A}(k))$. For $k=2$ the matrix is singular.

2. For $k=1$ we have $\det(\mathbf{A}(1)) = -1 \neq 0$ but $\det(\mathbf{A}_{[2]}) = 0$. After $R_2 \leftarrow R_2 - R_1$ ($\ell_{21} = 1, \ell_{31} = 0$) the entry in position $(2, 2)$ is zero, so we swap rows 2 and 3 together with their multipliers (now $\ell_{21} = 0, \ell_{31} = 1$):

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

Then $\ell_{32} = 0$ and

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$\mathbf{PA}(1) = \mathbf{LU} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

4.5 Eigenvalues and eigenvectors

4.5.1 Eigenvalues

Definition 4.23: eigenvalue and eigenvector

Let $Q \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar $\lambda \in \mathbb{R}$ is called an **eigenvalue** of Q if there exists a nonzero vector $v \in \mathbb{R}^n \setminus \{0\}$ such that

$$Qv = \lambda v.$$

In this case, v is called an **eigenvector** associated with λ .

- Eigenvalues are solutions of the **characteristic equation**

$$\det(Q - \lambda I) = 0.$$

- If Q is symmetric, then all its eigenvalues are real.
- The function $p(\lambda) = \det(Q - \lambda I)$ is a polynomial of degree n in λ , called the **characteristic polynomial** of Q .
- If λ is an eigenvalue, the eigenvectors associated with λ are the nonzero solutions v of the homogeneous system

$$(Q - \lambda I)v = 0.$$

- If v is an eigenvector associated with λ , then also αv is an eigenvector associated with λ , for every $\alpha \in \mathbb{R} \setminus \{0\}$. Eigenvectors are therefore never unique: we just give one of them (“for example”).

A square symmetric matrix $Q \in \mathbb{R}^{n \times n}$ is

- positive semidefinite if and only if all its eigenvalues are non-negative.
- positive definite if and only if all its eigenvalues are strictly positive.
- negative semidefinite if and only if all its eigenvalues are non-positive.
- negative definite if and only if all its eigenvalues are strictly negative.

- Recall that (semi)definiteness has been defined in the chapter on matrices through the sign of $x'Qx$: for instance, Q is positive semidefinite if $x'Qx \geq 0$ for all $x \in \mathbb{R}^n$. The box above gives an equivalent characterization in terms of eigenvalues.

- One direction is easy to see: if $\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then

$$\mathbf{v}'\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}'\mathbf{v} = \lambda \sum_{i=1}^n v_i^2, \quad \text{with} \quad \sum_{i=1}^n v_i^2 > 0,$$

so $\mathbf{v}'\mathbf{Q}\mathbf{v}$ and λ have the same sign. For example, if $\mathbf{Q}$ is positive semidefinite, then every eigenvalue satisfies $\lambda \geq 0$.

- A symmetric matrix which is neither positive semidefinite nor negative semidefinite is called **indefinite**: equivalently, it has at least one strictly positive and at least one strictly negative eigenvalue.

Example 4.54: eigenvalues and eigenvectors

Let us consider the matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The characteristic equation is:

$$\det(\mathbf{Q} - \lambda\mathbf{I}) = \det \begin{pmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Hence

$$\det(\mathbf{Q} - \lambda\mathbf{I}) = 0 \iff (\lambda - 1)(\lambda - 3) = 0,$$

and therefore the eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$

An eigenvector associated with $\lambda_1 = 1$ can be found by solving

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0},$$

which gives (for example)

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Similarly, an eigenvector associated with $\lambda_2 = 3$ satisfies

$$(\mathbf{Q} - 3\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0},$$

which gives (for example)

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore,

$$\mathbf{Q}\mathbf{v}_1 = 1\mathbf{v}_1, \quad \mathbf{Q}\mathbf{v}_2 = 3\mathbf{v}_2.$$

We can verify it directly:

$$\mathbf{Q} \mathbf{v}_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \mathbf{v}_1, \quad \mathbf{Q} \mathbf{v}_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} = 3 \mathbf{v}_2.$$

Since $\mathbf{Q}$ is symmetric and both eigenvalues are strictly positive, the matrix $\mathbf{Q}$ is **positive definite**.

4.5.2 Properties of eigenvalues

4.5.2.1 Triangular and diagonal matrices

Observation 4.24: eigenvalues of triangular and diagonal matrices

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be an upper triangular, a lower triangular or a diagonal matrix. Then the eigenvalues of $\mathbf{Q}$ are its diagonal entries $q_{11}, q_{22}, \dots, q_{nn}$.

Proof. If $\mathbf{Q}$ is triangular (or diagonal), then also $\mathbf{Q} - \lambda \mathbf{I}$ is triangular (or diagonal), with diagonal entries $q_{11} - \lambda, \dots, q_{nn} - \lambda$. Recall that the determinant of a triangular matrix is the product of its diagonal entries. Hence

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \prod_{i=1}^n (q_{ii} - \lambda),$$

which is equal to zero if and only if $\lambda = q_{ii}$ for some $i \in \{1, 2, \dots, n\}$. $\square$

Example 4.55: eigenvalues of a triangular matrix

Let us consider the upper triangular matrix

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{pmatrix}.$$

The characteristic equation is

$$\det(\mathbf{U} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & 1 & 3 \\ 0 & -1 - \lambda & 4 \\ 0 & 0 & 5 - \lambda \end{pmatrix} = (2 - \lambda)(-1 - \lambda)(5 - \lambda) = 0,$$

and therefore the eigenvalues are the diagonal entries:

$$\lambda_1 = 2, \quad \lambda_2 = -1, \quad \lambda_3 = 5.$$

4.5.2.2 Trace, determinant and eigenvalues

Definition 4.25: trace of a matrix

Let $Q \in \mathbb{R}^{n \times n}$ be a square matrix. The **trace** of Q is the sum of its diagonal entries:

$$\operatorname{tr}(Q) = \sum_{i=1}^n q_{ii} = q_{11} + q_{22} + \cdots + q_{nn}.$$

- The characteristic polynomial $p(\lambda) = \det(Q - \lambda I)$ has degree n . Over the complex numbers, a polynomial of degree n has exactly n roots $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$, provided that each root is counted as many times as its **multiplicity** (the number of times the factor $(\lambda_i - \lambda)$ appears in the factorization). Hence

$$p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda).$$

- The real roots are the eigenvalues of Definition 4.23. The roots which are not real are called **complex eigenvalues** of Q (their eigenvectors have complex entries). If Q is symmetric, all the roots are real.

Observation 4.26: determinant, trace and eigenvalues

Let $Q \in \mathbb{R}^{n \times n}$ and let $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$ be the roots of its characteristic polynomial, each repeated according to its multiplicity. Then

$$\det(Q) = \prod_{i=1}^n \lambda_i, \quad \operatorname{tr}(Q) = \sum_{i=1}^n \lambda_i.$$

Idea of the proof. Setting $\lambda = 0$ in $p(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ gives $\det(Q) = p(0) = \lambda_1 \lambda_2 \cdots \lambda_n$. For the trace, we compare the coefficients of λ^{n-1} : in the product $(\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ this coefficient is $(-1)^{n-1} \sum_{i=1}^n \lambda_i$, while in $\det(Q - \lambda I)$ the power λ^{n-1} only comes from the product of the diagonal entries $(q_{11} - \lambda) \cdots (q_{nn} - \lambda)$, and its coefficient is $(-1)^{n-1} \sum_{i=1}^n q_{ii}$. $\square$

Example 4.56: trace and determinant

- For the matrix of Example 4.54 we have $\lambda_1 = 1$ and $\lambda_2 = 3$, and indeed

$$\operatorname{tr} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 2 + 2 = 4 = 1 + 3, \quad \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 = 1 \cdot 3.$$

- Multiplicities must be taken into account. The diagonal matrix

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

has characteristic polynomial $(4 - \lambda)^2(-3 - \lambda)$: the eigenvalue 4 has multiplicity 2 and the eigenvalue -3 has multiplicity 1. Hence $\lambda_1 = \lambda_2 = 4$, $\lambda_3 = -3$, and

$$\operatorname{tr}(\mathbf{D}) = 4 + 4 - 3 = 5, \quad \det(\mathbf{D}) = 4 \cdot 4 \cdot (-3) = -48.$$

- Complex roots must be taken into account. For the (non symmetric) matrix

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{we have} \quad \det(\mathbf{R} - \lambda \mathbf{I}) = \lambda^2 + 1,$$

which has no real roots: $\mathbf{R}$ has no (real) eigenvalues. Its complex eigenvalues are $\lambda_1 = i$ and $\lambda_2 = -i$, and indeed

$$\operatorname{tr}(\mathbf{R}) = 0 = i + (-i), \quad \det(\mathbf{R}) = 1 = i \cdot (-i).$$

4.5.2.3 Eigenvectors of distinct eigenvalues

- Recall that the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$ implies $\alpha_1 = \dots = \alpha_k = 0$.

Observation 4.27: eigenvectors of distinct eigenvalues

Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be **distinct** eigenvalues of $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be associated eigenvectors ($\mathbf{Q}\mathbf{v}_i = \lambda_i \mathbf{v}_i$). Then $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent.

- In particular, if $\mathbf{Q} \in \mathbb{R}^{n \times n}$ has n distinct real eigenvalues, then one eigenvector for each eigenvalue gives n linearly independent vectors of $\mathbb{R}^n$.
- In Example 4.54, the eigenvectors $\mathbf{v}_1 = (1, 1)'$ and $\mathbf{v}_2 = (1, -1)'$ are linearly independent, since neither of them is a multiple of the other.

4.5.3 A complete example with a 3×3 matrix

Example 4.57: eigenvalues and eigenvectors of a 3×3 matrix

Let us consider the symmetric matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Characteristic polynomial. Expanding along the first row (Laplace expansion):

$$\begin{aligned}
 \det(\mathbf{Q} - \lambda \mathbf{I}) &= \det \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 1 & 3 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{pmatrix} \\
 &= (2 - \lambda)[(3 - \lambda)(2 - \lambda) - 1] - 1 \cdot [1 \cdot (2 - \lambda) - 1 \cdot 0] \\
 &= (2 - \lambda)[(3 - \lambda)(2 - \lambda) - 2] = (2 - \lambda)(\lambda^2 - 5\lambda + 4) \\
 &= (2 - \lambda)(\lambda - 1)(\lambda - 4).
 \end{aligned}$$

Therefore the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 4.$$

Eigenvector for $\lambda_1 = 1$. We solve

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} v_1 + v_2 = 0 \\ v_1 + 2v_2 + v_3 = 0 \\ v_2 + v_3 = 0 \end{cases}$$

From the first and the third equation, $v_1 = -v_2$ and $v_3 = -v_2$ (the second equation is then satisfied). Choosing $v_2 = -1$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Eigenvector for $\lambda_2 = 2$. We solve

$$(\mathbf{Q} - 2\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} v_2 = 0 \\ v_1 + v_2 + v_3 = 0 \end{cases}$$

hence $v_2 = 0$, $v_3 = -v_1$ and, for example,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Eigenvector for $\lambda_3 = 4$. We solve

$$(\mathbf{Q} - 4\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} -2v_1 + v_2 = 0 \\ v_1 - v_2 + v_3 = 0 \\ v_2 - 2v_3 = 0 \end{cases}$$

From the first and the third equation, $v_2 = 2v_1$ and $v_3 = v_2/2 = v_1$ (the second equation is then satisfied: $v_1 - 2v_1 + v_1 = 0$). Choosing $v_1 = 1$:

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Checks. By Observation 4.26,

$$\operatorname{tr}(\mathbf{Q}) = 2 + 3 + 2 = 7 = 1 + 2 + 4, \quad \det(\mathbf{Q}) = 2 \cdot (6 - 1) - 1 \cdot (2 - 0) = 8 = 1 \cdot 2 \cdot 4.$$

Since the three eigenvalues are distinct, by Observation 4.27 the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent. Finally, $\mathbf{Q}$ is symmetric with strictly positive eigenvalues, hence it is **positive definite**.

4.5.4 Sylvester's criterion

- Computing all the eigenvalues of a matrix can be difficult. For symmetric matrices, definiteness and semidefiniteness can also be checked by computing determinants of suitable square submatrices.

Definition 4.28: principal minors

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$.

- A **principal minor** of order k of $\mathbf{Q}$ is the determinant of the $k \times k$ submatrix obtained from $\mathbf{Q}$ by keeping the rows and the columns with the **same** indices $i_1 < i_2 < \dots < i_k$ (equivalently, by deleting the same $n - k$ rows and columns).
- The **leading principal minor** of order k is the principal minor obtained by keeping the first k rows and columns:

$$\Delta_k = \det \begin{pmatrix} q_{11} & \cdots & q_{1k} \\ \vdots & \ddots & \vdots \\ q_{k1} & \cdots & q_{kk} \end{pmatrix}, \quad k = 1, 2, \dots, n.$$

- A matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ has exactly n leading principal minors $\Delta_1 = q_{11}, \Delta_2, \dots, \Delta_n = \det(\mathbf{Q})$.
- For $n = 3$, the principal minors are: of order 1, q_{11}, q_{22}, q_{33} ; of order 2, the determinants of the submatrices with indices $\{1, 2\}, \{1, 3\}, \{2, 3\}$; of order 3, $\det(\mathbf{Q})$.

Theorem 4.29: Sylvester's criterion

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be a **symmetric** matrix with leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$. Then

- $\mathbf{Q}$ is **positive definite** if and only if

$$\Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\};$$

- $\mathbf{Q}$ is **negative definite** if and only if the leading principal minors alternate in sign starting with a negative one:

$$\Delta_1 < 0, \quad \Delta_2 > 0, \quad \Delta_3 < 0, \quad \dots$$

i.e.

$$(-1)^k \Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\}.$$

Observation 4.30: semidefinite matrices and principal minors

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then

- $\mathbf{Q}$ is **positive semidefinite** if and only if **all** its principal minors (not only the leading ones) are ≥ 0 ;
- $\mathbf{Q}$ is **negative semidefinite** if and only if every principal minor of order k has the sign of $(-1)^k$ or is zero, i.e., $(-1)^k (\text{principal minor of order } k) \geq 0$, for all k .

For semidefiniteness it is **not** enough to check $\Delta_k \geq 0$ for the leading principal minors only.

Example 4.58: a positive definite matrix

For the matrix of Example 4.57

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

the leading principal minors are

$$\Delta_1 = 2 > 0, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5 > 0, \quad \Delta_3 = \det(\mathbf{Q}) = 8 > 0.$$

By Sylvester's criterion, $\mathbf{Q}$ is positive definite, in agreement with its eigenvalues 1, 2, 4.

Example 4.59: an indefinite matrix

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have $\Delta_1 = 1 > 0$ and $\Delta_2 = 1 - 4 = -3 < 0$. Hence:

- $\mathbf{Q}$ is not positive definite ($\Delta_2 < 0$) and not negative definite ($\Delta_1 > 0$);
- $\mathbf{Q}$ is not positive semidefinite (the principal minor Δ_2 of order 2 is negative) and not negative semidefinite ($(-1)^2 \Delta_2 < 0$).

Therefore $\mathbf{Q}$ is **indefinite**. Indeed, recalling the definition via $\mathbf{x}'\mathbf{Q}\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 > 0, \quad \mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 - 4 + 1 = -2 < 0.$$

The eigenvalues are the roots of $(1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1)$, i.e., 3 and -1 : one positive and one negative.

Example 4.60: a singular positive semidefinite matrix

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The leading principal minors are $\Delta_1 = 1$, $\Delta_2 = 2 - 1 = 1$ and $\Delta_3 = \det(\mathbf{Q}) = 1 \cdot (2 - 1) - 1 \cdot (1 - 0) = 0$. Since $\Delta_3 = 0$, $\mathbf{Q}$ is **not** positive definite.

To decide whether it is positive semidefinite we check **all** principal minors:

- order 1: $q_{11} = 1$, $q_{22} = 2$, $q_{33} = 1$;
- order 2: $\det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = 1$, $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$, $\det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1$;
- order 3: $\det(\mathbf{Q}) = 0$.

All principal minors are ≥ 0 , hence $\mathbf{Q}$ is **positive semidefinite**. Indeed

$$\mathbf{x}'\mathbf{Q}\mathbf{x} = x_1^2 + 2x_1x_2 + 2x_2^2 + 2x_2x_3 + x_3^2 = (x_1 + x_2)^2 + (x_2 + x_3)^2 \geq 0,$$

and the eigenvalues are 0, 1, 3 ($\mathbf{v} = (1, -1, 1)'$ satisfies $\mathbf{Q}\mathbf{v} = \mathbf{0} = 0\mathbf{v}$).

Example 4.61: leading principal minors are not enough for semidefiniteness

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

The leading principal minors are $\Delta_1 = 0 \geq 0$ and $\Delta_2 = 0 \geq 0$. However $\mathbf{Q}$ is **not** positive semidefinite: for

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{we have} \quad \mathbf{x}'\mathbf{Q}\mathbf{x} = -1 < 0.$$

The principal minor $q_{22} = -1$ (which is not a leading one) detects this. In fact, the eigenvalues are 0 and -1 , and $\mathbf{Q}$ is negative semidefinite.

4.5.5 Exercises**Exercise 4.40**

Compute the eigenvalues of the matrix

$$\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

and one eigenvector associated with each eigenvalue.

Solution

The characteristic equation is

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5) = 0,$$

hence $\lambda_1 = 2$ and $\lambda_2 = 5$.

For $\lambda_1 = 2$:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff 2v_1 + v_2 = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

For $\lambda_2 = 5$:

$$(\mathbf{A} - 5\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff v_1 = v_2, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Check: $\mathbf{A}\mathbf{v}_1 = (4 - 2, 2 - 6)' = (2, -4)' = 2\mathbf{v}_1$ and $\mathbf{A}\mathbf{v}_2 = (5, 5)' = 5\mathbf{v}_2$.

Exercise 4.41

Compute the eigenvalues and one eigenvector for each eigenvalue of the symmetric matrix

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}.$$

Is $\mathbf{Q}$ positive definite?

Solution

The characteristic equation is

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6) = 0,$$

hence $\lambda_1 = 1$ and $\lambda_2 = 6$.

For $\lambda_1 = 1$:

$$\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff 2v_1 + v_2 = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

For $\lambda_2 = 6$:

$$\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff v_1 = 2v_2, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Since $\mathbf{Q}$ is symmetric and both eigenvalues are strictly positive, $\mathbf{Q}$ is **positive definite**. (Note also that $\mathbf{v}_1' \mathbf{v}_2 = 2 - 2 = 0$.)

Exercise 4.42

Without computing any determinant, find the eigenvalues of

$$U = \begin{pmatrix} 3 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

Then find one eigenvector for each eigenvalue and check the trace and the determinant.

Solution

U is upper triangular, hence its eigenvalues are its diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = -1, \quad \lambda_3 = 2.$$

- $\lambda_1 = 3$: $(U - 3I)v = 0$ gives $v_2 - 2v_3 = 0$, $-4v_2 + 4v_3 = 0$, $-v_3 = 0$, hence $v_2 = v_3 = 0$ and $v_1 = (1, 0, 0)'$.
- $\lambda_2 = -1$:

$$(U + I)v = \begin{pmatrix} 4 & 1 & -2 \\ 0 & 0 & 4 \\ 0 & 0 & 3 \end{pmatrix} v = 0 \implies v_3 = 0, \quad 4v_1 + v_2 = 0, \quad v_2 = \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix}.$$

- $\lambda_3 = 2$:

$$(U - 2I)v = \begin{pmatrix} 1 & 1 & -2 \\ 0 & -3 & 4 \\ 0 & 0 & 0 \end{pmatrix} v = 0 \implies v_2 = \frac{4}{3}v_3, \quad v_1 = -v_2 + 2v_3 = \frac{2}{3}v_3.$$

Choosing $v_3 = 3$: $v_3 = (2, 4, 3)'$.

Check: $\text{tr}(U) = 3 - 1 + 2 = 4 = \lambda_1 + \lambda_2 + \lambda_3$ and $\det(U) = 3 \cdot (-1) \cdot 2 = -6 = \lambda_1 \lambda_2 \lambda_3$.

Exercise 4.43

Compute the characteristic polynomial, the eigenvalues and one eigenvector for each eigenvalue of

$$Q = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 2 & 3 \\ 0 & 3 & -1 \end{pmatrix}.$$

Classify Q (definite, semidefinite or indefinite).

Solution

Expanding along the first row:

$$\begin{aligned}
 \det(\mathbf{Q} - \lambda \mathbf{I}) &= \det \begin{pmatrix} -1 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 3 \\ 0 & 3 & -1 - \lambda \end{pmatrix} \\
 &= (-1 - \lambda)[(2 - \lambda)(-1 - \lambda) - 9] - 1 \cdot [1 \cdot (-1 - \lambda) - 3 \cdot 0] \\
 &= (-1 - \lambda)[(2 - \lambda)(-1 - \lambda) - 9 - 1] = (-1 - \lambda)(\lambda^2 - \lambda - 12) \\
 &= (-1 - \lambda)(\lambda - 4)(\lambda + 3).
 \end{aligned}$$

The eigenvalues are $\lambda_1 = -1$, $\lambda_2 = 4$, $\lambda_3 = -3$.

- $\lambda_1 = -1$: $(\mathbf{Q} + \mathbf{I})\mathbf{v} = \mathbf{0}$ with $\mathbf{Q} + \mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 3 & 3 \\ 0 & 3 & 0 \end{pmatrix}$ gives $v_2 = 0$ and $v_1 + 3v_3 = 0$: $\mathbf{v}_1 = (3, 0, -1)'$.
- $\lambda_2 = 4$: $\mathbf{Q} - 4\mathbf{I} = \begin{pmatrix} -5 & 1 & 0 \\ 1 & -2 & 3 \\ 0 & 3 & -5 \end{pmatrix}$ gives $v_2 = 5v_1$, $v_3 = \frac{3}{5}v_2 = 3v_1$ (and $v_1 - 10v_1 + 9v_1 = 0$): $\mathbf{v}_2 = (1, 5, 3)'$.
- $\lambda_3 = -3$: $\mathbf{Q} + 3\mathbf{I} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 5 & 3 \\ 0 & 3 & 2 \end{pmatrix}$ gives $v_2 = -2v_1$, $v_3 = -\frac{3}{2}v_2 = 3v_1$ (and $v_1 - 10v_1 + 9v_1 = 0$): $\mathbf{v}_3 = (1, -2, 3)'$.

Check: $\text{tr}(\mathbf{Q}) = -1 + 2 - 1 = 0 = -1 + 4 - 3$ and $\det(\mathbf{Q}) = -1 \cdot (-2 - 9) - 1 \cdot (-1 - 0) = 12 = (-1) \cdot 4 \cdot (-3)$.

$\mathbf{Q}$ is symmetric with a positive and a negative eigenvalue, hence it is **indefinite**.

Exercise 4.44

The matrix

$$\mathbf{Q} = \begin{pmatrix} 4 & -1 & 1 \\ -1 & 4 & -1 \\ 1 & -1 & 4 \end{pmatrix}$$

has the eigenvalue $\lambda = 3$ with multiplicity 2. Using the trace, find the third eigenvalue; then check the result with the determinant and find an eigenvector for it.

Solution

The eigenvalues, counted with multiplicity, are $\lambda_1 = \lambda_2 = 3$ and λ_3 . Since the trace is the sum of the eigenvalues:

$$\text{tr}(\mathbf{Q}) = 4 + 4 + 4 = 12 = 3 + 3 + \lambda_3 \implies \lambda_3 = 6.$$

Check with the determinant (Laplace expansion along the first row):

$$\det(\mathbf{Q}) = 4(16 - 1) - (-1)(-4 + 1) + 1(1 - 4) = 60 - 3 - 3 = 54 = 3 \cdot 3 \cdot 6. \quad \checkmark$$

Eigenvector for $\lambda_3 = 6$:

$$(\mathbf{Q} - 6\mathbf{I})\mathbf{v} = \begin{pmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

Adding the first two equations: $-3v_1 - 3v_2 = 0$, i.e., $v_2 = -v_1$; then from the first equation $v_3 = 2v_1 + v_2 = v_1$. Hence $\mathbf{v} = (1, -1, 1)'$; indeed $\mathbf{Q}\mathbf{v} = (4 + 1 + 1, -1 - 4 - 1, 1 + 1 + 4)' = (6, -6, 6)' = 6\mathbf{v}$.

Exercise 4.45

Using the eigenvalues, classify the following symmetric matrices:

$$\text{a) } \mathbf{Q}_1 = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}, \quad \text{b) } \mathbf{Q}_2 = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \quad \text{c) } \mathbf{Q}_3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

Solution

- a) $(-3 - \lambda)^2 - 1 = \lambda^2 + 6\lambda + 8 = (\lambda + 2)(\lambda + 4)$: the eigenvalues are -2 and -4 , both strictly negative, hence $\mathbf{Q}_1$ is **negative definite**.
- b) $(1 - \lambda)^2 - 9 = \lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2)$: the eigenvalues are 4 and -2 , hence $\mathbf{Q}_2$ is **indefinite**.
- c) $(2 - \lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4)$: the eigenvalues are 0 and 4 , both non-negative, hence $\mathbf{Q}_3$ is **positive semidefinite** (but not positive definite, since 0 is an eigenvalue).

Exercise 4.46

Using Sylvester's criterion, prove that the matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

is positive definite.

Solution

$\mathbf{Q}$ is symmetric and its leading principal minors are

$$\Delta_1 = 2 > 0, \quad \Delta_2 = \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 > 0,$$

$$\Delta_3 = \det(\mathbf{Q}) = 2(4 - 1) - (-1)(-2 - 0) + 0 = 6 - 2 = 4 > 0.$$

All the leading principal minors are strictly positive, hence $\mathbf{Q}$ is **positive definite**. (Here Sylvester's criterion is convenient: the eigenvalues of $\mathbf{Q}$ are 2 and $2 \pm \sqrt{2}$, which are not integers.)

Exercise 4.47

Using Sylvester's criterion, classify

$$\mathbf{Q} = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -3 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

Solution

$\mathbf{Q}$ is symmetric and its leading principal minors are

$$\Delta_1 = -2 < 0, \quad \Delta_2 = \det \begin{pmatrix} -2 & 1 \\ 1 & -3 \end{pmatrix} = 6 - 1 = 5 > 0,$$

$$\Delta_3 = \det(\mathbf{Q}) = -2(6 - 1) - 1(-2 - 0) = -10 + 2 = -8 < 0.$$

The leading principal minors alternate in sign starting with a negative one, hence $\mathbf{Q}$ is **negative definite**. (Indeed, its eigenvalues are -1 , -2 and -4 .)

Exercise 4.48

Using the principal minors, decide whether the following symmetric matrices are positive semidefinite:

$$\text{a) } \mathbf{Q}_a = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{b) } \mathbf{Q}_b = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Solution

a) The leading principal minors are $\Delta_1 = 2$, $\Delta_2 = 4 - 4 = 0$, $\Delta_3 = \det(\mathbf{Q}_a) = 0$: $\mathbf{Q}_a$ is not positive definite. We check **all** principal minors:

– order 1: 2, 2, 1;

– order 2: $\det \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} = 0$, $\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 2$, $\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 2$;

– order 3: $\det(\mathbf{Q}_a) = 0$.

All are ≥ 0 : $\mathbf{Q}_a$ is **positive semidefinite**. Indeed $\mathbf{x}'\mathbf{Q}_a\mathbf{x} = 2(x_1 - x_2)^2 + x_3^2 \geq 0$, and the eigenvalues are 0, 1, 4.

b) The leading principal minors are $\Delta_1 = 1$, $\Delta_2 = 1 - 1 = 0$, $\Delta_3 = \det(\mathbf{Q}_b) = -1 \cdot 0 = 0$: they are all ≥ 0 , but this is **not** enough. The principal minor of order 1 given by the

entry $q_{33} = -1$ is negative, hence $\mathbf{Q}_b$ is **not** positive semidefinite: for $\mathbf{x} = (0, 0, 1)'$ we have $\mathbf{x}'\mathbf{Q}_b\mathbf{x} = -1 < 0$. Since for $\mathbf{x} = (1, 0, 0)'$ we have $\mathbf{x}'\mathbf{Q}_b\mathbf{x} = 1 > 0$, the matrix is **indefinite** (its eigenvalues are $2, 0, -1$).

Exercise 4.49

Let $k \in \mathbb{R}$ and

$$\mathbf{Q}(k) = \begin{pmatrix} 2 & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

For which values of k is $\mathbf{Q}(k)$ positive definite? For which values is it positive semidefinite?

Solution

$\mathbf{Q}(k)$ is symmetric for every k . Its leading principal minors are

$$\Delta_1 = 2, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & k \end{pmatrix} = 2k - 1, \quad \Delta_3 = 2(2k - 1) - 1 \cdot (2 - 0) = 4k - 4.$$

By Sylvester's criterion, $\mathbf{Q}(k)$ is positive definite if and only if

$$2 > 0, \quad 2k - 1 > 0 \Leftrightarrow k > \frac{1}{2}, \quad 4k - 4 > 0 \Leftrightarrow k > 1,$$

that is, if and only if $k > 1$.

For positive semidefiniteness we need all principal minors ≥ 0 : order 1: $2, k, 2$; order 2: $2k - 1, \det \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 4, 2k - 1$; order 3: $4k - 4$. They are all ≥ 0 if and only if $k \geq 0, k \geq \frac{1}{2}$ and $k \geq 1$, i.e., if and only if $k \geq 1$. For $k = 1$ the matrix is positive semidefinite but not positive definite: its eigenvalues are $0, 2, 3$.

Part III

Norms



Norms

5.1 Introduction to norms

- A **norm** is a mathematical function that assigns a non-negative length or size to vectors. While there are many different types of norms, they all share three essential properties that ensure they behave consistently and meaningfully.

Definition: Norm.

A function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a **norm** if it satisfies the following three properties for all vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ and all scalars $\lambda \in \mathbb{R}$:

1. Non-negativity and definiteness:

$$\|\mathbf{p}\| \geq 0 \quad \text{and} \quad \|\mathbf{p}\| = 0 \iff \mathbf{p} = \mathbf{0}$$

The norm is always non-negative, and it equals zero if and only if the vector is the zero vector.

2. Absolute homogeneity (or scalability):

$$\|\lambda \mathbf{p}\| = |\lambda| \|\mathbf{p}\|$$

Scaling a vector by a scalar λ scales its norm by the absolute value $|\lambda|$.

3. Triangle inequality (or subadditivity):

$$\|\mathbf{p} + \mathbf{w}\| \leq \|\mathbf{p}\| + \|\mathbf{w}\|$$

The norm of the sum of two vectors is at most the sum of their norms.

- **Non-negativity and definiteness** give the norm a reliable meaning as “size” or “length.” They ensure that the norm is never negative and that the only vector with zero length is the zero vector.

- **Absolute homogeneity** ensures that scaling a vector scales its length proportionally. This matches intuition: stretching a vector makes it longer in the same proportion, while reversing its direction does not change its length.
- **Triangle inequality** captures the idea that taking a detour cannot be shorter than going directly. In other words, combining two displacements cannot produce a length larger than the sum of their lengths, reflecting the “straight line is shortest” intuition behind distance.
- Different norms may emphasize different aspects of a vector (e.g., its largest component, the sum of absolute values, or the geometric length), but they all respect these fundamental axioms.

5.2 ℓ_1 norm

Definition 5.1: ℓ_1 norm

The ℓ_1 **norm** (also called **Manhattan norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \quad (5.1)$$

- The ℓ_1 **norm** represents the sum of the absolute values of all components.
- In $\mathbb{R}^2$, the ℓ_1 norm corresponds to the **Manhattan distance** (the distance a taxi would travel on a grid-like street network).
- The ℓ_1 norm is widely used in **robust statistics** and **sparse optimization** (e.g., Lasso regression).

Example 5.1: ℓ_1 norm

Consider the vectors:

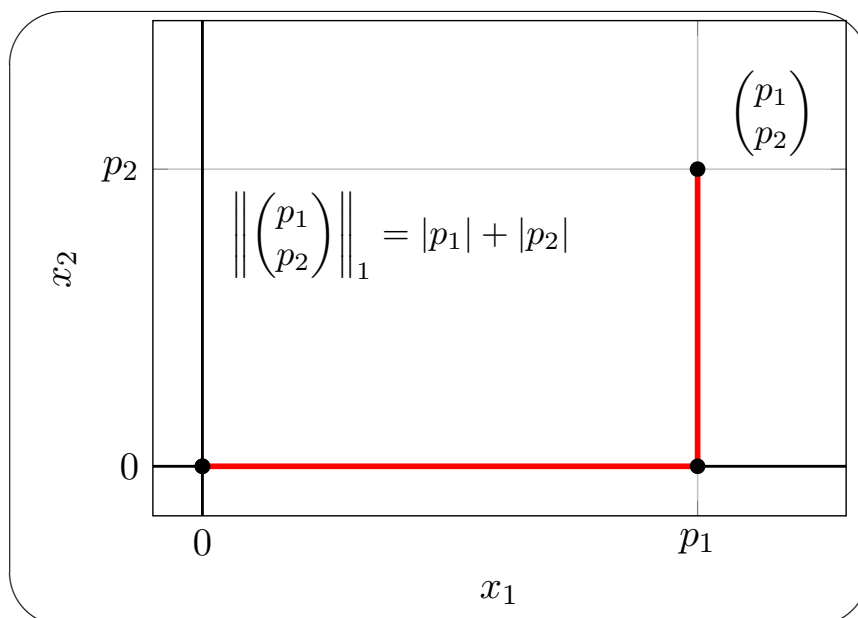
$$\mathbf{p} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_1 norms are:

$$\left\| \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right\|_1 = |3| + |-4| = 3 + 4 = 7,$$

$$\left\| \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \right\|_1 = |1| + |2| + |-2| + |3| = 1 + 2 + 2 + 3 = 8$$

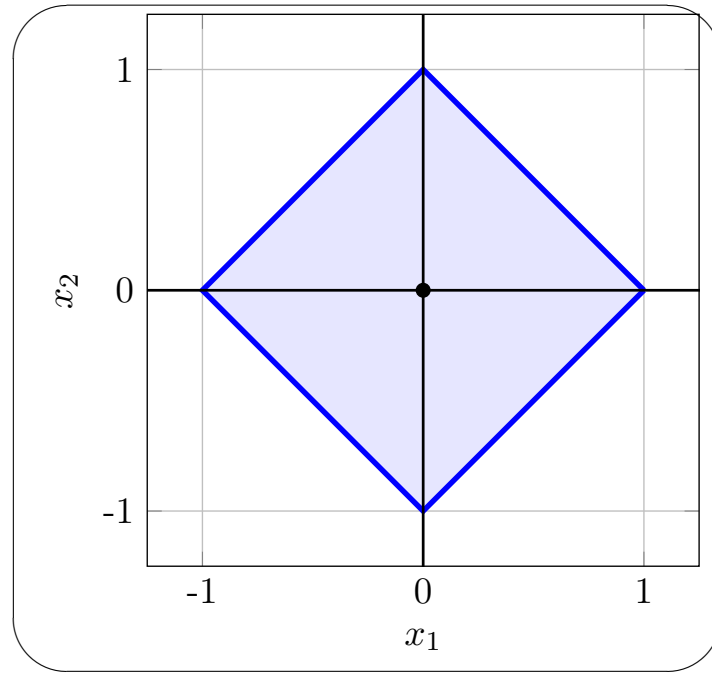
- The ℓ_1 **norm** of a column vector in $\mathbb{R}^n$ represents the **sum of the absolute values of its components**. In $\mathbb{R}^2$, this corresponds to the **Manhattan distance** (the distance traveled along grid lines), as illustrated by the following picture:



- In two dimensions, all the points at ℓ_1 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_1 \leq 1\}$$

are the points that lie on or inside a **diamond shape** (square rotated by 45°) centered at the origin.



5.2.1 Properties

- The ℓ_1 norm satisfies the three fundamental properties that define a norm.

Observation 5.2: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_1 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_1 = 0 \iff \mathbf{p} = \mathbf{0} \quad (5.2)$$

Proof. From the definition $\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$.

1. **Non-negativity:** Since $|p_j| \geq 0$ for all $j \in \{1, 2, \dots, n\}$, and the sum of non-negative numbers is non-negative, we have

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \geq 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^n.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_1 = 0$, then $\sum_{j=1}^n |p_j| = 0$. Since each $|p_j| \geq 0$, the sum equals zero only if $|p_j| = 0$ for all $j \in \{1, 2, \dots, n\}$, hence $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, i.e., $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, so

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |0| = 0.$$

□

Observation 5.3: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_1 = |\lambda| \|\mathbf{p}\|_1 \quad (5.3)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\lambda \mathbf{p}\|_1 &= \sum_{j=1}^n |\lambda p_j| \\ &= \sum_{j=1}^n |\lambda| |p_j| \quad (\text{property of absolute value}) \\ &= |\lambda| \sum_{j=1}^n |p_j| \\ &= |\lambda| \|\mathbf{p}\|_1 \end{aligned}$$

□

Observation 5.4: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_1 \leq \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1 \quad (5.4)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_1 &= \sum_{j=1}^n |p_j + w_j| \\ &\leq \sum_{j=1}^n (|p_j| + |w_j|) \quad (\text{triangle inequality for absolute value}) \\ &= \sum_{j=1}^n |p_j| + \sum_{j=1}^n |w_j| \\ &= \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1 \end{aligned}$$

□

5.3 ℓ_2 norm

Definition 5.5: ℓ_2 norm

The ℓ_2 **norm** (also called **Euclidean norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_2 = \underbrace{\sqrt{\sum_{j=1}^n p_j^2}}_{\sqrt{\mathbf{p}' \mathbf{p}}} \quad (5.5)$$

- When the subscript is omitted, i.e., when we write $\|\mathbf{p}\|$ instead of $\|\mathbf{p}\|_2$, the ℓ_2 norm is implied by default.

Example 5.2: ℓ_2 norm

- Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

Their norms are:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}, \quad \left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

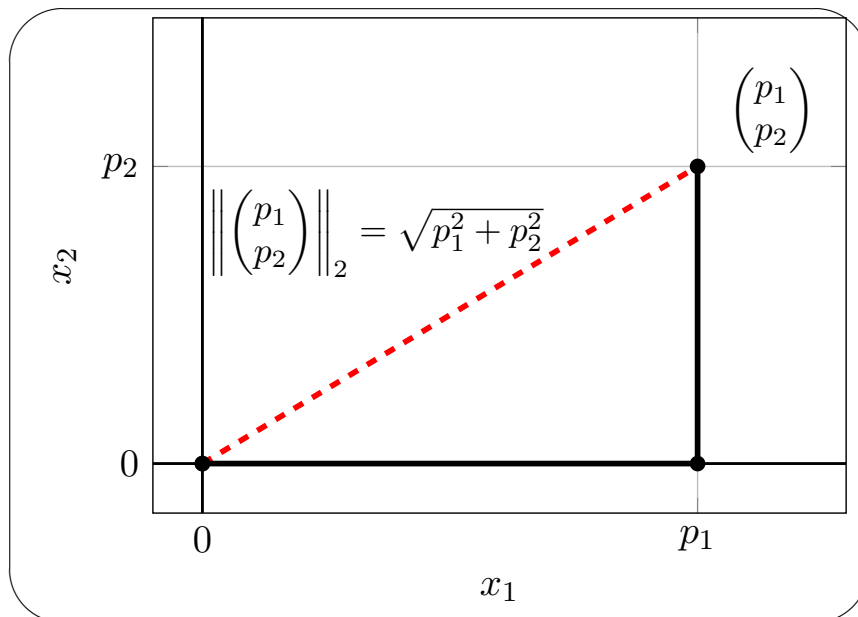
- Consider the four vertices of the diamond-shaped ℓ_1 unit ball in $\mathbb{R}^2$:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Their norms are:

$$\begin{aligned} \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|_2 &= \sqrt{1^2 + 0^2} = 1, & \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|_2 &= \sqrt{0^2 + 1^2} = 1, \\ \left\| \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\|_2 &= \sqrt{(-1)^2 + 0^2} = 1, & \left\| \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\|_2 &= \sqrt{0^2 + (-1)^2} = 1 \end{aligned}$$

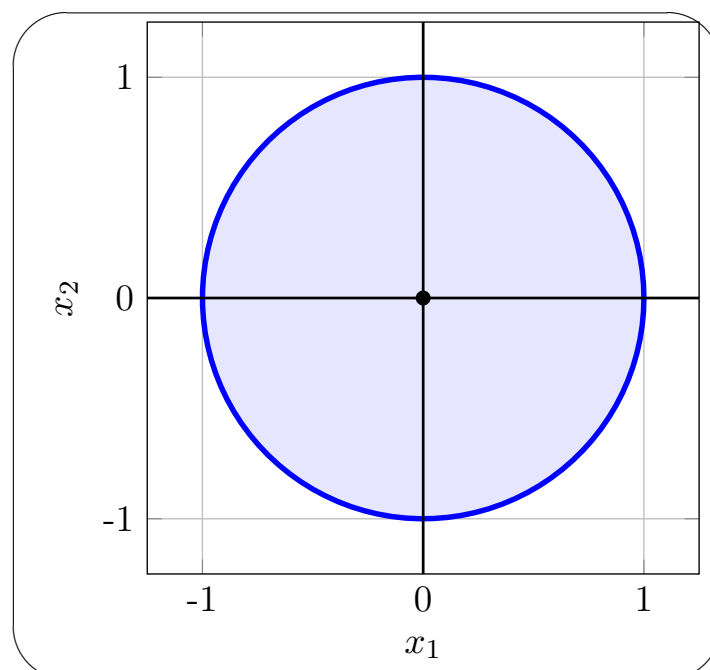
- The ℓ_2 **norm** of a column vector in $\mathbb{R}^n$ represents its **Euclidean distance from the origin**. This interpretation follows from the **Pythagorean theorem**, as illustrated in $\mathbb{R}^2$ by the following picture:



- In two dimensions, all the points at ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_2 \leq 1\}$$

are the points that lie on or inside the **unit circle** centered at the origin.



Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the norm of the **difference** of the two vectors is:

$$\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j - w_j)^2}$$

This quantity corresponds both to the **Euclidean distance** between the two points $\mathbf{p}$ and $\mathbf{w}$, and to the **length of the diagonal** of the parallelogram spanned by those points, starting from one and pointing to the other.

Example 5.3: ℓ_2 norm the difference of vectors

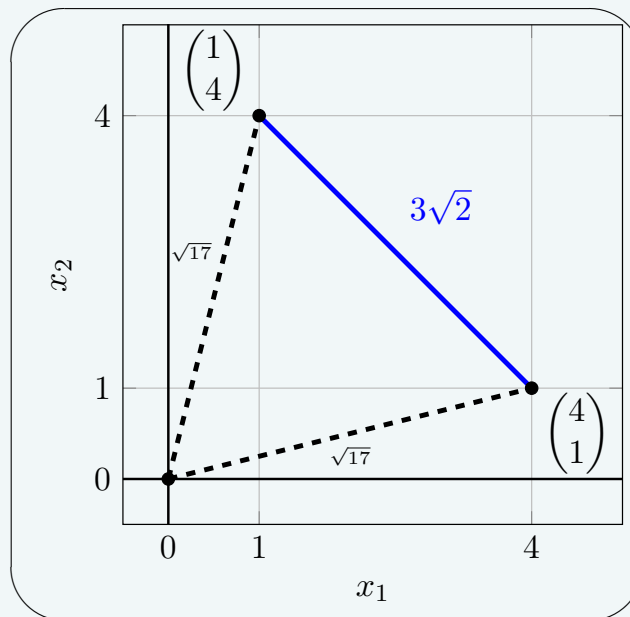
Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

The norm of the difference of the two vectors is

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1-4)^2 + (4-1)^2} = \sqrt{(-3)^2 + 3^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

Graphically, we have:



Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n,$$

the norm of the **sum** of the two vectors is:

$$\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j + w_j)^2}$$

This quantity represents the **length of the diagonal** of the parallelogram spanned by the vectors $\mathbf{p}$ and $\mathbf{w}$, both originating from the origin.

Example 5.4: ℓ_2 norm of the sum of vectors

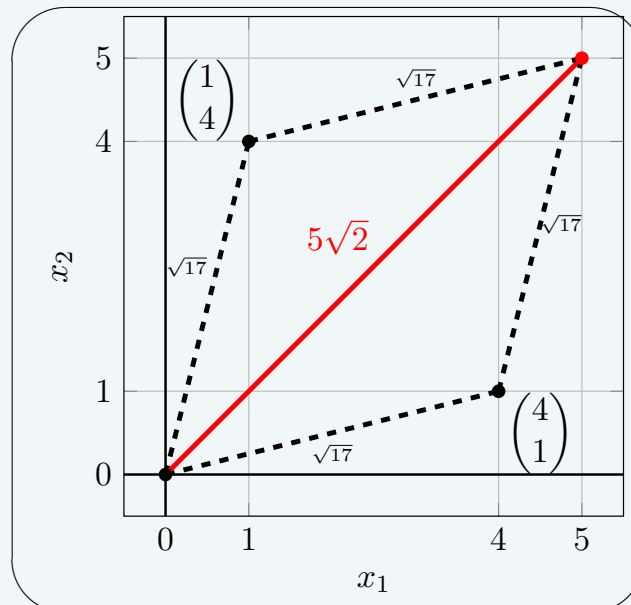
Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

The norm of the sum of the two vectors is:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

Graphically, we have:



5.3.1 Cauchy–Schwarz inequality

Observation 5.6: Cauchy–Schwarz inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\mathbf{p}' \mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (5.6)$$

Proof. If $\mathbf{p} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$ (or both), the inequality clearly holds.

Assume $\mathbf{p} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$. Consider the vector:

$$\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}, \quad \text{with } \alpha, \beta \in \mathbb{R}$$

Then, for every $\alpha, \beta \in \mathbb{R}$, we have:

$$\begin{aligned} \underbrace{\mathbf{u}' \mathbf{u}}_{\geq 0} &= (\alpha \mathbf{p} + \beta \mathbf{w})' (\alpha \mathbf{p} + \beta \mathbf{w}) \\ &= \alpha^2 \mathbf{p}' \mathbf{p} + 2\alpha\beta \mathbf{p}' \mathbf{w} + \beta^2 \mathbf{w}' \mathbf{w} \geq 0 \end{aligned}$$

Now choose:

$$\alpha = \mathbf{w}' \mathbf{w}, \quad \beta = -\mathbf{p}' \mathbf{w}$$

Substituting, we get:

$$\begin{aligned} &(\mathbf{w}' \mathbf{w})^2 \mathbf{p}' \mathbf{p} - 2(\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) + (\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) \\ &= \mathbf{w}' \mathbf{w} ((\mathbf{w}' \mathbf{w}) \mathbf{p}' \mathbf{p} - (\mathbf{p}' \mathbf{w})^2) \geq 0 \end{aligned}$$

Since $\mathbf{w}' \mathbf{w} = \|\mathbf{w}\|_2^2 > 0$, dividing both sides yields:

$$(\mathbf{p}' \mathbf{w})^2 \leq \mathbf{p}' \mathbf{p} \mathbf{w}' \mathbf{w} \implies |\mathbf{p}' \mathbf{w}| \leq \sqrt{\mathbf{p}' \mathbf{p} \mathbf{w}' \mathbf{w}} = \sqrt{\mathbf{p}' \mathbf{p}} \sqrt{\mathbf{w}' \mathbf{w}} = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$$

□

5.3.2 Properties

- The ℓ_2 norm satisfies the three fundamental properties that define a norm.

Observation 5.7: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_2 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_2 = 0 \iff \mathbf{p} = \mathbf{0} \quad (5.7)$$

Proof. From the definition $\|\mathbf{p}\|_2 = \sqrt{\mathbf{p}' \mathbf{p}}$.

1. **Non-negativity:** Since $\mathbf{p}' \mathbf{p} = \sum_{j=1}^n p_j^2 \geq 0$ (as a sum of squares), and the square root function yields non-negative values, we have $\|\mathbf{p}\|_2 \geq 0$ for all $\mathbf{p} \in \mathbb{R}^n$.

2. Definiteness:

($\Rightarrow$) If $\|\mathbf{p}\|_2 = 0$, then $\sqrt{\mathbf{p}'\mathbf{p}} = 0$, hence $\mathbf{p}'\mathbf{p} = 0$, i.e.

$$\sum_{j=1}^n p_j^2 = 0.$$

Since each $p_j^2 \geq 0$, the sum equals zero only if $p_j^2 = 0$ for all $j \in \{1, 2, \dots, n\}$, hence $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, i.e., $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $\mathbf{p}'\mathbf{p} = 0$, so $\|\mathbf{p}\|_2 = \sqrt{0} = 0$.

□

Observation 5.8: Absolute homogeneity

For every column vector $\mathbf{p} \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$, we have:

$$\|\lambda \mathbf{p}\|_2 = |\lambda| \|\mathbf{p}\|_2 \quad (5.8)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_2 = \sqrt{\sum_{j=1}^n (\lambda p_j)^2} = \sqrt{\sum_{j=1}^n \lambda^2 p_j^2} = |\lambda| \sqrt{\sum_{j=1}^n p_j^2} = |\lambda| \|\mathbf{p}\|_2$$

□

Observation 5.9: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2 \quad (5.9)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_2^2 &= (\mathbf{p} + \mathbf{w})' (\mathbf{p} + \mathbf{w}) = \mathbf{p}'\mathbf{p} + 2\mathbf{p}'\mathbf{w} + \mathbf{w}'\mathbf{w} = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2\mathbf{p}'\mathbf{w} \\ &\leq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2\|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (\text{by Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_2 + \|\mathbf{w}\|_2)^2 \end{aligned}$$

Accordingly, we obtain:

$$\|\mathbf{p} + \mathbf{w}\|_2^2 \leq (\|\mathbf{p}\|_2 + \|\mathbf{w}\|_2)^2 \implies \|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2$$

□

- In $\mathbb{R}^2$, the triangle inequality states that **for any triangle the sum of the lengths of any two sides must be greater than or equal to the length of the remaining side**.

Example 5.5: triangle inequality

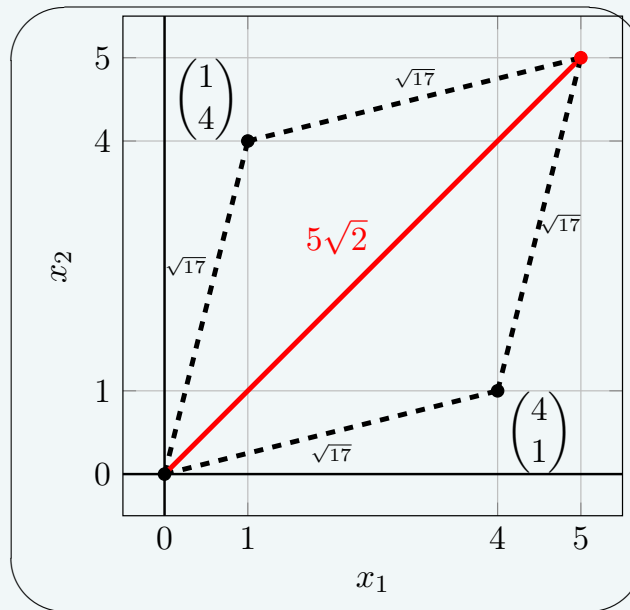
Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1+4)^2 + (4+1)^2} = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

$$5\sqrt{2} \leq \sqrt{17} + \sqrt{17}$$



- Another form of the triangle inequality is:

$$\|\mathbf{r}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2 + \|\mathbf{s}\|_2 \quad \text{that is} \quad \|\mathbf{r}\|_2 - \|\mathbf{s}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2, \quad \forall \mathbf{r}, \mathbf{s} \in \mathbb{R}^n \quad (5.10)$$

To obtain it, it suffices to set in (5.9):

$$\mathbf{p} = \mathbf{r} - \mathbf{s}, \quad \mathbf{w} = \mathbf{s}$$

Observation 5.10: reverse triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$||\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2 \quad (5.11)$$

Proof. For every two vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} - \mathbf{w}\|_2^2 &= (\mathbf{p} - \mathbf{w})' (\mathbf{p} - \mathbf{w}) = \mathbf{p}' \mathbf{p} - 2 \mathbf{p}' \mathbf{w} + \mathbf{w}' \mathbf{w} = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \mathbf{p}' \mathbf{w} \\ &\geq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (\text{by Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2)^2 \end{aligned}$$

Accordingly, we obtain:

$$(\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2)^2 \leq \|\mathbf{p} - \mathbf{w}\|_2^2 \implies ||\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$$

□

- In $\mathbb{R}^2$, the reverse triangle inequality states that **for any triangle the length of any side must be greater than or equal to the difference between the lengths of the other two sides.**

Example 5.6: reverse triangle inequality

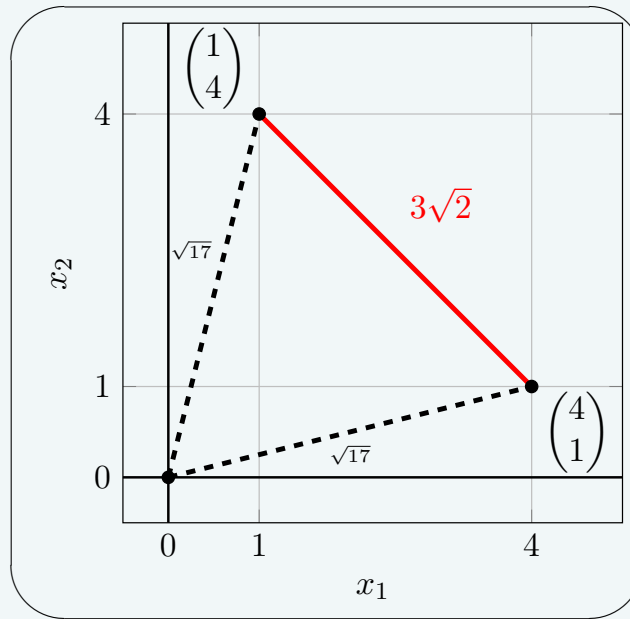
Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1-4)^2 + (4-1)^2} = 3\sqrt{2}$$

$$|\sqrt{17} - \sqrt{17}| \leq 3\sqrt{2}$$



5.4 Generalized ℓ_2 norm

Definition 5.11: generalized ℓ_2 norm

Given a symmetric positive definite matrix

$$\underbrace{\begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{21} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{n1} & q_{n2} & \cdots & q_{nn} \end{pmatrix}}_Q \in \mathbb{R}^{n \times n}$$

the **generalized ℓ_2 norm** of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_p \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|p\|_Q = \sqrt{\underbrace{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j}_{\sqrt{p' Q p}}} \quad (5.12)$$

- The **generalized ℓ_2 norm** extends the standard Euclidean norm by incorporating a positive definite matrix $\mathbf{Q}$, which weights and potentially correlates the components of the vector.
- When $\mathbf{Q}$ is the identity matrix $\mathbf{I}$, the generalized ℓ_2 norm reduces to the standard ℓ_2 norm.
- The matrix $\mathbf{Q}$ must be **positive definite** to ensure that $\mathbf{p}'\mathbf{Q}\mathbf{p} \geq 0$ for all $\mathbf{p} \in \mathbb{R}^n$, making the square root well-defined, and that $\mathbf{p}'\mathbf{Q}\mathbf{p} = 0$ only for $\mathbf{p} = \mathbf{0}$.
- In two dimensions, all the points at generalized ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\}$$

are the points that lie on or inside an **ellipse** centered at the origin, whose size, shape, and rotation are determined by the matrix $\mathbf{Q}$. In particular, the eigenvectors of $\mathbf{Q}$ determine the principal axes and the orientation, while the eigenvalues determine the lengths of the semi-axes.

Example 5.7: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = 4, \quad \lambda_2 = 1$$

and the eigenvectors are the standard basis vectors:

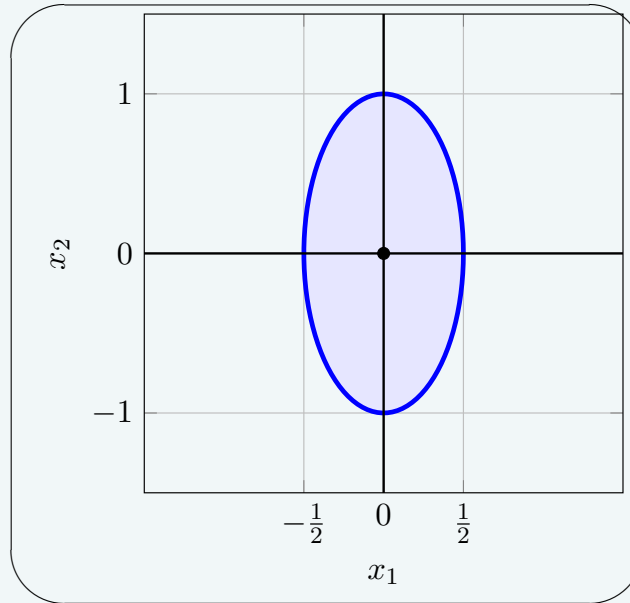
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = 4), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = 1)$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1}} = 1$.
- Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\begin{aligned} \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{4x_1^2 + x_2^2} \leq 1 \right\} \\ &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \frac{x_1^2}{(1/2)^2} + \frac{x_2^2}{1^2} \leq 1 \right\} \end{aligned}$$



Example 5.8: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{9} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = \frac{1}{9}, \quad \lambda_2 = \frac{1}{4}$$

and the eigenvectors are the standard basis vectors:

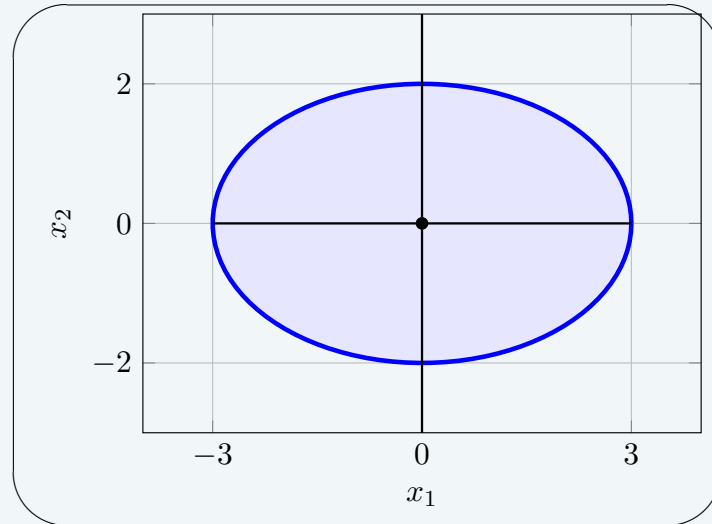
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = \frac{1}{9}), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = \frac{1}{4})$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1/9}} = 3$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1/4}} = 2$.
- Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{\frac{x_1^2}{9} + \frac{x_2^2}{4}} \leq 1 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \frac{x_1^2}{3^2} + \frac{x_2^2}{2^2} \leq 1 \right\}$$



Example 5.9: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following symmetric positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The eigenvalues of $\mathbf{Q}$ are found by solving the characteristic equation:

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

which gives:

$$\lambda_1 = 1, \quad \lambda_2 = 3$$

The corresponding eigenvectors are:

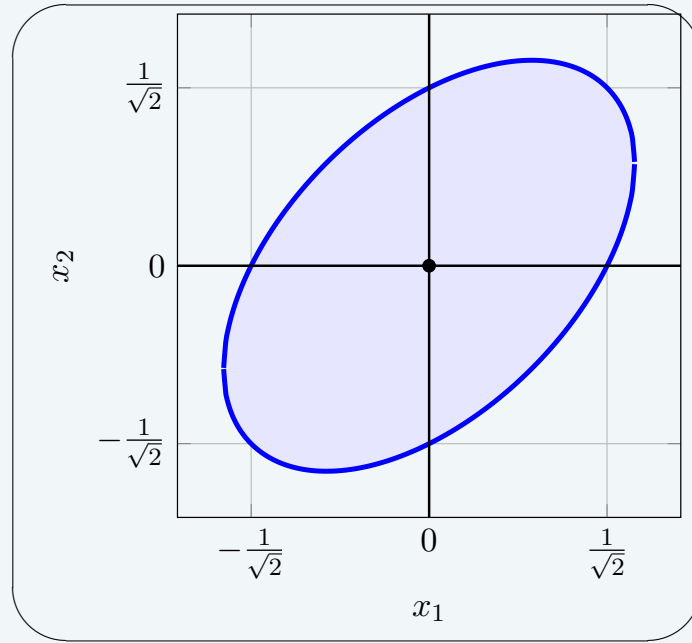
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_1 = 1), \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\text{for } \lambda_2 = 3)$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1}} = 1$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{3}}$.
- Since the eigenvectors are **not aligned** with the coordinate axes, the ellipse is **rotated**.
- The principal axes coincide with the directions of the eigenvectors.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\begin{aligned} \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{2x_1^2 - 2x_1x_2 + 2x_2^2} \leq 1 \right\} \\ &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : 2x_1^2 - 2x_1x_2 + 2x_2^2 \leq 1 \right\} \end{aligned}$$



5.4.1 Generalized Cauchy–Schwarz inequality

Observation 5.12: generalized Cauchy–Schwarz inequality

For every symmetric positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\mathbf{p}' \mathbf{Q} \mathbf{w}| \leq \|\mathbf{p}\|_{\mathbf{Q}} \|\mathbf{w}\|_{\mathbf{Q}} \quad (5.13)$$

Proof. If $\mathbf{p} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$ (or both), the inequality clearly holds.

Assume $\mathbf{p} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$. Consider the vector:

$$\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}, \quad \text{with } \alpha, \beta \in \mathbb{R}$$

Then, for every $\alpha, \beta \in \mathbb{R}$, we have:

$$\begin{aligned} \underbrace{\mathbf{u}' \mathbf{Q} \mathbf{u}}_{\geq 0} &= (\alpha \mathbf{p} + \beta \mathbf{w})' \mathbf{Q} (\alpha \mathbf{p} + \beta \mathbf{w}) \\ &= \alpha^2 \mathbf{p}' \mathbf{Q} \mathbf{p} + 2\alpha\beta \mathbf{p}' \mathbf{Q} \mathbf{w} + \beta^2 \mathbf{w}' \mathbf{Q} \mathbf{w} \geq 0 \end{aligned}$$

where we used $\mathbf{w}' \mathbf{Q} \mathbf{p} = \mathbf{p}' \mathbf{Q} \mathbf{w}$, which holds since $\mathbf{Q}$ is symmetric.

Now choose:

$$\alpha = \mathbf{w}' \mathbf{Q} \mathbf{w}, \quad \beta = -\mathbf{p}' \mathbf{Q} \mathbf{w}$$

Substituting, we get:

$$\begin{aligned} &(\mathbf{w}' \mathbf{Q} \mathbf{w})^2 \mathbf{p}' \mathbf{Q} \mathbf{p} - 2(\mathbf{p}' \mathbf{Q} \mathbf{w})^2 (\mathbf{w}' \mathbf{Q} \mathbf{w}) + (\mathbf{p}' \mathbf{Q} \mathbf{w})^2 (\mathbf{w}' \mathbf{Q} \mathbf{w}) \\ &= \mathbf{w}' \mathbf{Q} \mathbf{w} ((\mathbf{w}' \mathbf{Q} \mathbf{w}) \mathbf{p}' \mathbf{Q} \mathbf{p} - (\mathbf{p}' \mathbf{Q} \mathbf{w})^2) \geq 0 \end{aligned}$$

Since $\mathbf{w}' \mathbf{Q} \mathbf{w} = \|\mathbf{w}\|_Q^2 > 0$, dividing both sides yields:

$$(\mathbf{p}' \mathbf{Q} \mathbf{w})^2 \leq \mathbf{p}' \mathbf{Q} \mathbf{p} \cdot \mathbf{w}' \mathbf{Q} \mathbf{w} \implies |\mathbf{p}' \mathbf{Q} \mathbf{w}| \leq \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} \sqrt{\mathbf{w}' \mathbf{Q} \mathbf{w}} = \|\mathbf{p}\|_Q \|\mathbf{w}\|_Q$$

□

5.4.2 Properties

- The generalized ℓ_2 norm satisfies the three fundamental properties that define a norm.

Observation 5.13: Non-negativity and definiteness

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_Q \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_Q = 0 \iff \mathbf{p} = \mathbf{0} \quad (5.14)$$

Proof. By definition, $\|\mathbf{p}\|_Q = \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}}$. Since $\mathbf{Q}$ is positive definite, we have $\mathbf{p}' \mathbf{Q} \mathbf{p} > 0$ for all $\mathbf{p} \neq \mathbf{0}$, and $\mathbf{0}' \mathbf{Q} \mathbf{0} = 0$.

1. **Non-negativity:** For all $\mathbf{p} \in \mathbb{R}^n$, we have $\mathbf{p}' \mathbf{Q} \mathbf{p} \geq 0$. Therefore

$$\|\mathbf{p}\|_Q = \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} \geq 0.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_Q = 0$, then

$$\sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} = 0 \implies \mathbf{p}' \mathbf{Q} \mathbf{p} = 0,$$

which implies $\mathbf{p} = \mathbf{0}$ (by positive definiteness of $\mathbf{Q}$).

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then

$$\mathbf{p}' \mathbf{Q} \mathbf{p} = \mathbf{0}' \mathbf{Q} \mathbf{0} = 0,$$

and therefore $\|\mathbf{p}\|_Q = \sqrt{0} = 0$.

□

Observation 5.14: Absolute homogeneity

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$, scalar $\lambda \in \mathbb{R}$, and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_Q = |\lambda| \|\mathbf{p}\|_Q \quad (5.15)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\lambda \mathbf{p}\|_Q &= \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} \lambda p_i \lambda p_j} \\ &= \sqrt{\sum_{i=1}^n \sum_{j=1}^n \lambda^2 q_{ij} p_i p_j} \\ &= |\lambda| \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j} \\ &= |\lambda| \|\mathbf{p}\|_Q\end{aligned}$$

□

Observation 5.15: Triangle inequality

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and pairs of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_Q \leq \|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q \quad (5.16)$$

Proof. For every $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} + \mathbf{w}\|_Q^2 &= (\mathbf{p} + \mathbf{w})' \mathbf{Q} (\mathbf{p} + \mathbf{w}) = \mathbf{p}' \mathbf{Q} \mathbf{p} + 2 \mathbf{p}' \mathbf{Q} \mathbf{w} + \mathbf{w}' \mathbf{Q} \mathbf{w} \\ &= \|\mathbf{p}\|_Q^2 + \|\mathbf{w}\|_Q^2 + 2 \mathbf{p}' \mathbf{Q} \mathbf{w} \\ &\leq \|\mathbf{p}\|_Q^2 + \|\mathbf{w}\|_Q^2 + 2 \|\mathbf{p}\|_Q \|\mathbf{w}\|_Q \quad (\text{by gen. Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q)^2\end{aligned}$$

Accordingly, we obtain:

$$\|\mathbf{p} + \mathbf{w}\|_Q^2 \leq (\|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q)^2 \implies \|\mathbf{p} + \mathbf{w}\|_Q \leq \|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q$$

□

5.5 ℓ_∞ norm

Definition 5.16: ℓ_∞ norm

The ℓ_∞ norm (also called **maximum norm**, **supremum norm**, or **Chebyshev norm**)

of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_\infty = \max \{ |p_j| : j \in \{1, 2, \dots, n\} \} \quad (5.17)$$

- The ℓ_∞ **norm** represents the maximum absolute value among all components.
- The ℓ_∞ norm is half the side length of the smallest hypercube (aligned with the axes and centered at the origin) that contains the vector.

Example 5.10: ℓ_∞ norm

Consider the vectors:

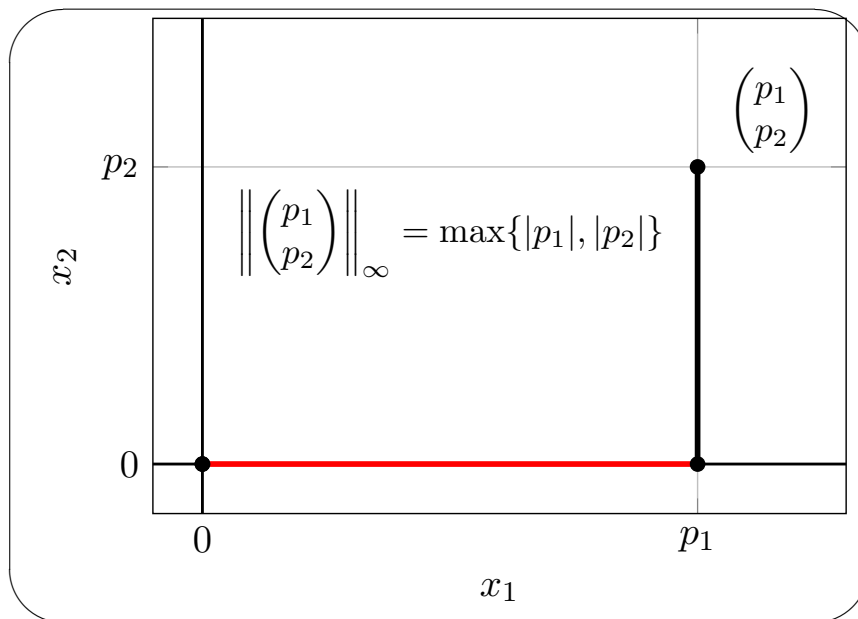
$$\mathbf{p} = \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_∞ norms are:

$$\left\| \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|3|, |-7|, |2|\} = \max\{3, 7, 2\} = 7,$$

$$\left\| \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|-1|, |5|, |-5|, |2|\} = \max\{1, 5, 5, 2\} = 5$$

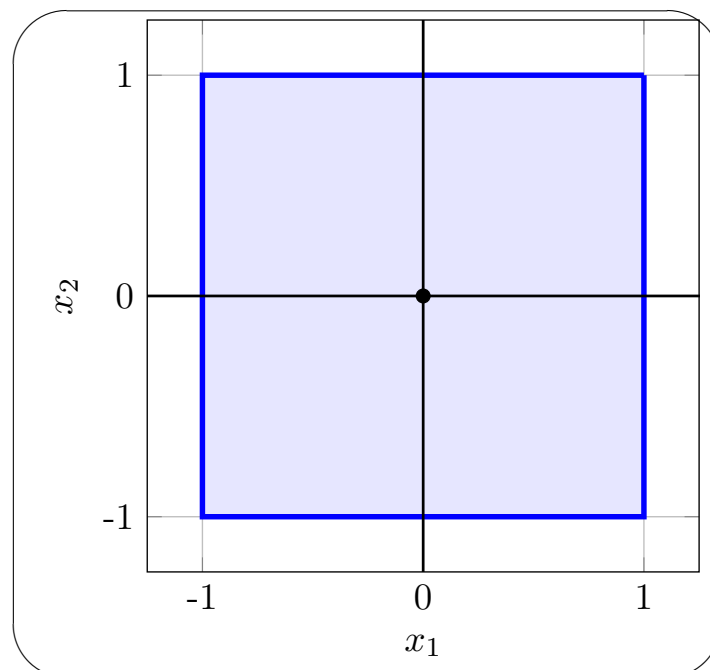
- The ℓ_∞ **norm** of a column vector in $\mathbb{R}^n$ represents the **maximum absolute value of its components**. In $\mathbb{R}^2$, this corresponds to the **Chebyshev distance** (the maximum of the absolute differences in coordinates), as illustrated by the following picture:



- In two dimensions, all the points at ℓ_∞ norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_\infty \leq 1\}$$

are the points that lie on or inside a **square** (axis-aligned) centered at the origin.



5.5.1 Properties

- The ℓ_∞ norm satisfies the three fundamental properties that define a norm.

Observation 5.17: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_\infty \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_\infty = 0 \iff \mathbf{p} = \mathbf{0} \quad (5.18)$$

Proof. From the definition $\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\}$.

1. **Non-negativity:** Since $|p_j| \geq 0$ for all $j \in \{1, 2, \dots, n\}$, and the maximum of non-negative numbers is non-negative, we have

$$\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \geq 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^n.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_\infty = 0$, then

$$\max\{|p_j| : j \in \{1, 2, \dots, n\}\} = 0.$$

Since

$$|p_j| \leq \max\{|p_k| : k \in \{1, 2, \dots, n\}\} \quad \text{for all } j \in \{1, 2, \dots, n\},$$

it follows that

$$|p_j| \leq 0 \quad \text{for all } j \in \{1, 2, \dots, n\}.$$

But $|p_j| \geq 0$ always holds, hence $|p_j| = 0$ for all j , i.e. $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, and therefore $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, so

$$\|\mathbf{p}\|_\infty = \max\{|0| : j \in \{1, 2, \dots, n\}\} = 0.$$

□

Observation 5.18: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_\infty = |\lambda| \|\mathbf{p}\|_\infty \quad (5.19)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\lambda \mathbf{p}\|_\infty &= \max\{|\lambda p_j| : j \in \{1, 2, \dots, n\}\} \\ &= \max\{|\lambda| |p_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{property of absolute value}) \\ &= |\lambda| \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{factor constant out of max}) \\ &= |\lambda| \|\mathbf{p}\|_\infty \end{aligned}$$

□

Observation 5.19: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_\infty \leq \|\mathbf{p}\|_\infty + \|\mathbf{w}\|_\infty \quad (5.20)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_\infty &= \max\{|p_j + w_j| : j \in \{1, 2, \dots, n\}\} \\ &\leq \max\{|p_j| + |w_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{triangle inequality for absolute value}) \\ &\leq \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \\ &\quad + \max\{|w_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{max distribution}) \\ &= \|\mathbf{p}\|_\infty + \|\mathbf{w}\|_\infty \end{aligned}$$

where in the second inequality we used the fact that

$$|p_j| \leq \max\{|p_k| : k \in \{1, 2, \dots, n\}\}$$

and

$$|w_j| \leq \max\{|w_k| : k \in \{1, 2, \dots, n\}\} \quad \text{for all } j \in \{1, 2, \dots, n\}.$$

□

5.6 Comparison of the ℓ_1 , ℓ_2 and ℓ_∞ norms

- The ℓ_1 , ℓ_2 and ℓ_∞ norms of the same vector are, in general, different numbers. However, they are related by simple inequalities.

Observation 5.20: Inequalities between norms

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty \quad (5.21)$$

and

$$\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2 \quad (5.22)$$

Proof. Let $k \in \{1, 2, \dots, n\}$ be an index such that $|p_k| = \|\mathbf{p}\|_\infty$.

1. $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$: since all the terms p_j^2 are non-negative, we have

$$\|\mathbf{p}\|_\infty^2 = p_k^2 \leq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$.

2. $\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$: expanding the square of the sum, we have

$$\|\mathbf{p}\|_1^2 = \left(\sum_{j=1}^n |p_j| \right)^2 = \sum_{j=1}^n |p_j|^2 + \underbrace{\sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n |p_i| |p_j|}_{\geq 0} \geq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$.

3. $\|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty$: since $|p_j| \leq \|\mathbf{p}\|_\infty$ for all $j \in \{1, 2, \dots, n\}$, we have

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \leq \sum_{j=1}^n \|\mathbf{p}\|_\infty = n \|\mathbf{p}\|_\infty$$

4. $\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$: consider the column vectors

$$\mathbf{a} = \begin{pmatrix} |p_1| \\ |p_2| \\ \vdots \\ |p_n| \end{pmatrix} \in \mathbb{R}^n \quad \text{and} \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n$$

for which $\mathbf{1}' \mathbf{a} = \|\mathbf{p}\|_1$, $\|\mathbf{a}\|_2 = \|\mathbf{p}\|_2$ and $\|\mathbf{1}\|_2 = \sqrt{n}$. By the Cauchy-Schwarz inequality (5.6), we get

$$\|\mathbf{p}\|_1 = \mathbf{1}' \mathbf{a} \leq \|\mathbf{1}\|_2 \|\mathbf{a}\|_2 = \sqrt{n} \|\mathbf{p}\|_2$$

□

Example 5.11: inequalities between norms

Consider the vector:

$$\mathbf{p} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

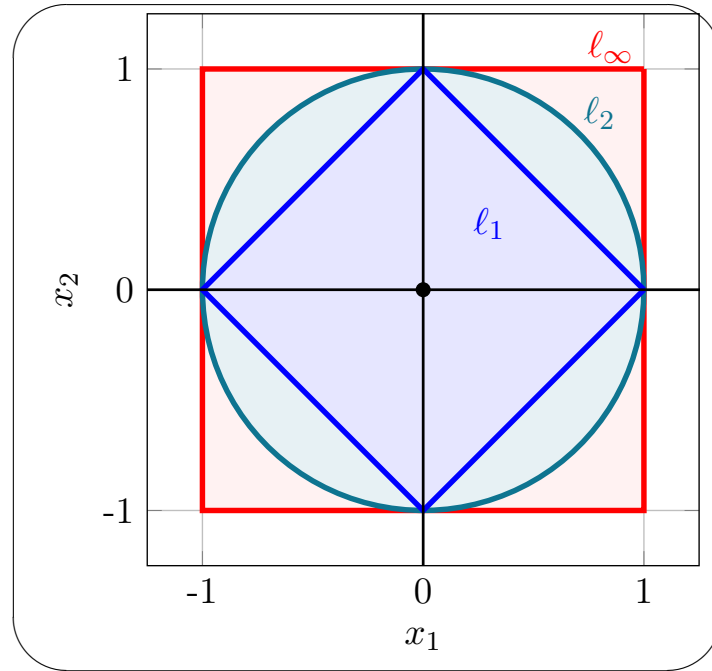
Its norms are:

$$\|\mathbf{p}\|_\infty = \max\{1, 2, 2\} = 2, \quad \|\mathbf{p}\|_2 = \sqrt{1 + 4 + 4} = 3, \quad \|\mathbf{p}\|_1 = 1 + 2 + 2 = 5$$

and, with $n = 3$, we have:

$$\underbrace{2}_{\|\mathbf{p}\|_\infty} \leq \underbrace{3}_{\|\mathbf{p}\|_2} \leq \underbrace{5}_{\|\mathbf{p}\|_1} \leq \underbrace{6}_{3\|\mathbf{p}\|_\infty} \quad \text{and} \quad \underbrace{5}_{\|\mathbf{p}\|_1} \leq \underbrace{3\sqrt{3}}_{\sqrt{3}\|\mathbf{p}\|_2} \approx 5.196$$

- In two dimensions, the inequalities $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ mean that the three unit balls are nested: the ℓ_1 diamond lies inside the ℓ_2 unit circle, which in turn lies inside the ℓ_∞ square.



5.7 Exercises

Exercise 5.1

Compute the ℓ_1 , ℓ_2 and ℓ_∞ norms of the following column vectors:

$$\mathbf{p} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{u} = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^4$$

For each vector, check that $\|\cdot\|_\infty \leq \|\cdot\|_2 \leq \|\cdot\|_1$.

Solution

Using the definitions, we have:

$$\begin{aligned} \|\mathbf{p}\|_1 &= |2| + |-3| + |6| = 11, & \|\mathbf{p}\|_2 &= \sqrt{4 + 9 + 36} = 7, & \|\mathbf{p}\|_\infty &= \max\{2, 3, 6\} = 6 \\ \|\mathbf{w}\|_1 &= |4| + |0| + |-3| = 7, & \|\mathbf{w}\|_2 &= \sqrt{16 + 0 + 9} = 5, & \|\mathbf{w}\|_\infty &= \max\{4, 0, 3\} = 4 \\ \|\mathbf{u}\|_1 &= 1 + 1 + 1 + 1 = 4, & \|\mathbf{u}\|_2 &= \sqrt{1 + 1 + 1 + 1} = 2, & \|\mathbf{u}\|_\infty &= \max\{1, 1, 1, 1\} = 1 \end{aligned}$$

Accordingly:

$$6 \leq 7 \leq 11, \quad 4 \leq 5 \leq 7, \quad 1 \leq 2 \leq 4$$

Exercise 5.2

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \in \mathbb{R}^3 \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix} \in \mathbb{R}^3$$

Compute the distance between $\mathbf{p}$ and $\mathbf{w}$ measured with the ℓ_1 , ℓ_2 and ℓ_∞ norms, i.e., $\|\mathbf{p} - \mathbf{w}\|_1$, $\|\mathbf{p} - \mathbf{w}\|_2$ and $\|\mathbf{p} - \mathbf{w}\|_\infty$. Then explain why $\|\mathbf{w} - \mathbf{p}\| = \|\mathbf{p} - \mathbf{w}\|$ for every norm.

Solution

The difference of the two vectors is

$$\mathbf{p} - \mathbf{w} = \begin{pmatrix} 1 - 3 \\ 2 - (-1) \\ -1 - 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

hence:

$$\|\mathbf{p} - \mathbf{w}\|_1 = 2 + 3 + 6 = 11, \quad \|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{4 + 9 + 36} = 7, \quad \|\mathbf{p} - \mathbf{w}\|_\infty = \max\{2, 3, 6\} = 6$$

Since $\mathbf{w} - \mathbf{p} = (-1)(\mathbf{p} - \mathbf{w})$, by absolute homogeneity (with $\lambda = -1$) we get, for every norm,

$$\|\mathbf{w} - \mathbf{p}\| = |-1| \|\mathbf{p} - \mathbf{w}\| = \|\mathbf{p} - \mathbf{w}\|$$

Exercise 5.3

Consider the column vector

$$\mathbf{p} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

1. Find the vector $\mathbf{u}$ with the same direction and orientation as $\mathbf{p}$ such that $\|\mathbf{u}\|_2 = 1$ (i.e., normalize $\mathbf{p}$ with respect to the ℓ_2 norm), and check the result.
2. Normalize $\mathbf{p}$ with respect to the ℓ_1 norm and with respect to the ℓ_∞ norm.
3. Explain why the zero vector cannot be normalized.

Solution

1. We have $\|\mathbf{p}\|_2 = \sqrt{4 + 1 + 4} = 3$. For any norm, the vector $\mathbf{u} = \frac{1}{\|\mathbf{p}\|} \mathbf{p}$ has, by absolute homogeneity, $\|\mathbf{u}\| = \frac{1}{\|\mathbf{p}\|} \|\mathbf{p}\| = 1$. Hence

$$\mathbf{u} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix}, \quad \|\mathbf{u}\|_2 = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{1} = 1$$

2. We have $\|\mathbf{p}\|_1 = 2 + 1 + 2 = 5$ and $\|\mathbf{p}\|_\infty = 2$, hence

$$\frac{1}{5}\mathbf{p} = \begin{pmatrix} \frac{2}{5} \\ -\frac{1}{5} \\ \frac{2}{5} \end{pmatrix}, \quad \left\| \frac{1}{5}\mathbf{p} \right\|_1 = \frac{2}{5} + \frac{1}{5} + \frac{2}{5} = 1, \quad \frac{1}{2}\mathbf{p} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 1 \end{pmatrix}, \quad \left\| \frac{1}{2}\mathbf{p} \right\|_\infty = 1$$

3. By definiteness, $\|\mathbf{0}\| = 0$, so we cannot divide by it. Moreover, $\|\lambda\mathbf{0}\| = \|\mathbf{0}\| = 0 \neq 1$ for every $\lambda \in \mathbb{R}$.

Exercise 5.4

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \in \mathbb{R}^3 \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \in \mathbb{R}^3$$

1. Verify the triangle inequality $\|\mathbf{p} + \mathbf{w}\| \leq \|\mathbf{p}\| + \|\mathbf{w}\|$ for the ℓ_1 , ℓ_2 and ℓ_∞ norms.
2. Verify the reverse triangle inequality $|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$.

Solution

We have

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{p} - \mathbf{w} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$$

1.
 - ℓ_1 : $\|\mathbf{p}\|_1 = 5$, $\|\mathbf{w}\|_1 = 5$, $\|\mathbf{p} + \mathbf{w}\|_1 = 6$, and $6 \leq 5 + 5 = 10$.
 - ℓ_2 : $\|\mathbf{p}\|_2 = \sqrt{1+4+4} = 3$, $\|\mathbf{w}\|_2 = \sqrt{4+4+1} = 3$, $\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{9+0+9} = 3\sqrt{2} \approx 4.243$, and $3\sqrt{2} \leq 3 + 3 = 6$.
 - ℓ_∞ : $\|\mathbf{p}\|_\infty = 2$, $\|\mathbf{w}\|_\infty = 2$, $\|\mathbf{p} + \mathbf{w}\|_\infty = 3$, and $3 \leq 2 + 2 = 4$.
2. $\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{1+16+1} = \sqrt{18} = 3\sqrt{2}$, and

$$|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| = |3 - 3| = 0 \leq 3\sqrt{2}$$

Exercise 5.5

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -2 \\ -4 \\ -6 \end{pmatrix}$$

Verify the Cauchy–Schwarz inequality $|\mathbf{p}'\mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$ for the pair $\mathbf{p}, \mathbf{w}$ and for the pair $\mathbf{p}, \mathbf{u}$. In which case does equality hold?

Solution

We have $\|\mathbf{p}\|_2 = \sqrt{1+4+9} = \sqrt{14}$.

- Pair $\mathbf{p}, \mathbf{w}$: $\mathbf{p}'\mathbf{w} = 4 - 10 + 18 = 12$ and $\|\mathbf{w}\|_2 = \sqrt{16+25+36} = \sqrt{77}$, hence

$$|\mathbf{p}'\mathbf{w}| = 12 \leq \sqrt{14}\sqrt{77} = \sqrt{1078} \approx 32.83$$

The inequality is strict.

- Pair $\mathbf{p}, \mathbf{u}$: $\mathbf{p}'\mathbf{u} = -2 - 8 - 18 = -28$ and $\|\mathbf{u}\|_2 = \sqrt{4+16+36} = \sqrt{56} = 2\sqrt{14}$, hence

$$|\mathbf{p}'\mathbf{u}| = 28 = \sqrt{14} \cdot 2\sqrt{14} = \|\mathbf{p}\|_2 \|\mathbf{u}\|_2$$

Equality holds: indeed $\mathbf{u} = -2\mathbf{p}$, i.e., $\mathbf{u}$ is a scalar multiple of $\mathbf{p}$.

Exercise 5.6

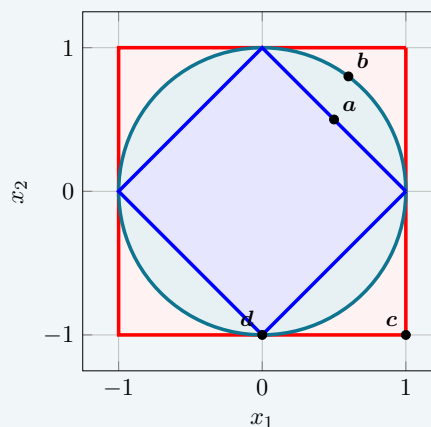
Consider the points of $\mathbb{R}^2$

$$\mathbf{a} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

1. Describe and sketch the unit balls $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$ of the ℓ_1 , ℓ_2 and ℓ_∞ norms.
2. For each point, establish whether it lies inside, on the boundary, or outside each of the three unit balls.

Solution

1. The ℓ_1 unit ball $|x_1| + |x_2| \leq 1$ is a diamond (a square rotated by 45°) with vertices $(\pm 1, 0)$ and $(0, \pm 1)$; the ℓ_2 unit ball $x_1^2 + x_2^2 \leq 1$ is the disk bounded by the unit circle; the ℓ_∞ unit ball $\max\{|x_1|, |x_2|\} \leq 1$ is the axis-aligned square with vertices $(\pm 1, \pm 1)$.



2. We compute the three norms of each point (value < 1 : inside; $= 1$: on the boundary; > 1 : outside):

	ℓ_1	ℓ_2	ℓ_∞
a	1 (boundary)	$\frac{\sqrt{2}}{2} \approx 0.707$ (inside)	$\frac{1}{2}$ (inside)
b	$\frac{7}{5}$ (outside)	$\sqrt{\frac{9}{25} + \frac{16}{25}} = 1$ (boundary)	$\frac{4}{5}$ (inside)
c	2 (outside)	$\sqrt{2}$ (outside)	1 (boundary)
d	1 (boundary)	1 (boundary)	1 (boundary)

Exercise 5.7

Consider the matrix

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

1. Using the definition, prove that $\mathbf{Q}$ is positive definite, so that $\|\mathbf{x}\|_{\mathbf{Q}} = \sqrt{\mathbf{x}' \mathbf{Q} \mathbf{x}}$ is a norm on $\mathbb{R}^2$.
2. Compute $\|\mathbf{p}\|_{\mathbf{Q}}$, $\|\mathbf{w}\|_{\mathbf{Q}}$ and $\|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}}$ for $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
3. Verify the triangle inequality and the generalized Cauchy–Schwarz inequality for $\mathbf{p}$ and $\mathbf{w}$.

Solution

1. $\mathbf{Q}$ is symmetric and, for every $\mathbf{x} \in \mathbb{R}^2$,

$$\mathbf{x}' \mathbf{Q} \mathbf{x} = 5x_1^2 + 4x_1x_2 + 2x_2^2 = x_1^2 + (2x_1 + x_2)^2 + x_2^2 \geq 0$$

The sum of squares is zero only if $x_1 = 0$ and $x_2 = 0$; hence $\mathbf{x}' \mathbf{Q} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, i.e., $\mathbf{Q}$ is positive definite, and $\|\cdot\|_{\mathbf{Q}}$ is a generalized ℓ_2 norm.

2. Using $\|\mathbf{x}\|_{\mathbf{Q}}^2 = 5x_1^2 + 4x_1x_2 + 2x_2^2$:

$$\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{5}, \quad \|\mathbf{w}\|_{\mathbf{Q}} = \sqrt{2}, \quad \|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}} = \left\| \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\|_{\mathbf{Q}} = \sqrt{5 + 4 + 2} = \sqrt{11}$$

3. Triangle inequality: $\sqrt{11} \approx 3.317 \leq \sqrt{5} + \sqrt{2} \approx 2.236 + 1.414 = 3.650$.

Generalized Cauchy–Schwarz inequality: $\mathbf{p}' \mathbf{Q} \mathbf{w} = q_{12} = 2$, and

$$|\mathbf{p}' \mathbf{Q} \mathbf{w}| = 2 \leq \sqrt{5} \sqrt{2} = \sqrt{10} \approx 3.162$$

Note that in the standard ℓ_2 norm the same vectors are orthogonal ($\mathbf{p}' \mathbf{w} = 0$), while $\mathbf{p}' \mathbf{Q} \mathbf{w} \neq 0$.

Exercise 5.8

Consider again the symmetric positive definite matrix $\mathbf{Q} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$.

1. Compute the eigenvalues and the eigenvectors of $\mathbf{Q}$.
2. Describe the ellipse $\{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}' \mathbf{Q} \mathbf{x} = 1\}$: its semi-axes (lengths and directions), the endpoints of the semi-axes, and its intersections with the coordinate axes. Sketch the set $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\}$.

Solution

1. The characteristic equation is

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6) = 0$$

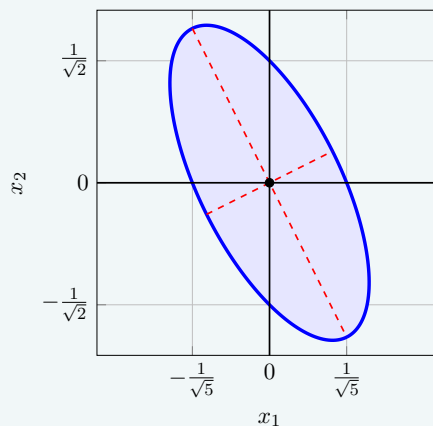
hence $\lambda_1 = 1$ and $\lambda_2 = 6$ (both positive, as expected). The corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \left(\mathbf{Q} \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} = 1 \cdot \mathbf{v}_1 \right), \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \left(\mathbf{Q} \mathbf{v}_2 = \begin{pmatrix} 12 \\ 6 \end{pmatrix} = 6 \mathbf{v}_2 \right)$$

2. The ellipse $5x_1^2 + 4x_1x_2 + 2x_2^2 = 1$ is centered at the origin and rotated (its principal axes are the directions of $\mathbf{v}_1$ and $\mathbf{v}_2$, not the coordinate axes).

- Semi-axis along $\mathbf{v}_1$: length $\frac{1}{\sqrt{\lambda_1}} = 1$, endpoints $\pm \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|_2} = \pm \left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right)$.
- Semi-axis along $\mathbf{v}_2$: length $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{6}}$, endpoints $\pm \frac{1}{\sqrt{6}} \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|_2} = \pm \left(\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}} \right)$.
- Intersections with the axes: for $x_2 = 0$, $5x_1^2 = 1$, i.e., $\left(\pm \frac{1}{\sqrt{5}}, 0 \right)$; for $x_1 = 0$, $2x_2^2 = 1$, i.e., $\left(0, \pm \frac{1}{\sqrt{2}} \right)$.

For instance, $\left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right)$ lies on the ellipse: $5 \cdot \frac{1}{5} + 4 \cdot \left(-\frac{2}{5} \right) + 2 \cdot \frac{4}{5} = 1$.



Exercise 5.9

Prove that, for every column vector $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty \quad \text{and} \quad \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1$$

For which vectors does equality hold in each inequality?

Solution

- Since $x_j^2 = |x_j|^2 \leq \|\mathbf{x}\|_\infty^2$ for all $j \in \{1, 2, \dots, n\}$, we have

$$\|\mathbf{x}\|_2^2 = \sum_{j=1}^n x_j^2 \leq \sum_{j=1}^n \|\mathbf{x}\|_\infty^2 = n \|\mathbf{x}\|_\infty^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty$. Equality holds if and only if $x_j^2 = \|\mathbf{x}\|_\infty^2$ for all j , i.e., all components have the same absolute value (e.g., $\mathbf{x} = (1, -1, 1, -1)'$ of Exercise 5.7: $2 = \sqrt{4} \cdot 1$).

- Let k be an index such that $|x_k| = \|\mathbf{x}\|_\infty$. Then

$$\|\mathbf{x}\|_\infty = |x_k| \leq |x_k| + \sum_{j \neq k} |x_j| = \|\mathbf{x}\|_1$$

Equality holds if and only if $\sum_{j \neq k} |x_j| = 0$, i.e., $\mathbf{x}$ has at most one non-zero component.

Exercise 5.10

Let $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ with $\mathbf{w} \neq \mathbf{0}$. Prove that equality holds in the Cauchy–Schwarz inequality, i.e.,

$$|\mathbf{p}' \mathbf{w}| = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$$

if and only if $\mathbf{p} = \lambda \mathbf{w}$ for some scalar $\lambda \in \mathbb{R}$.

Hint: use the vector $\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}$, with $\alpha = \mathbf{w}' \mathbf{w}$ and $\beta = -\mathbf{p}' \mathbf{w}$, introduced in the proof of the Cauchy–Schwarz inequality.

Solution

($\Leftarrow$) If $\mathbf{p} = \lambda \mathbf{w}$, then

$$|\mathbf{p}' \mathbf{w}| = |\lambda \mathbf{w}' \mathbf{w}| = |\lambda| \|\mathbf{w}\|_2^2 \quad \text{and} \quad \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 = |\lambda| \|\mathbf{w}\|_2 \|\mathbf{w}\|_2 = |\lambda| \|\mathbf{w}\|_2^2$$

($\Rightarrow$) As shown in the proof of the Cauchy–Schwarz inequality, with $\alpha = \mathbf{w}' \mathbf{w}$ and $\beta = -\mathbf{p}' \mathbf{w}$ we have

$$\|\mathbf{u}\|_2^2 = \mathbf{u}' \mathbf{u} = \mathbf{w}' \mathbf{w} ((\mathbf{w}' \mathbf{w}) (\mathbf{p}' \mathbf{p}) - (\mathbf{p}' \mathbf{w})^2)$$

If $|\mathbf{p}' \mathbf{w}| = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$, then $(\mathbf{p}' \mathbf{w})^2 = (\mathbf{p}' \mathbf{p}) (\mathbf{w}' \mathbf{w})$, so the term in parentheses is zero and $\|\mathbf{u}\|_2 = 0$. By definiteness, $\mathbf{u} = \mathbf{0}$, i.e.,

$$(\mathbf{w}' \mathbf{w}) \mathbf{p} - (\mathbf{p}' \mathbf{w}) \mathbf{w} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{p} = \lambda \mathbf{w} \quad \text{with} \quad \lambda = \frac{\mathbf{p}' \mathbf{w}}{\mathbf{w}' \mathbf{w}}$$

where we divided by $\mathbf{w}'\mathbf{w} = \|\mathbf{w}\|_2^2 > 0$.

For instance, in Exercise 5.7, $\mathbf{u} = -2\mathbf{p}$ and equality holds, while $\mathbf{w}$ is not a multiple of $\mathbf{p}$ and the inequality is strict.

Exercise 5.11

For $\mathbf{x} \in \mathbb{R}^2$, consider the functions

$$f(\mathbf{x}) = |x_1| + 2|x_2|, \quad g(\mathbf{x}) = |x_1|, \quad h(\mathbf{x}) = x_1^2 + x_2^2$$

1. Prove that f is a norm and sketch the set $\{\mathbf{x} \in \mathbb{R}^2 : f(\mathbf{x}) \leq 1\}$.
2. Show, with a counterexample, that g and h are not norms.

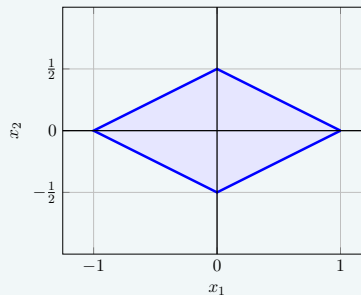
Solution

1. We check the three properties.

- Non-negativity and definiteness: $f(\mathbf{x}) \geq 0$ as a sum of non-negative terms; $f(\mathbf{x}) = 0$ if and only if $|x_1| = 0$ and $|x_2| = 0$, i.e., $\mathbf{x} = \mathbf{0}$.
- Absolute homogeneity: $f(\lambda\mathbf{x}) = |\lambda x_1| + 2|\lambda x_2| = |\lambda|(|x_1| + 2|x_2|) = |\lambda|f(\mathbf{x})$.
- Triangle inequality: by the triangle inequality for the absolute value,

$$f(\mathbf{x} + \mathbf{y}) = |x_1 + y_1| + 2|x_2 + y_2| \leq |x_1| + |y_1| + 2|x_2| + 2|y_2| = f(\mathbf{x}) + f(\mathbf{y})$$

The set $|x_1| + 2|x_2| \leq 1$ is a diamond with vertices $(\pm 1, 0)$ and $(0, \pm \frac{1}{2})$:



2. g violates definiteness: $g((0, 1)') = 0$ but $(0, 1)' \neq \mathbf{0}$.

h violates absolute homogeneity: for $\mathbf{x} = (1, 0)'$ and $\lambda = 2$, $h(2\mathbf{x}) = 4 \neq 2 = |\lambda| h(\mathbf{x})$. It also violates the triangle inequality: $h((1, 0)' + (1, 0)') = 4 > 1 + 1$. (Note that $h(\mathbf{x}) = \|\mathbf{x}\|_2^2$: the square of a norm is not a norm.)

Exercise 5.12

Let $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ be such that $\|\mathbf{p}\|_2 = 3$ and $\|\mathbf{w}\|_2 = 4$.

1. Find the smallest and the largest possible values of $\|\mathbf{p} + \mathbf{w}\|_2$, and give vectors attaining them.
2. Compute $\|\mathbf{p} + \mathbf{w}\|_2$ if $\mathbf{p}'\mathbf{w} = 0$.
3. Compute $\|\mathbf{p} + \mathbf{w}\|_2$ and $\|\mathbf{p} - \mathbf{w}\|_2$ if $\mathbf{p}'\mathbf{w} = 6$.
4. Is it possible that $\mathbf{p}'\mathbf{w} = 13$?

Solution

1. By the triangle inequality, $\|\mathbf{p} + \mathbf{w}\|_2 \leq 3 + 4 = 7$. By the reverse triangle inequality applied to $\mathbf{p}$ and $-\mathbf{w}$ (note that $\|-\mathbf{w}\|_2 = \|\mathbf{w}\|_2$),

$$\|\mathbf{p} + \mathbf{w}\|_2 = \|\mathbf{p} - (-\mathbf{w})\|_2 \geq |\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| = |3 - 4| = 1$$

Both bounds are attained: if $\mathbf{w} = \frac{4}{3}\mathbf{p}$, then $\|\mathbf{p} + \mathbf{w}\|_2 = \frac{7}{3} \cdot 3 = 7$; if $\mathbf{w} = -\frac{4}{3}\mathbf{p}$, then $\|\mathbf{p} + \mathbf{w}\|_2 = \frac{1}{3} \cdot 3 = 1$. For instance, $\mathbf{p} = (3, 0)'$ and $\mathbf{w} = (\pm 4, 0)'$.

2. Using $\|\mathbf{p} + \mathbf{w}\|_2^2 = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2\mathbf{p}'\mathbf{w}$, we get $\|\mathbf{p} + \mathbf{w}\|_2^2 = 9 + 16 + 0 = 25$, hence $\|\mathbf{p} + \mathbf{w}\|_2 = 5$ (Pythagorean theorem).
3. $\|\mathbf{p} + \mathbf{w}\|_2^2 = 9 + 16 + 12 = 37$ and $\|\mathbf{p} - \mathbf{w}\|_2^2 = 9 + 16 - 12 = 13$, hence $\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{37}$ and $\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{13}$.
4. No: by the Cauchy–Schwarz inequality, $|\mathbf{p}'\mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 = 12 < 13$.

Part IV

Linear systems



System of linear equations

6.1 Existence and uniqueness of solutions

Consider a system of m linear equations with n variables:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

where:

- $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix (m equations, n variables)
- $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of variables
- $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the right-hand side vector
- $(\mathbf{A}|\mathbf{b}) \in \mathbb{R}^{m \times (n+1)}$ is the augmented matrix

Rouché–Capelli theorem. The existence and uniqueness of solutions to the system $\mathbf{A} \mathbf{x} = \mathbf{b}$ depend on the rank of the coefficient matrix $\mathbf{A}$ and the rank of the augmented matrix $(\mathbf{A}|\mathbf{b})$.

Case 1 - No solution (inconsistent system):

$$\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}|\mathbf{b})$$

The system is inconsistent and has no solution.

Case 2 - Unique solution:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = n$$

The system has a unique solution.

Case 3 - Infinite solutions:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) < n$$

The system has infinitely many solutions with $n - \text{rank}(\mathbf{A})$ degrees of freedom (free variables). If $r = \text{rank}(\mathbf{A})$, the solutions can be described by means of $n - r$ **free parameters**.

- Since $(\mathbf{A}|\mathbf{b})$ is obtained from $\mathbf{A}$ by adding one column, we always have

$$\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A}|\mathbf{b}) \leq \text{rank}(\mathbf{A}) + 1,$$

so the three cases above cover all the possibilities. In particular, the system has at least one solution (it is **consistent**) if and only if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b})$.

- Since $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$, a unique solution is possible only if $m \geq n$.
- If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, recall that $\text{rank}(\mathbf{A}) = n$ if and only if $\det(\mathbf{A}) \neq 0$. Hence, if $\det(\mathbf{A}) \neq 0$, then also $\text{rank}(\mathbf{A}|\mathbf{b}) = n$ and the system has a unique solution, for every right-hand side $\mathbf{b}$.

6.2 Gaussian Elimination Method

The **Gaussian elimination method** is an algorithm for solving systems of linear equations by transforming the augmented matrix $(A|b)$ into **row echelon form** (upper triangular form) through elementary row operations.

- Given the system of linear equations:

$$A x = b$$

we form the **augmented matrix**:

$$(A|b)$$

- The method consists of two phases:

Phase 1 - Forward elimination: Transform the augmented matrix into upper triangular form using elementary row operations:

- Swap two rows
- Multiply a row by a non-zero scalar
- Add a multiple of one row to another row

The goal is to create zeros below the diagonal, obtaining:

$$\left(\begin{array}{cccc|c} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} & \tilde{b}_1 \\ 0 & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} & \tilde{b}_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \tilde{a}_{nn} & \tilde{b}_n \end{array} \right)$$

Phase 2 - Backward substitution: Solve the upper triangular system from bottom to top (assuming $\tilde{a}_{ii} \neq 0$ for all i):

$$\begin{aligned} x_n &= \frac{\tilde{b}_n}{\tilde{a}_{nn}} \\ x_{n-1} &= \frac{\tilde{b}_{n-1} - \tilde{a}_{n-1,n} x_n}{\tilde{a}_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{\tilde{b}_i - \sum_{j=i+1}^n \tilde{a}_{ij} x_j}{\tilde{a}_{ii}} \end{aligned}$$

Elementary row operations:

- $R_i \leftrightarrow R_j$: swap row i with row j
- $R_i \leftarrow k R_i$: multiply row i by scalar $k \neq 0$
- $R_i \leftarrow R_i + k R_j$: add k times row j to row i

These operations do not change the solution set of the system.

Example 6.1: Solving a system using Gaussian elimination - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

Augmented matrix:

$$(A|b) = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 4 & 3 & 3 & 10 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

Step 1: Eliminate x_1 from rows 2 and 3

$R_2 \leftarrow R_2 - 2R_1$ (subtract 2 times row 1 from row 2):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

$R_3 \leftarrow R_3 - 4R_1$ (subtract 4 times row 1 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & 5 & 8 \end{array} \right)$$

Step 2: Eliminate x_2 from row 3

$R_3 \leftarrow R_3 - 3R_2$ (subtract 3 times row 2 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right)$$

Upper triangular form achieved! The corresponding system is:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 1x_2 + 1x_3 = 2 \\ 2x_3 = 2 \end{cases}$$

Example 6.2: Solving a system using Gaussian elimination - Part 2**Backward substitution:**

From the third equation:

$$2x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - x_3 = 2 - 1 = 1$$

From the first equation:

$$2x_1 + x_2 + x_3 = 4 \implies 2x_1 = 4 - x_2 - x_3 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A} \mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

6.3 Gaussian elimination and the Rouché–Capelli theorem

- Gaussian elimination can be applied to any system $\mathbf{A} \mathbf{x} = \mathbf{b}$, with $\mathbf{A} \in \mathbb{R}^{m \times n}$. In general, forward elimination transforms $(\mathbf{A}|\mathbf{b})$ into **row echelon form**: in each nonzero row, the first nonzero entry (called **pivot**) is strictly to the right of the pivot of the row above, and the rows of zeros (if any) are at the bottom.
- Elementary row operations do not change the solution set of the system, and they do not change the rank of $\mathbf{A}$ nor the rank of $(\mathbf{A}|\mathbf{b})$.
- The rank of a matrix in row echelon form is equal to the number of its nonzero rows. Therefore, at the end of forward elimination we can read $\text{rank}(\mathbf{A})$ and $\text{rank}(\mathbf{A}|\mathbf{b})$ and apply the Rouché–Capelli theorem:
 - a row of the form $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ corresponds to the impossible equation $0 = c$: then $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}|\mathbf{b})$ and the system has no solution;
 - otherwise, the variables whose columns contain a pivot are computed by backward substitution, and the remaining $n - r$ variables are the **free parameters**.

When the system has infinitely many solutions with one free parameter $t \in \mathbb{R}$, the solutions can be written in **parametric form**

$$\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}, \quad t \in \mathbb{R},$$

where $\mathbf{x}_0$ is a particular solution (obtained for $t = 0$) and $\mathbf{v}$ is a nonzero vector such that $\mathbf{A} \mathbf{v} = \mathbf{0}$.

Example 6.3: Rouché–Capelli: unique solution

Consider the system of linear equations:

$$\begin{cases} 1 x_2 + 1 x_3 = 1 \\ 1 x_1 + 1 x_2 + 1 x_3 = 2 \\ 2 x_1 + 1 x_2 + 3 x_3 = 1 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

The entry in position $(1, 1)$ is zero, so we first swap rows 1 and 2.

$R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

$R_3 \leftarrow R_3 - 2 R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 1 & -3 \end{array} \right)$$

$R_3 \leftarrow R_3 + R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & -2 \end{array} \right)$$

There are 3 nonzero rows both in $\mathbf{A}$ and in $(\mathbf{A}|\mathbf{b})$: $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 3 = n$, hence the solution is unique. By backward substitution:

$$2 x_3 = -2 \implies x_3 = -1, \quad x_2 = 1 - x_3 = 2, \quad x_1 = 2 - x_2 - x_3 = 2 - 2 + 1 = 1,$$

that is,

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

Example 6.4: Rouché–Capelli: infinitely many solutions

Consider the system of linear equations:

$$\begin{cases} 1 x_1 + 2 x_2 - 1 x_3 = 1 \\ 2 x_1 + 5 x_2 + 1 x_3 = 4 \\ 3 x_1 + 7 x_2 = 5 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

$$R_2 \leftarrow R_2 - 2 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

$$R_3 \leftarrow R_3 - 3 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{array} \right)$$

$$R_3 \leftarrow R_3 - R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

There are 2 nonzero rows both in $\mathbf{A}$ and in $(\mathbf{A}|\mathbf{b})$: $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 2 < 3 = n$, hence there are infinitely many solutions with $n - r = 3 - 2 = 1$ free parameter. The pivots are in the columns of x_1 and x_2 , so x_3 is free: we set $x_3 = t$, $t \in \mathbb{R}$. By backward substitution:

$$x_2 = 2 - 3x_3 = 2 - 3t, \quad x_1 = 1 - 2x_2 + x_3 = 1 - 4 + 6t + t = -3 + 7t.$$

In parametric form:

$$\mathbf{x} = \begin{pmatrix} -3 + 7t \\ 2 - 3t \\ t \end{pmatrix} = \underbrace{\begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}}_{\mathbf{x}_0} + t \underbrace{\begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}}_{\mathbf{v}}, \quad t \in \mathbb{R}.$$

Verification: $\mathbf{A} \mathbf{x}_0 = (-3 + 4, -6 + 10, -9 + 14)' = (1, 4, 5)' = \mathbf{b}$ and $\mathbf{A} \mathbf{v} = (7 - 6 - 1, 14 - 15 + 1, 21 - 21)' = \mathbf{0}$, hence $\mathbf{A}(\mathbf{x}_0 + t \mathbf{v}) = \mathbf{b}$ for every $t \in \mathbb{R}$ ✓

Example 6.5: Rouché–Capelli: no solution

We change only the last right-hand side of the previous system:

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 = 1 \\ 2x_1 + 5x_2 + 1x_3 = 4 \\ 3x_1 + 7x_2 = 6 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 6 \end{array} \right)$$

With the same row operations $R_2 \leftarrow R_2 - 2 R_1$, $R_3 \leftarrow R_3 - 3 R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right)$$

$$R_3 \leftarrow R_3 - R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row corresponds to the equation $0x_1 + 0x_2 + 0x_3 = 1$, which is impossible. Indeed $\mathbf{A}$ has 2 nonzero rows while $(\mathbf{A}|\mathbf{b})$ has 3:

$$\text{rank}(\mathbf{A}) = 2 < 3 = \text{rank}(\mathbf{A}|\mathbf{b}),$$

hence the system is inconsistent and has **no solution**.

6.4 Homogeneous systems

Definition 6.1: homogeneous system

A system of linear equations $\mathbf{A} \mathbf{x} = \mathbf{b}$ with $\mathbf{A} \in \mathbb{R}^{m \times n}$ is called **homogeneous** if $\mathbf{b} = \mathbf{0}$, i.e., if it has the form

$$\mathbf{A} \mathbf{x} = \mathbf{0}.$$

Observation 6.2: solutions of a homogeneous system

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then:

- the homogeneous system $\mathbf{A} \mathbf{x} = \mathbf{0}$ always has the solution $\mathbf{x} = \mathbf{0}$, called the **trivial solution**;
- it has nontrivial solutions ($\mathbf{x} \neq \mathbf{0}$) if and only if $\text{rank}(\mathbf{A}) < n$; in this case it has infinitely many solutions, with $n - \text{rank}(\mathbf{A})$ free parameters;
- if $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, it has nontrivial solutions if and only if $\det(\mathbf{A}) = 0$.

Proof. Adding a column of zeros does not change the rank, hence $\text{rank}(\mathbf{A}|\mathbf{0}) = \text{rank}(\mathbf{A})$ and the system is always consistent (as shown by $\mathbf{x} = \mathbf{0}$). By the Rouché–Capelli theorem, the solution $\mathbf{x} = \mathbf{0}$ is the unique one if and only if $\text{rank}(\mathbf{A}) = n$; otherwise there are infinitely many solutions with $n - \text{rank}(\mathbf{A})$ free parameters. For a square matrix, $\text{rank}(\mathbf{A}) < n$ if and only if $\det(\mathbf{A}) = 0$. $\square$

- In particular, if $m < n$ (fewer equations than variables), then $\text{rank}(\mathbf{A}) \leq m < n$ and the homogeneous system always has nontrivial solutions.
- If $\mathbf{x}_0$ is a solution of $\mathbf{A} \mathbf{x} = \mathbf{b}$ and $\mathbf{v}$ is a solution of $\mathbf{A} \mathbf{x} = \mathbf{0}$, then $\mathbf{A}(\mathbf{x}_0 + \mathbf{v}) = \mathbf{b} + \mathbf{0} = \mathbf{b}$. This explains the parametric form $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$ of Example 6.4.

Example 6.6: a homogeneous system

Consider the homogeneous system with the coefficient matrix of Example 6.4:

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 = 0 \\ 2x_1 + 5x_2 + 1x_3 = 0 \\ 3x_1 + 7x_2 = 0 \end{cases}$$

By the Sarrus rule,

$$\det(\mathbf{A}) = 1 \cdot 5 \cdot 0 + 2 \cdot 1 \cdot 3 + (-1) \cdot 2 \cdot 7 - (-1) \cdot 5 \cdot 3 - 2 \cdot 2 \cdot 0 - 1 \cdot 1 \cdot 7 = 0 + 6 - 14 + 15 - 0 - 7 = 0,$$

so there are nontrivial solutions. With the same row operations as in Example 6.4 (the last column stays equal to zero):

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Setting $x_3 = t$: $x_2 = -3t$ and $x_1 = -2x_2 + x_3 = 6t + t = 7t$. Hence the solutions are

$$\mathbf{x} = t \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R},$$

and the trivial solution is obtained for $t = 0$.

6.5 LU Factorization Method

The **LU factorization** (or **LU decomposition**) is a method for solving systems of linear equations by decomposing the coefficient matrix $\mathbf{A}$ into the product of two triangular matrices:

$$\mathbf{A} = \mathbf{L} \mathbf{U}$$

where $\mathbf{L}$ is a **lower triangular matrix** (with 1's on the diagonal) and $\mathbf{U}$ is an **upper triangular matrix**.

- Given the system of linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

if we have the LU factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$, we can substitute:

$$\mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$$

- We introduce the **auxiliary variable** $\mathbf{y}$ defined as:

$$\mathbf{y} := \mathbf{U} \mathbf{x}$$

so that $\mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$ becomes $\mathbf{L} \mathbf{y} = \mathbf{b}$.

Why does this work? We are splitting the original system $\mathbf{A} \mathbf{x} = \mathbf{b}$ into two simpler triangular systems:

1. Find $\mathbf{y}$ such that $\mathbf{L} \mathbf{y} = \mathbf{b}$.
2. Find $\mathbf{x}$ such that $\mathbf{U} \mathbf{x} = \mathbf{y}$.

If both steps succeed, then $\mathbf{A} \mathbf{x} = \mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{L} \mathbf{y} = \mathbf{b}$, so $\mathbf{x}$ is indeed a solution of the original system. Since both $\mathbf{L}$ and $\mathbf{U}$ are triangular, each of the two systems can be solved in $O(n^2)$ operations by simple substitution.

- We then solve the system in two steps:

Step 1 - Forward substitution: Solve $\mathbf{L} \mathbf{y} = \mathbf{b}$ for $\mathbf{y}$

Since $\mathbf{L}$ is lower triangular, we can solve this system easily from top to bottom:

$$\begin{aligned} y_1 &= b_1 \\ y_2 &= b_2 - \ell_{21} y_1 \\ y_3 &= b_3 - \ell_{31} y_1 - \ell_{32} y_2 \\ &\vdots \\ y_i &= b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \end{aligned}$$

Step 2 - Backward substitution: Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$ for $\mathbf{x}$

Since $\mathbf{U}$ is upper triangular, we can solve this system easily from bottom to top:

$$\begin{aligned} x_n &= \frac{y_n}{u_{nn}} \\ x_{n-1} &= \frac{y_{n-1} - u_{n-1,n} x_n}{u_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{y_i - \sum_{j=i+1}^n u_{ij} x_j}{u_{ii}} \end{aligned}$$

Advantages of LU factorization:

- Once the factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$ is computed, we can solve the system for different right-hand sides $\mathbf{b}$ efficiently
- Both forward and backward substitution require only $O(n^2)$ operations
- The factorization itself requires $O(n^3)$ operations, but it needs to be done only once

Example 6.7: Solving a system using LU factorization - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

In matrix form: $\mathbf{A} \mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

Given the LU factorization of $\mathbf{A}$:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Verification:

$$\mathbf{L} \mathbf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} = \mathbf{A} \quad \checkmark$$

Example 6.8: Solving a system using LU factorization - Part 2

Step 1 - Forward substitution: Solve $\mathbf{L} \mathbf{y} = \mathbf{b}$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

From the first equation:

$$y_1 = 4$$

From the second equation:

$$2y_1 + y_2 = 10 \implies y_2 = 10 - 2 \cdot 4 = 10 - 8 = 2$$

From the third equation:

$$4y_1 + 3y_2 + y_3 = 24 \implies y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 24 - 16 - 6 = 2$$

Thus:

$$\mathbf{y} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

Step 2 - Backward substitution: Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

From the third equation:

$$2x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - 1 = 1$$

From the first equation:

$$2x_1 + x_2 + x_3 = 4 \implies 2x_1 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A} \mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

6.6 Cramer's rule

Cramer's rule is an explicit formula for the solution of a system of m linear equations with m variables (valid whenever the system has a unique solution).

- Given a column vector $\mathbf{b} \in \mathbb{R}^{m \times 1}$ of m rows, a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$ of m rows and m columns and a column vector $\mathbf{x} \in \mathbb{R}^{m \times 1}$ of m rows containing the m variables:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

if $\det(\mathbf{A}) \neq 0$, the solution $\tilde{\mathbf{x}}$ of the system of m linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

is given by the formula:

$$\tilde{x}_j = \frac{\det(\mathbf{A}_j)}{\det(\mathbf{A})}, \quad \forall j \in \{1, 2, \dots, m\} \quad (6.1)$$

where $\mathbf{A}_j$ is the matrix formed by replacing the j -th column of $\mathbf{A}$ by the column vector $\mathbf{b}$.

6.6.1 Systems of two equations and two variables

With $m = 2$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = a_{11} a_{22} - a_{12} a_{21}$$

$$\det(\mathbf{A}_1) = b_1 a_{22} - a_{12} b_2$$

$$\det(\mathbf{A}_2) = a_{11} b_2 - b_1 a_{21}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} a_{11} x_1 + a_{12} x_2 = b_1 \\ a_{21} x_1 + a_{22} x_2 = b_2 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})} \right)$$

Example 6.9: solution of a system of two equations and two variables

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 8 & 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 19 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ 19 & 2 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & 2 \\ 8 & 19 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot 2 - 1 \cdot 8 = -10$$

$$\det(\mathbf{A}_1) = 2 \cdot 2 - 1 \cdot 19 = -15$$

$$\det(\mathbf{A}_2) = (-1) \cdot 19 - 2 \cdot 8 = -35$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 + x_2 = 2 \\ 8x_1 + 2x_2 = 19 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{-15}{-10}, \frac{-35}{-10} \right) = \left(\frac{3}{2}, \frac{7}{2} \right)$$

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} -2 & -1 \\ 0 & -1 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & -2 \\ 1 & 0 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot (-1) - (-1) \cdot 1 = 2$$

$$\det(\mathbf{A}_1) = (-2) \cdot (-1) - (-1) \cdot 0 = 2$$

$$\det(\mathbf{A}_2) = (-1) \cdot 0 - (-2) \cdot 1 = 2$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 & - & x_2 & = & -2 \\ x_1 & - & x_2 & = & 0 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{2}{2}, \frac{2}{2} \right) = (1, 1)$$

6.6.2 Systems of three equations and three variables

With $m = 3$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{pmatrix} \quad \mathbf{A}_3 = \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{pmatrix}$$

and (Sarrus rule)

$$\det(\mathbf{A}) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

$$\det(\mathbf{A}_1) = b_1a_{22}a_{33} + a_{12}a_{23}b_3 + a_{13}b_2a_{32} - a_{13}a_{22}b_3 - a_{12}b_2a_{33} - b_1a_{23}a_{32}$$

$$\det(\mathbf{A}_2) = a_{11}b_2a_{33} + b_1a_{23}a_{31} + a_{13}a_{21}b_3 - a_{13}b_2a_{31} - b_1a_{21}a_{33} - a_{11}a_{23}b_3$$

$$\det(\mathbf{A}_3) = a_{11}a_{22}b_3 + a_{12}b_2a_{31} + b_1a_{21}a_{32} - b_1a_{22}a_{31} - a_{12}a_{21}b_3 - a_{11}b_2a_{32}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_3)}{\det(\mathbf{A})} \right)$$

Example 6.10: solution of a system of three equations and three variables

We solve with Cramer's rule the system of Example 6.1:

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

We have

$$\mathbf{A}_1 = \begin{pmatrix} 4 & 1 & 1 \\ 10 & 3 & 3 \\ 24 & 7 & 9 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} 2 & 4 & 1 \\ 4 & 10 & 3 \\ 8 & 24 & 9 \end{pmatrix} \quad \mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 4 \\ 4 & 3 & 10 \\ 8 & 7 & 24 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = 54 + 24 + 28 - 24 - 36 - 42 = 4$$

$$\det(\mathbf{A}_1) = 108 + 72 + 70 - 72 - 90 - 84 = 4$$

$$\det(\mathbf{A}_2) = 180 + 96 + 96 - 80 - 144 - 144 = 4$$

$$\det(\mathbf{A}_3) = 144 + 80 + 112 - 96 - 96 - 140 = 4$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{4}{4}, \frac{4}{4}, \frac{4}{4} \right) = (1, 1, 1),$$

which is the solution found with Gaussian elimination.

6.6.3 Computational cost

- Cramer's rule gives an explicit formula, which is very useful for small systems ($m = 2$ or $m = 3$) and for theoretical purposes. However, it is **not** a practical method for large systems.
- To apply Cramer's rule we need $m + 1$ determinants of order m : $\det(\mathbf{A})$, $\det(\mathbf{A}_1)$, $\dots$, $\det(\mathbf{A}_m)$. If each determinant is computed with the Laplace expansion, the number of arithmetic operations grows roughly like $m!$ for each determinant.
- Gaussian elimination, instead, solves the system with a number of operations which grows roughly like m^3 (about $\frac{2}{3}m^3$ operations).
- For example, with $m = 10$ we have $10! = 3\,628\,800$, while $\frac{2}{3} \cdot 10^3 \approx 667$; with $m = 20$, $20!$ is larger than $2 \cdot 10^{18}$, while $\frac{2}{3} \cdot 20^3 \approx 5\,333$.
- Even if the determinants are computed in a more efficient way (for instance with Gaussian elimination itself), Cramer's rule still needs $m + 1$ of them, and it remains more expensive than solving the system directly with Gaussian elimination.

6.7 Exercises

Exercise 6.1

Solve with Gaussian elimination the system

$$\begin{cases} 1x_1 + 1x_2 + 1x_3 = 6 \\ 2x_1 + 3x_2 + 1x_3 = 11 \\ 1x_1 - 1x_2 + 2x_3 = 5 \end{cases}$$

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & -1 & 2 & 5 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & 1 & -1 \end{array} \right)$$

$R_3 \leftarrow R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{array} \right)$$

Backward substitution:

$$-x_3 = -3 \implies x_3 = 3, \quad x_2 = -1 + x_3 = 2, \quad x_1 = 6 - x_2 - x_3 = 1.$$

Hence $\mathbf{x} = (1, 2, 3)'$. Verification: $2 + 6 + 3 = 11$ and $1 - 2 + 6 = 5$ ✓

Exercise 6.2

Solve with Gaussian elimination the system

$$\begin{cases} 2x_2 + 1x_3 = -1 \\ 1x_1 - 1x_2 + 2x_3 = 5 \\ 2x_1 + 1x_2 + 1x_3 = 4 \end{cases}$$

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 0 & 2 & 1 & -1 \\ 1 & -1 & 2 & 5 \\ 2 & 1 & 1 & 4 \end{array} \right)$$

The entry in position $(1, 1)$ is zero. $R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 1 & -1 \\ 2 & 1 & 1 & 4 \end{array} \right)$$

$R_3 \leftarrow R_3 - 2 R_1$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 1 & -1 \\ 0 & 3 & -3 & -6 \end{array} \right)$$

$R_3 \leftarrow \frac{1}{3} R_3$, then $R_2 \leftrightarrow R_3$ (to avoid fractions):

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & -1 & -2 \\ 0 & 2 & 1 & -1 \end{array} \right)$$

$R_3 \leftarrow R_3 - 2 R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 3 & 3 \end{array} \right)$$

Backward substitution:

$$x_3 = 1, \quad x_2 = -2 + x_3 = -1, \quad x_1 = 5 + x_2 - 2 x_3 = 5 - 1 - 2 = 2.$$

Hence $\mathbf{x} = (2, -1, 1)'$. Verification: $-2 + 1 = -1$, $2 + 1 + 2 = 5$, $4 - 1 + 1 = 4 \checkmark$

Exercise 6.3

Solve with Gaussian elimination the system below and write its solutions in the parametric form $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$:

$$\begin{cases} 1 x_1 - 1 x_2 + 2 x_3 = 1 \\ 2 x_1 - 1 x_2 + 5 x_3 = 4 \\ 1 x_1 \quad \quad + 3 x_3 = 3 \end{cases}$$

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & -1 & 5 & 4 \\ 1 & 0 & 3 & 3 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2 R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 2 < 3$: infinitely many solutions with $3 - 2 = 1$ free parameter.
Setting $x_3 = t$:

$$x_2 = 2 - t, \quad x_1 = 1 + x_2 - 2x_3 = 1 + 2 - t - 2t = 3 - 3t,$$

$$\mathbf{x} = \begin{pmatrix} 3 - 3t \\ 2 - t \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Exercise 6.4

Solve the system of 2 equations in 4 variables

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 + 1x_4 = 2 \\ 2x_1 + 4x_2 - 1x_3 + 3x_4 = 5 \end{cases}$$

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 2 & 4 & -1 & 3 & 5 \end{array} \right) \quad R_2 \leftarrow R_2 - 2R_1 : \quad \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right)$$

$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 2 < 4$: infinitely many solutions with $4 - 2 = 2$ free parameters.
The pivots are in the columns of x_1 and x_3 , so $x_2 = s$ and $x_4 = t$ are free:

$$x_3 = 1 - x_4 = 1 - t, \quad x_1 = 2 - 2x_2 + x_3 - x_4 = 2 - 2s + 1 - t - t = 3 - 2s - 2t,$$

$$\mathbf{x} = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

Exercise 6.5

Show that the following system has no solution:

$$\begin{cases} 1x_1 + 2x_2 + 1x_3 = 1 \\ 2x_1 + 3x_2 + 1x_3 = 2 \\ 3x_1 + 5x_2 + 2x_3 = 4 \end{cases}$$

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 3 & 1 & 2 \\ 3 & 5 & 2 & 4 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2 R_1$ and $R_3 \leftarrow R_3 - 3 R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 1 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row reads $0 = 1$: $\text{rank}(\mathbf{A}) = 2 < 3 = \text{rank}(\mathbf{A}|\mathbf{b})$, hence by the Rouché–Capelli theorem the system has **no solution**. (Note that the third row of $\mathbf{A}$ is the sum of the first two, while $4 \neq 1 + 2$.)

Exercise 6.6

The following augmented matrices are already in row echelon form. Using the Rouché–Capelli theorem, say whether each system has a unique solution, infinitely many solutions or no solution (variables x_1, x_2, x_3). Then solve the systems which have solutions.

$$\text{a) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 5 & 3 \end{array} \right) \quad \text{b) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad \text{c) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{array} \right)$$

Solution

a) $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 3 = n$: **unique solution**. $x_3 = \frac{3}{5}$, $x_2 = 2 - 4 x_3 = -\frac{2}{5}$, $x_1 = 1 - 2 x_2 - 3 x_3 = 1 + \frac{4}{5} - \frac{9}{5} = 0$. Hence $\mathbf{x} = (0, -\frac{2}{5}, \frac{3}{5})'$.

b) $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 2 < 3$: **infinitely many solutions** with 1 free parameter. The pivots are in the columns of x_1 and x_3 , so $x_2 = t$ is free: $x_3 = 2$, $x_1 = 1 - 2t - 3 \cdot 2 = -5 - 2t$, i.e., $\mathbf{x} = (-5, 0, 2)' + t(-2, 1, 0)'$, $t \in \mathbb{R}$.

c) $\text{rank}(\mathbf{A}) = 2 < 3 = \text{rank}(\mathbf{A}|\mathbf{b})$ (the last row reads $0 = 4$): **no solution**.

Exercise 6.7

Discuss, as $k \in \mathbb{R}$ varies, the number of solutions of the system

$$\begin{cases} 1 x_1 + 1 x_2 + k x_3 = 1 \\ 1 x_1 + k x_2 + 1 x_3 = 1 \\ k x_1 + 1 x_2 + 1 x_3 = 1 \end{cases}$$

and solve it whenever it has solutions.

Solution

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 1 & k & 1 & 1 \\ k & 1 & 1 & 1 \end{array} \right)$$

$R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - k R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & k-1 & 1-k & 0 \\ 0 & 1-k & 1-k^2 & 1-k \end{array} \right)$$

$R_3 \leftarrow R_3 + R_2$ (note that $(1-k) + (1-k^2) = (1-k)(2+k)$):

$$\left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & k-1 & 1-k & 0 \\ 0 & 0 & (1-k)(2+k) & 1-k \end{array} \right)$$

- **Case $k \neq 1$ and $k \neq -2$:** three pivots, $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 3$, **unique solution**. Dividing by $1-k \neq 0$: $(2+k)x_3 = 1$, i.e., $x_3 = \frac{1}{k+2}$; the second row gives $(k-1)(x_2 - x_3) = 0$, i.e., $x_2 = x_3 = \frac{1}{k+2}$; finally $x_1 = 1 - x_2 - k x_3 = 1 - \frac{1+k}{k+2} = \frac{1}{k+2}$. Hence

$$\mathbf{x} = \frac{1}{k+2} (1, 1, 1)'$$

- **Case $k = 1$:** the second and third rows become zero rows. $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 1 < 3$: **infinitely many solutions** with 2 free parameters. The system reduces to $x_1 + x_2 + x_3 = 1$; with $x_2 = s$, $x_3 = t$: $\mathbf{x} = (1 - s - t, s, t)'$, $s, t \in \mathbb{R}$.
- **Case $k = -2$:** the last row becomes $(0 \ 0 \ 0 \mid 3)$, i.e., $0 = 3$. $\text{rank}(\mathbf{A}) = 2 < 3 = \text{rank}(\mathbf{A}|\mathbf{b})$: **no solution**.

(Consistently, $\det(\mathbf{A}) = -(k-1)^2(k+2)$ vanishes exactly for $k = 1$ and $k = -2$.)

Exercise 6.8

Find the values of $k \in \mathbb{R}$ for which the homogeneous system

$$\begin{cases} 1x_1 + 2x_2 + 1x_3 = 0 \\ 2x_1 + kx_2 + 3x_3 = 0 \\ 1x_1 + 1x_2 + 2x_3 = 0 \end{cases}$$

has nontrivial solutions, and compute them.

Solution

The matrix $\mathbf{A}$ is square, so there are nontrivial solutions if and only if $\det(\mathbf{A}) = 0$. By the Sarrus rule:

$$\det(\mathbf{A}) = 2k + 6 + 2 - k - 8 - 3 = k - 3,$$

hence nontrivial solutions exist if and only if $k = 3$ (for $k \neq 3$ the only solution is $\mathbf{x} = \mathbf{0}$).

For $k = 3$, $R_2 \leftarrow R_2 - 2 R_1$ and $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 3 & 3 & 0 \\ 1 & 1 & 2 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Setting $x_3 = t$: $x_2 = x_3 = t$ and $x_1 = -2 x_2 - x_3 = -3t$. Hence

$$\mathbf{x} = t (-3, 1, 1)', \quad t \in \mathbb{R}.$$

Exercise 6.9

Solve with Cramer's rule the system

$$\begin{cases} 3 x_1 + 2 x_2 = 7 \\ 1 x_1 - 1 x_2 = -1 \end{cases}$$

Solution

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \quad \mathbf{A}_1 = \begin{pmatrix} 7 & 2 \\ -1 & -1 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} 3 & 7 \\ 1 & -1 \end{pmatrix}$$

$$\det(\mathbf{A}) = 3 \cdot (-1) - 2 \cdot 1 = -5$$

$$\det(\mathbf{A}_1) = 7 \cdot (-1) - 2 \cdot (-1) = -5$$

$$\det(\mathbf{A}_2) = 3 \cdot (-1) - 7 \cdot 1 = -10$$

Since $\det(\mathbf{A}) \neq 0$:

$$(\tilde{x}_1, \tilde{x}_2) = \left(\frac{-5}{-5}, \frac{-10}{-5} \right) = (1, 2).$$

Verification: $3 + 4 = 7$ and $1 - 2 = -1$ ✓

Exercise 6.10

Solve with Cramer's rule the system

$$\begin{cases} 1 x_1 + 1 x_2 + 1 x_3 = 6 \\ 1 x_1 - 1 x_2 + 2 x_3 = 5 \\ 2 x_1 + 1 x_2 - 1 x_3 = 1 \end{cases}$$

Solution

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{pmatrix} \quad \mathbf{A}_1 = \begin{pmatrix} 6 & 1 & 1 \\ 5 & -1 & 2 \\ 1 & 1 & -1 \end{pmatrix}$$

$$\mathbf{A}_2 = \begin{pmatrix} 1 & 6 & 1 \\ 1 & 5 & 2 \\ 2 & 1 & -1 \end{pmatrix} \quad \mathbf{A}_3 = \begin{pmatrix} 1 & 1 & 6 \\ 1 & -1 & 5 \\ 2 & 1 & 1 \end{pmatrix}$$

By the Sarrus rule:

$$\det(\mathbf{A}) = 1 + 4 + 1 - (-2) - (-1) - 2 = 7$$

$$\det(\mathbf{A}_1) = 6 + 2 + 5 - (-1) - (-5) - 12 = 7$$

$$\det(\mathbf{A}_2) = -5 + 24 + 1 - 10 - (-6) - 2 = 14$$

$$\det(\mathbf{A}_3) = -1 + 10 + 6 - (-12) - 1 - 5 = 21$$

Since $\det(\mathbf{A}) \neq 0$:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{7}{7}, \frac{14}{7}, \frac{21}{7} \right) = (1, 2, 3).$$

Verification: $1 + 2 + 3 = 6$, $1 - 2 + 6 = 5$, $2 + 2 - 3 = 1$ ✓

Exercise 6.11

Given the LU factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$ with

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix},$$

solve the system $\mathbf{A} \mathbf{x} = \mathbf{b}$ with $\mathbf{b} = (-1, 0, 13)'$.

Solution

Step 1 - Forward substitution: solve $\mathbf{L} \mathbf{y} = \mathbf{b}$:

$$y_1 = -1, \quad y_2 = 0 - 3 y_1 = 3, \quad y_3 = 13 - (-1) y_1 - 2 y_2 = 13 - 1 - 6 = 6.$$

Step 2 - Backward substitution: solve $\mathbf{U} \mathbf{x} = \mathbf{y}$ with $\mathbf{y} = (-1, 3, 6)'$:

$$3 x_3 = 6 \implies x_3 = 2, \quad x_2 = 3 - 2 x_3 = -1,$$

$$2 x_1 = -1 - x_2 + x_3 = -1 + 1 + 2 = 2 \implies x_1 = 1.$$

Hence $\mathbf{x} = (1, -1, 2)'$.

Verification:

$$\mathbf{A} = \mathbf{L} \mathbf{U} = \begin{pmatrix} 2 & 1 & -1 \\ 6 & 4 & -1 \\ -2 & 1 & 8 \end{pmatrix}, \quad \mathbf{A} \mathbf{x} = \begin{pmatrix} 2 - 1 - 2 \\ 6 - 4 - 2 \\ -2 - 1 + 16 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 13 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

Further topics

A.1 Absolute value

A.1.1 Definition

Definition A.1: absolute value

The **absolute value** $|a|$ of a real number a is the following non-negative real number:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} \quad (\text{A.1})$$

From the definition of absolute value it immediately follows that:

$$|a| < \varepsilon \iff -\varepsilon < a < \varepsilon, \quad \forall \varepsilon > 0, a \in \mathbb{R} \quad (\text{A.2})$$

and, similarly, for non-strict inequalities:

$$|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon, \quad \forall \varepsilon \geq 0, a \in \mathbb{R} \quad (\text{A.3})$$

A.1.2 Triangle inequality

Observation A.2: triangle inequality

$$|b + c| \leq |b| + |c|, \quad \forall b, c \in \mathbb{R} \quad (\text{A.4})$$

Proof. We write the two relations:

$$-|b| \leq b \leq |b|, \quad -|c| \leq c \leq |c|$$

and add member to member:

$$-(|b| + |c|) \leq b + c \leq |b| + |c|$$

So for (A.3), with $\varepsilon = |b| + |c| \geq 0$ and $a = b + c$, it follows (A.4). $\square$

- The triangle inequality is also used in the following form:

$$|d - e| \leq |d - f| + |e - f| \quad \forall d, e, f \in \mathbb{R} \quad (\text{A.5})$$

To obtain it, it suffices to set in (A.4):

$$b = d - f, \quad c = f - e$$

and we get:

$$|d - f + f - e| = |d - e| \leq |d - f| + |f - e| = |d - f| + |e - f|$$

since:

$$|f - e| = |e - f| \quad \forall e, f \in \mathbb{R}$$

- Finally, another form of the triangle inequality is:

$$|g| \leq |g - h| + |h| \quad \text{that is} \quad |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R} \quad (\text{A.6})$$

To obtain it, it suffices to set in (A.4):

$$b = g - h, \quad c = h$$

Observation A.3: reverse triangle inequality

$$||g| - |h|| \leq |g - h|, \quad \forall g, h \in \mathbb{R} \quad (\text{A.7})$$

Proof. From (A.6), we have:

$$|g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Similarly exchanging g for h in (A.6) we get:

$$|h| - |g| \leq |h - g| = |g - h| \quad \text{that is} \quad |g| - |h| \geq -|g - h|$$

then we have

$$-(|g - h|) \leq |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Accordingly from (A.3), with $\varepsilon = |g - h| \geq 0$ and $a = |g| - |h|$, we get the reverse triangle inequality. □

- The triangle inequality (A.4) can easily be generalized to the case of k addends:

$$\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i|. \quad (\text{A.8})$$

- The following immediate properties also hold:

$$|b \, c| = |b| \, |c|, \quad \left| \frac{b}{c} \right| = \frac{|b|}{|c|}, \quad |-b| = |b| \quad \forall b, c \in \mathbb{R} \, (c \neq 0 \text{ in the quotient}). \quad (\text{A.9})$$

A.1.3 Exercises

Exercise A.1

Using the definition of absolute value, compute:

$$\text{a) } |-3| + |2-7| - |-4| \left| \frac{1}{2} \right|, \quad \text{b) } |3-\pi|, \quad \text{c) } \left| \frac{1}{3} - \frac{1}{2} \right|$$

Solution

a) Since $-3 < 0$, $2-7 = -5 < 0$, $-4 < 0$ and $\frac{1}{2} \geq 0$:

$$|-3| + |2-7| - |-4| \left| \frac{1}{2} \right| = 3 + 5 - 4 \cdot \frac{1}{2} = 6$$

b) Since $\pi > 3$, we have $3-\pi < 0$ and then $|3-\pi| = -(3-\pi) = \pi-3$.

c) Since $\frac{1}{3} - \frac{1}{2} = -\frac{1}{6} < 0$, we have $\left| \frac{1}{3} - \frac{1}{2} \right| = \frac{1}{6}$.

Exercise A.2

Prove that for every real number b we have:

$$\text{a) } |b| \geq 0 \text{ and } |b| = 0 \iff b = 0, \quad \text{b) } -|b| \leq b \leq |b|, \quad \text{c) } |b|^2 = b^2$$

Solution

We distinguish the two cases of the definition of absolute value.

- If $b \geq 0$, then $|b| = b \geq 0$; moreover $|b| = 0$ if and only if $b = 0$; we have $-|b| = -b \leq 0 \leq b = |b|$; finally $|b|^2 = b^2$.
- If $b < 0$, then $|b| = -b > 0$ (in particular $|b| \neq 0$); we have $-|b| = b < 0 < -b = |b|$; finally $|b|^2 = (-b)^2 = b^2$.

In both cases a), b) and c) hold.

Exercise A.3

Find all real numbers x such that:

$$\text{a) } |x-2| < 3, \quad \text{b) } |2x+1| \leq 5, \quad \text{c) } |x-4| \geq 1$$

Solution

a) From $|a| < \varepsilon \iff -\varepsilon < a < \varepsilon$ with $a = x-2$ and $\varepsilon = 3$:

$$-3 < x-2 < 3 \iff -1 < x < 5$$

b) From $|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon$ with $a = 2x + 1$ and $\varepsilon = 5$:

$$-5 \leq 2x + 1 \leq 5 \iff -6 \leq 2x \leq 4 \iff -3 \leq x \leq 2$$

c) The inequality $|x-4| \geq 1$ is false exactly when $|x-4| < 1$, that is when $-1 < x-4 < 1$, i.e., $3 < x < 5$. Then the solutions are:

$$x \leq 3 \quad \text{or} \quad x \geq 5$$

Exercise A.4

Find all real numbers x such that $|x - 1| = |x + 3|$.

Solution

Two real numbers have the same absolute value if and only if they are equal or opposite. Then:

- $x - 1 = x + 3$ gives $-1 = 3$, which is impossible;
- $x - 1 = -(x + 3)$ gives $2x = -2$, that is $x = -1$.

The only solution is $x = -1$. Check: $|-1 - 1| = 2 = |-1 + 3|$.

Exercise A.5

Find all real numbers x such that $|x| + |x - 2| = 4$.

Solution

The signs of x and $x - 2$ change at $x = 0$ and $x = 2$, so we distinguish three cases:

- $x < 0$: $|x| = -x$ and $|x - 2| = 2 - x$, then $-2x + 2 = 4$, i.e., $x = -1$, which satisfies $x < 0$;
- $0 \leq x < 2$: $|x| = x$ and $|x - 2| = 2 - x$, then $x + 2 - x = 2 = 4$, which is impossible;
- $x \geq 2$: $|x| = x$ and $|x - 2| = x - 2$, then $2x - 2 = 4$, i.e., $x = 3$, which satisfies $x \geq 2$.

The solutions are $x = -1$ and $x = 3$.

Exercise A.6

Prove that for every pair of real numbers b and c :

$$|b + c| = |b| + |c| \iff bc \geq 0$$

Solution

Both $|b + c|$ and $|b| + |c|$ are non-negative, and two non-negative numbers are equal if and only if their squares are equal. Using Exercise A.1.3 c) and $|bc| = |b||c|$, we have:

$$|b + c|^2 = (b + c)^2 = b^2 + 2bc + c^2, \quad (|b| + |c|)^2 = b^2 + 2|bc| + c^2$$

Then $|b + c| = |b| + |c|$ if and only if $bc = |bc|$, that is (by definition of absolute value) if and only if $b \geq 0$.

Exercise A.7

Verify the triangle inequality $|b + c| \leq |b| + |c|$ and the reverse triangle inequality $||b| - |c|| \leq |b - c|$ for $b = 5$ and $c = -3$.

Solution

We have $|b| = 5$ and $|c| = 3$.

- Triangle inequality: $|b + c| = |5 - 3| = 2 \leq 8 = |b| + |c|$.
- Reverse triangle inequality: $||b| - |c|| = |5 - 3| = 2 \leq 8 = |5 - (-3)| = |b - c|$.

The triangle inequality holds with the strict inequality, in agreement with Exercise A.1.3, since $bc = -15 < 0$.

Exercise A.8

Let x be a real number such that $|x - 3| \leq 1$. Prove that:

$$\text{a) } |x| \leq 4, \quad \text{b) } |x^2 - 9| \leq 7$$

and show that the bound in b) cannot be improved.

Solution

- a) From the form $|g| \leq |g - h| + |h|$ of the triangle inequality, with $g = x$ and $h = 3$:

$$|x| \leq |x - 3| + |3| \leq 1 + 3 = 4$$

- b) We have $x^2 - 9 = (x - 3)(x + 3)$ and then $|x^2 - 9| = |x - 3| |x + 3|$. From the triangle inequality:

$$|x + 3| = |(x - 3) + 6| \leq |x - 3| + 6 \leq 7$$

$$\text{and therefore } |x^2 - 9| = |x - 3| |x + 3| \leq 1 \cdot 7 = 7.$$

The bound cannot be improved: for $x = 4$ (which satisfies $|4 - 3| = 1$) we have $|16 - 9| = 7$.

A.2 Modular arithmetic

A.2.1 Euclid's division

Lemma A.4: Euclid's division (or division with remainder)

For every pair of integer numbers n and m , with $m \neq 0$, there exist unique integer numbers q and r such that

$$n = qm + r \quad \text{and} \quad 0 \leq r < |m| \quad (\text{A.10})$$

- The value n is called the **dividend**, the value m is called the **divisor**, the value q is called the **quotient** and the value r is called the **remainder**.
- The standard convention is that the remainder is non negative but there are other conventions where the remainder r can be negative.

Proof.

- To prove *existence*, we consider the set

$$S = \{ n - km : k \in \mathbb{Z} \text{ and } n - km \geq 0 \}$$

The set S is nonempty, since for $k = -|n|m$ we have $n - km = n + |n|m^2 \geq n + |n| \geq 0$. Since S is a nonempty subset of the nonnegative integers, the well-ordering principle guarantees that there exists a smallest element $r \in S$, meaning there exists some integer q such that

$$r = n - qm, \quad \text{with } r \geq 0$$

By construction, we must have $r < |m|$, since if $r \geq |m|$, then we could write

$$r' = r - |m| = n - \left(q + \frac{|m|}{m}\right)m \geq 0$$

which would imply $r' \in S$, contradicting the minimality of r .

- To prove *uniqueness*, suppose there exist two pairs (q_1, r_1) and (q_2, r_2) such that

$$n = q_1m + r_1 = q_2m + r_2$$

with $0 \leq r_1, r_2 < |m|$. Subtracting the two equations, we have

$$(q_1 - q_2)m = r_2 - r_1$$

Since $|r_2 - r_1| < |m|$ and m divides $r_2 - r_1$ (which equals the left-hand side), the only possibility is $r_1 = r_2$, which implies $q_1 = q_2$, ensuring uniqueness.

□

Definition A.5: multiple

An integer number n is a **multiple** of an integer number m if there exists an integer number q such that

$$n = q \cdot m \quad (\text{A.11})$$

Definition A.6: factor

An integer number m is a **factor** of an integer number n if there exists an integer number q such that

$$n = q \cdot m \quad (\text{A.12})$$

- An integer number n is **divisible** by an integer number m if m is a factor of n .

A.2.2 Modulo and modulo congruence

Given two integer numbers n and m , with $m \neq 0$, n **modulo** m (abbreviated in $n \bmod m$) is the **remainder** r of the Euclidean division (or division with remainder) of n by m , where n is the **dividend** and m is the **divisor**.

- It follows from the definition that:

$$0 \leq n \bmod m < |m| \quad \forall n, m \in \mathbb{Z}, m \neq 0$$

Two natural numbers n and k are **congruent** modulo a natural number $m \geq 1$ if

$$n \bmod m = k \bmod m$$

and we write

$$n \equiv k \pmod{m}$$

- The parentheses mean that $(\bmod m)$ applies to the entire equation, not just to the right-hand side (here, k).
- Equivalently, we have $n \equiv k \pmod{m}$
 1. if n and k have the same remainder when divided by m .
 2. if m is a divisor of $|n-k|$, i.e., if there exists a natural number q such that $|n-k| = q \cdot m$ or equivalently if there exists an integer number q such that $n - k = q \cdot m$

Example A.1: congruence modulo 5

For example, 23 and 13 are congruent modulo 5 and we have $23 \equiv 13 \pmod{5}$

A.2.3 Exercises

Exercise A.9

Find the quotient q and the remainder r (with $0 \leq r < |m|$) of Euclid's division of n by m in the following cases:

- a) $n = 47, m = 5,$ b) $n = -47, m = 5,$
 c) $n = 47, m = -5,$ d) $n = -47, m = -5$

Solution

In each case we look for the integers q and r such that $n = qm + r$ and $0 \leq r < 5$:

- a) $47 = 9 \cdot 5 + 2$, then $q = 9$ and $r = 2$.
 b) $-47 = (-10) \cdot 5 + 3$, then $q = -10$ and $r = 3$. Note that $-47 = (-9) \cdot 5 - 2$ is not Euclid's division, since $-2 < 0$.
 c) $47 = (-9) \cdot (-5) + 2$, then $q = -9$ and $r = 2$.
 d) $-47 = 10 \cdot (-5) + 3$, then $q = 10$ and $r = 3$.

Exercise A.10

Compute:

- a) $100 \bmod 7,$ b) $-100 \bmod 7,$ c) $2025 \bmod 9,$ d) $7 \bmod 12$

Solution

By definition, $n \bmod m$ is the remainder of Euclid's division of n by m :

- a) $100 = 14 \cdot 7 + 2$, then $100 \bmod 7 = 2$.
 b) $-100 = (-15) \cdot 7 + 5$, then $-100 \bmod 7 = 5$.
 c) $2025 = 225 \cdot 9 + 0$, then $2025 \bmod 9 = 0$ (2025 is a multiple of 9).
 d) $7 = 0 \cdot 12 + 7$, then $7 \bmod 12 = 7$.

Exercise A.11

Establish whether the following congruences hold, using both the definition (same remainder) and the divisibility of the difference:

- a) $38 \equiv 14 \pmod{6},$ b) $17 \equiv 4 \pmod{5},$ c) $100 \equiv 1 \pmod{11}$

Solution

- a) $38 = 6 \cdot 6 + 2$ and $14 = 2 \cdot 6 + 2$ have the same remainder 2; equivalently $38 - 14 = 24 = 4 \cdot 6$. The congruence holds.

- b) $17 = 3 \cdot 5 + 2$ and $4 = 0 \cdot 5 + 4$ have different remainders; equivalently $17 - 4 = 13$ is not a multiple of 5. The congruence does not hold.
- c) $100 = 9 \cdot 11 + 1$ and $1 = 0 \cdot 11 + 1$ have the same remainder 1; equivalently $100 - 1 = 99 = 9 \cdot 11$. The congruence holds.

Exercise A.12

Let $m \geq 1$ be a natural number. Prove that for every pair of integer numbers n and k :

$$n \bmod m = k \bmod m \iff \text{there exists an integer } q \text{ such that } n - k = q \cdot m$$

Solution

($\Rightarrow$) Let $r = n \bmod m = k \bmod m$. By Euclid's division there exist integers q_1 and q_2 such that $n = q_1 m + r$ and $k = q_2 m + r$. Subtracting:

$$n - k = (q_1 - q_2) m$$

so the claim holds with $q = q_1 - q_2$.

($\Leftarrow$) Let $n - k = q m$ and let $k = q_2 m + r$ with $0 \leq r < m$ (Euclid's division of k by m). Then:

$$n = k + q m = (q + q_2) m + r \quad \text{with} \quad 0 \leq r < m$$

By the uniqueness in Euclid's division, r is the remainder of the division of n by m , that is $n \bmod m = r = k \bmod m$.

Exercise A.13

Let $m \geq 1$, n, n', k, k' be natural numbers such that $n \equiv n' \pmod{m}$ and $k \equiv k' \pmod{m}$. Prove that:

$$\text{a) } n + k \equiv n' + k' \pmod{m}, \quad \text{b) } n k \equiv n' k' \pmod{m}$$

Then compute $(123 \cdot 456 + 789) \bmod 10$ without computing the product.

Solution

By Exercise A.2.3 there exist integers a and b such that $n = n' + a m$ and $k = k' + b m$.

a) $n + k = n' + k' + (a + b) m$, then $(n + k) - (n' + k')$ is a multiple of m .

b) $n k = (n' + a m) (k' + b m) = n' k' + (n' b + k' a + a b m) m$, then $n k - n' k'$ is a multiple of m .

Since $123 \equiv 3$, $456 \equiv 6$ and $789 \equiv 9 \pmod{10}$, we have:

$$123 \cdot 456 + 789 \equiv 3 \cdot 6 + 9 = 27 \equiv 7 \pmod{10}$$

then $(123 \cdot 456 + 789) \bmod 10 = 7$. Check: $123 \cdot 456 + 789 = 56088 + 789 = 56877$.

Exercise A.14

Compute:

$$\text{a) } 2^{10} \bmod 7, \quad \text{b) } 3^{100} \bmod 4, \quad \text{c) } 5^{21} \bmod 6$$

Solution

We use Exercise A.2.3 b) repeatedly: if $n \equiv n' \pmod{m}$, then $n^p \equiv (n')^p \pmod{m}$ for every natural number $p \geq 1$.

a) $2^3 = 8 \equiv 1 \pmod{7}$, then $2^{10} = (2^3)^3 \cdot 2 \equiv 1^3 \cdot 2 = 2 \pmod{7}$. Then $2^{10} \bmod 7 = 2$ (check: $1024 = 146 \cdot 7 + 2$).

b) $3^2 = 9 \equiv 1 \pmod{4}$, then $3^{100} = (3^2)^{50} \equiv 1^{50} = 1 \pmod{4}$. Then $3^{100} \bmod 4 = 1$.

c) $5^2 = 25 \equiv 1 \pmod{6}$, then $5^{21} = (5^2)^{10} \cdot 5 \equiv 1^{10} \cdot 5 = 5 \pmod{6}$. Then $5^{21} \bmod 6 = 5$.

Exercise A.15

Prove that for every integer number n we have $n^2 \bmod 4 \in \{0, 1\}$. Deduce that 2023 is not the square of an integer number.

Solution

By Euclid's division of n by 2, every integer n is either even, $n = 2t$, or odd, $n = 2t + 1$, with t integer:

- if $n = 2t$, then $n^2 = 4t^2 = 4t^2 + 0$, so $n^2 \bmod 4 = 0$;
- if $n = 2t + 1$, then $n^2 = 4t^2 + 4t + 1 = 4(t^2 + t) + 1$, so $n^2 \bmod 4 = 1$.

Since $2023 = 505 \cdot 4 + 3$, we have $2023 \bmod 4 = 3 \notin \{0, 1\}$, then 2023 is not the square of an integer number.

Exercise A.16

Prove that every natural number is congruent modulo 9 to the sum of its decimal digits. Use this result to compute $123456789 \bmod 9$ and $20251007 \bmod 9$.

Solution

A natural number n with decimal digits $c_p, c_{p-1}, \dots, c_1, c_0$ can be written as:

$$n = \sum_{j=0}^p c_j 10^j$$

Since $10 = 1 \cdot 9 + 1$, we have $10 \equiv 1 \pmod{9}$ and then (Exercise A.2.3) $10^j \equiv 1 \pmod{9}$

for every $j \geq 1$ (and trivially for $j = 0$). Using Exercise A.2.3 for each term of the sum:

$$n = \sum_{j=0}^p c_j 10^j \equiv \sum_{j=0}^p c_j \pmod{9}$$

- $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45 = 5 \cdot 9$, then $123456789 \bmod 9 = 0$.
- $2 + 0 + 2 + 5 + 1 + 0 + 0 + 7 = 17 = 1 \cdot 9 + 8$, then $20251007 \bmod 9 = 8$.